The Lonely Runner Problem on Pair-Sum Lattices: A Lossless Finitization and a Conditional Scaling-Closure Theorem for the Covering Obstruction

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Abstract

We prove an exact identity that reduces the maximum loneliness of a Lonely Runner instance to a finite maximum over cyclic groups: for every speed vector V of n positive integers, $M(V) = \max_{\{u,w\}} \tau(u,w)$, where $\tau(u,w)$ is the optimal loneliness on the pair-sum lattice of denominator $N = v_u + v_w$. The analytic core of the argument is folklore (it appears in a comment on Tao's 2015 blog post); the contributions claimed here are the sharpening that sum lattices alone always suffice, the exact equality, and the lossless reformulation of the Lonely Runner Conjecture (LRC-$_n$) as the finite statement (TAU-$_n$). The reformulation is a *per-instance* finitization: for each fixed speed vector V it replaces the one-parameter supremum defining $M(V)$ by a finite maximum over the $\binom{n}{2}$ pair-sum lattices of denominators $N = v_u + v_w$, but the denominators grow with V and no uniform bound on N is claimed or implied. The reformulation turns the conjecture into a family of covering problems: a pair certifies iff $n-1$ explicit bad sets $B_w = \{k \in \mathbb{Z}_N : (n+1)\,\|wk\|_N < N\}$ fail to cover $\mathbb{Z}_N$. We develop this covering geometry: an exact size law, a class-group reduction at prime moduli, a lift law, and complete classifications of the unit coverings at the prime moduli $7, 11, 13$ (three bad sets) and $7, 13, 17, 19, 37$ (four bad sets)—at every such prime, covering is governed by a short explicit list of translate classes in a cyclic group. Empirically, these five primes form the entire unit-capable *rigid zone* at $n = 5$ within the verified range (primes ≤ 150, composites ≤ 111), a striking but unproved finiteness statement. On the full corpus of 3,712,576 primitive speed sets with $n = 5$ and speeds ≤ 56, proved lemmas close 75.6% of instances (78.2% including a coincidence-counting lemma), per-modulus verification closes all of them, and the residual obstruction is localized to a single named class (mixed order signatures at composite moduli). The proved-coverage rate declines as the corpus grows (93.4% $\to$ 78.2% over $V = 24 \to 56$ on the augmented battery, crossing below the continuation threshold at $V = 56$); the trend does not establish that coverage stabilizes, and its asymptotics are unresolved. We report the decline, the localization, and the exact verification record as the paper's empirical findings. The paper's culminating result is stated as a *conditional* theorem (Theorem 2.1): under three named open hypotheses—the multi-fiber survivor inequality Hq (the $\pm$-representative form; it subsumes the dilate-sparsity input H2m as its one-step case), the distinct-family volume law H1g with its uniform margin, and the resonance budget H2r—the fiber cascade closes the covering obstruction at the frontier cells $(1K, T{-}3U)$ of every rung $T \geq 8$ at $N = p^2$ within a depth that is T-flat in practice and never exceeds $T + 3$ in the worst case. The reduction chain and every other input (normal form, reflection fiber, chain-class theorem, renormalization lemma, lossless equivalence) are proved or machine-validated, and each constant's provenance—derived, measured, or fitted, with the data it is fit to—is disclosed in the hypothesis ledger. That boundary between the proved and the hypothesized has a structural type, verified statement-by-statement in §9.7: every proved result in this program is a closure, invariance, exhaustiveness, or machine-verification statement, while every remaining obligation is a uniform growth-in-k margin over a trivial or measured baseline—a statement type no turn of the program has proved. This type mismatch is why the paper stops where it stops. The one candidate instrument shaped to produce that missing type was a transfer-matrix or automaton model of the strand census. It was executed as a proof attempt under a pre-committed success criterion and closed by theorem *in its single-anchor scope* (for automata built on the cascade's bijective drift) (Lemma 9.13). The cascade's single-anchor constrained transfer matrices are partial permutations with a spectral radius of exactly 0 or 1 at every scale. The cyclic product is diagonal, with trace equal to the survivor count itself, and the missing margin is provably external to the single-anchor lossless dynamics. The reformulation is bounded by its own fidelity at the single-anchor level: the losslessness that makes it a faithful rewrite is the property that withholds the decay a margin would require for that operator. The general strand-census operator branches and is not covered by the lemma. This is a structural paper: it does not solve or reduce the Lonely Runner Conjecture, proves no new cases of it, and claims nothing beyond the verified ranges stated in the text; the three open hypotheses Hq$_\pm$, H1g, H2r of Theorem 2.1 are not silently established anywhere.

1 Introduction

1.1 The conjecture and its status

Let $n \geq 2$ and let $V = (v_1, \ldots, v_n)$ be distinct positive integer speeds. Runners start together at the origin of the unit circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ at time 0; runner p occupies $v_p t \bmod 1$ at time t. Write $\|x\|$ for the distance of a real number to the nearest integer. The *loneliness* of the stationary observer (runner 0) at time t is

$$g(t) \;=\; \min_{1 \leq p \leq n} \|v_p t\|, \qquad M(V) \;=\; \max_{t \in \mathbb{T}} g(t).$$

The Lonely Runner Conjecture (LRC-n), posed by Wills in 1967 and independently by Cusick in the language of view obstruction [1, 2, 3], asserts $M(V) \geq 1/(n{+}1)$ for every V with $\gcd(V) = 1$; by translation symmetry this is the statement that every runner on the track becomes lonely at some time. The conjecture is known for at most seven runners (Bohman–Bohman–Holzman and Barajas–Serra; see the survey [1] for history and precise references). The frontier has since moved entirely into computer-assisted territory: eight runners [4], nine and ten runners [5], eleven through thirteen runners [6], and fourteen and fifteen runners [7], the last with public certificates, resting on the linearly-exponential checking bounds of Malikiosis–Santos–Schymura [8].

Two neighboring literatures study the fine structure of the problem rather than new rungs. The *spectrum* line computes exact optimal values $\mathrm{ML} = p/q$ and their denominators [9, 10, 11, 12], including tight-instance classifications [13]. The *relations* line localizes counterexamples and tight instances to low-weight integer hyperplanes in parameter space [14]. The closest formulation in spirit is Rifford's [15]: he studies the *time* it takes a runner to become lonely, shows that bounding that time by a number of rounds is equivalent to a covering problem in dimension $n - 2$ on the continuous torus, formulates a covering conjecture there and proves it for $n \leq 6$, and bounds the one-round gap of loneliness by $1/(2m - 1)$. What the two formulations share is the move from runners to coverings; what separates them is the domain and the mode. Rifford's covering problem is continuous and its conjecture is a sufficient condition for a time bound, while the present reduction is finite and exact—an identity, not an inequality—reducing the threshold question itself to coverings of the cyclic groups $\mathbb{Z}_{v_u+v_w}$ by multiplicative bad sets. That finiteness is what makes the arithmetic visible: nothing corresponding to the rigid zone, the translate-class classifications, or the kernel world appears in, or is excluded by, the continuous formulation, and none of our results overlap his. The present paper is a reformulation paper whose subject is the geometry of the obstruction itself, at small moduli, in residue language.

1.2 What this paper does

The starting point is a classical observation: the optimal time is rational, and its denominator is the sum or difference of two speeds. In the form relevant here—"the maximum is attained at an intersection of two sine curves, so the optimal time is rational with denominator the sum or difference of two integer velocities"—this appears in a comment on Tao's blog post of 13 May 2015 [16]. We treat that technique as known folklore and cite it as such. What the folklore does not record, and what this paper builds on, is the following sharpening and its consequences.

1. **The sum-only equality theorem** (§3). At a global optimum with $0 < M < \frac{1}{2}$ the binding runners can be chosen two-sided, which forces the optimal time onto the lattice of a *sum* $v_u + v_w$; difference denominators never carry the optimum. On that lattice the two binding runners are exact reflections of each other at *every* lattice time (Lemma 3.4), so the n-runner minimum there equals an $(n{-}1)$-runner minimum. Together these give the exact identity $M(V) = \max_{\{u,w\}} \tau(u, w)$ of Theorem 3.2—not an inequality—and the lossless equivalence of Corollary 3.3: LRC-n holds iff (TAU-n) does, a *finite* statement about explicit cyclic groups $\mathbb{Z}_{v_u+v_w}$. The finitization is exact: no information is lost, and nothing about the classical difficulty of the conjecture is claimed to become easier.

2. **The covering framework** (§4). Pair certification becomes a covering question: a pair fails iff its $n-1$ bad sets $B_w = \{k : (n+1)\|wk\|_N < N\}$ cover $\mathbb{Z}_N$. Five empirically distinct obstruction classes from the program's earlier experiments collapse into *additive orders of one covering phenomenon* (Table 4). The framework yields proved lemmas: an exact size law (Lemma 4.1), a polygonal witness lemma (Lemma 4.2), moat counting (Lemma 4.3), a coincidence-counting lemma (Lemma 4.4), a lift law (Lemma 4.5), and a class-group reduction at prime moduli (Lemma 4.6).

3. **The rigid zone and its classifications** (§5). At each rung n there is a small set of *unit-capable* moduli—those N for which $n-1$ unit bad sets can cover $\mathbb{Z}_N$: $\{7, 11, 13\}$ at $n = 4$ and $\{7, 13, 17, 19, 37\}$ at $n = 5$ within the verified range. We classify the covering configurations completely at every one of these primes (Theorems 5.2–5.8): each classification is a short explicit list of translate classes in a cyclic group (antipodal classes, domino tilings, perfect matchings, and at 19 and 37 the two classes $\{0, 1, 2, 7\}$, $\{0, 2, 4, 6\}$ and the single class $\{0, 2, 12, 15\}$). This is the paper's central structural finding: *at rigid primes, the covering obstruction is governed by a small translate-class structure.* The measured coverage rates of §10 are a consequence of this structure, not the primary result.

4. **The measurement** (§10). On all 3,712,576 primitive 5-sets with speeds ≤ 56: proved lemmas close 75.6% (78.2% with the coincidence lemma), per-modulus verification closes 100%, and every set has a certifying pair at a never-covering configuration. The proved-coverage rate declines as the corpus grows (93.4% $\to$ 78.2% over $V = 24 \to 56$ on the augmented battery, crossing below the continuation threshold at $V = 56$); the trend does not establish stabilization, and the decline is reported as a finding, with the residual localized to one named class (mixed order signatures at composite moduli).

5. **The conditional scaling-closure theorem** (§2, §9). The frontier cells $(1K, T-3U)$ at $N = p^2$—beyond the reach of enumeration—are attacked structurally: the fiber cascade of §8 reduces, exactly, to iterated dilate-intersections of census solution sets (Lemma 9.1), and the closure question becomes a pool-versus-cut arithmetic. What is proved: the pool law, the spacing lemma, the k-cancellation, the free reflection fiber (Lemma 9.6), the chain-class dilate theorem (Theorem 9.9), and the renormalization lemma (Lemma 9.10). What is measured: the pool saturation at two rungs (the 840 identity is a $p = 17$ accident, disclosed as such), the zone-resolved cut quality, and the $T = 8$ volume law with margin growing in k. What remains is consolidated into *one* named conditional theorem (Theorem 2.1) with a six-entry hypothesis ledger, a free-parameter disclosure (c_1 data-bounded, $\Lambda_0 \approx 4$ data-chosen zone cutoff, γ_0 data-bounded, the fitted R retired), and sampled estimates carrying their intervals and *projected* labels. The open obligations are reduced to named objects: the strand census (for H1g) and the sharp pair-line constants (for H2m). The boundary between those two lists has a structural type, and it explains where the paper stops: every proved result in the program—everything enumerated above and everything in the sections that follow—is a closure, invariance, exhaustiveness, or machine-verification statement, while every remaining obligation (the H1g margin over the trivial bound, the volume law's growth in k, the chain class's gap from measured to proved constants) is a uniform growth-in-k margin over a trivial or measured baseline. No result in this paper proves a statement of the second type; §9.7 verifies this statement-by-statement, including the one candidate that reads like an exception, the cut law's "k-cancellation," which cancels the k-dependence—a k-invariance, the exact opposite of a growth-in-k margin. The empirical half of the reading is thereby settled; the structural half (that a classification toolkit cannot produce the missing type) is a hypothesis about tool reach. If it is right, the reformulation cannot close the conjecture in its current form: it closes the covering obstruction exactly where classification suffices, and the residual is a growth-in-k margin of a type that has resisted every census-shaped attack the program has tried—a structural gap, not a technical one. The one margin-shaped candidate was then

executed as a proof attempt on the committed cells under a pre-committed success criterion ($\rho \leq 1-\epsilon$, ϵ independent of k), with the state space and the whole-cascade comparison target fixed beforehand. It closed by theorem for the lossless state space, not by measurement: the constrained transfer matrices of the lossless cascade are partial permutations (Lemma 9.13), every spectral quantity in the framework is exactly 0 or 1, the cyclic product is diagonal—its trace is the cascade's own survivor count—and the survival-fraction deficit the criterion demanded is, in this language, precisely the survivor inequality Hq and the volume law H1g restated. The instrument is the census in different notation; it consumes the hypotheses it was commissioned to prove. That is the mechanism behind the type mismatch: losslessness forbids generated decay, and the reformulation is bounded by its own fidelity. (The closure theorem covers automata built on the cascade's bijective drift—the lossless state space; instruments that do not inherit the drift are outside its scope, as §9.7 records.)

6. **Open problems, and the post-submission structural additions** (§11). The pair-selection law, the kernel world, the composite micro-cases $49 = 7^2$, $77 = 7 \cdot 11$, $91 = 7 \cdot 13$, rigid-zone termination, and whether the decay is fundamental. The post-submission additions carry the composite side to its current shape: the group-ring translation and the four-family census (§6), the pq tensor side (§7), the discharge of the foot-count input (§5.5), and the higher rungs (§8): the general-T fiber machinery, the $T = 8$-critical census with its eleven p-independent families, the $T = 7$ core–tail census, and the shear step re-established against growing pinned sets, answering the paper's third-rung question affirmatively for the census+shear method; and, the scaling-closure record of §9—the derivation, calibration, and epistemic status behind Theorem 2.1. The final revision records the disposition of the third bridge of §11: the zonotope–Ehrhart instrument engaged as a full strand and closed by the same type mismatch (stable arithmetic, false geometric reading), with the negative half of the record—the falsified strengthenings, the exact witnesses, and the certification ladder—published in the companion paper [22].

Table 1: Status of the principal results. "Unconditional" means proved outright (possibly with a machine-verified finite check); "conditional" means the statement depends on a named open hypothesis or a fitted constant, as recorded in the ledger of Theorem 2.1.

result	kind	status
Pair-sum equality $M(V) = \max \tau$ (Thm 3.2)	identity	unconditional
Lossless reformulation LRC $\Leftrightarrow$ TAU (Cor 3.3)	equivalence	unconditional
Rigid-zone classifications $K_7, K_{11}, K_{13}, K_{17}, K_{19}, K_{37}$	finite list	unconditional (finite check)
Group-ring / tensor translations (Thm 6.1, Thm 7.1)	identity	unconditional
Foot-count lemma E (§5.5)	lemma	unconditional
Rigid-zone *finiteness* beyond 37	empirical	measured only
Scaling-closure theorem (Thm 2.1)	theorem	conditional (Hq$_\pm$, H1g, H2r)
$T = 8$-critical eleven-family census	finite list	unconditional (finite check)
$T = 7$ tail decay $183 \to 43$	pattern	measured only

1.3 What this paper does not claim

Because the subject is a famous open problem, the boundary of the claims is stated as carefully as the claims themselves. The paper does not solve, reduce, or weaken the Lonely Runner Conjecture; the equivalence of Corollary 3.3 is exact, so the full difficulty of LRC-$_n$ is present in (TAU-$_n$). It proves no new cases of the conjecture at any n; the rung results of [4, 5, 6, 7] are untouched and unaffected. The rigid-zone map of §5 is an empirical finding about a verified range (primes ≤ 150, composites ≤ 111), not a theorem of finiteness; the successor conjecture (Conjecture 5.18) is stated as a conjecture. The coverage measurements of §10 are measurements;

their scaling beyond the tested range is explicitly open. Finally, the analytic core of the equality theorem is folklore [16]; only the sharpening, the exact form, and everything built on them are claimed as new, and even there the correct prior, given how actively the spectrum line computes exact optima [12], is that pieces of this may exist in another language somewhere in the literature not reached in this search.

One boundary statement is specific to the conditional theorem. Theorem 2.1 is exactly what its name says: its three substantive hypotheses ($\mathrm{Hq}_{\pm}$, H1g, H2r) are open, are not established anywhere in this paper, and no consequence that requires them is claimed unconditionally. If they close, the reformulation yields a uniform in-T closure of the p^2 covering obstruction at the frontier cell *as a covering statement*; it does not by itself yield the conjecture at the rung, because (TAU-n) requires a certifying pair for every V at its own pair-sum moduli, and the p^2 frontier cell is one cell type among the prime, pq, and remaining cells. If any hypothesis fails, the reformulation's scope is precisely the bounded statement of Theorem 2.1—a conditional theorem with named hypotheses, not a proof. The measured calibration ($T = 8, 9, 10$ at $p \leq 73$; committed ground-truth cells at $p \leq 61$) is stated with its ranges and is not extrapolated beyond them except where explicitly labeled *projected*.

1.4 Notation

Throughout, $n \geq 2$ is the number of speeds, $T = n + 1$ is the loneliness target denominator, and V has $\gcd(V) = 1$ (dividing by the gcd does not change M). For a modulus N we write $\|x\|_N = \min(x \bmod N,\ N - (x \bmod N))$, $\mathrm{ord}_N(w) = N/\gcd(w, N)$ for the additive order of w in $\mathbb{Z}_N$, and $h = \lfloor (N-1)/T \rfloor$ for the covering threshold index of §4. The letter k always denotes a lattice time index (never a fiber count). Computations are exact integer/rational arithmetic wherever a theorem or table entry asserts an exact value; the Wilson intervals, the fitted Λ_0, the projected pools of Table 10, and displayed asymptotics are statistical or asymptotic estimates and are labeled as such in the free-parameter disclosure. The verification record is the subject of §12.

Table 2: Recurring symbols and the section where each is fixed. Where a symbol is reused, the binding listed here is the one in force in the section cited.

symbol	meaning	fixed in
n, T	speed count; $T = n + 1$ loneliness denominator	§1
$V,\ M$	speed tuple; optimal loneliness $\max_t g(t)$	§3
N	modulus (p, p^2, or pq)	§4
p, q	primes at composite modulus $N = pq$	§7
k	lattice time index (never a fiber count)	§1
m	moving-arc count at the frontier cell ($m = T - 3$)	§9
h	covering threshold index $\lfloor (N-1)/T \rfloor$	§4
D	canonical-difference bound $(p-1)/2$	§9
ov	overlap budget $\Sigma s - p$	§9
σ, τ	sorted cyclic order; pair-sum lattice translate	§9, §3
Γ	$\mathrm{Hq}_{\pm}$ survivor constant	§2
c_1	H1g volume constant	§2
γ_0	H2m cut-quality constant	§2
Λ_0	H2r resonance zone cutoff (fitted)	§2
ρ	zonotope covering radius $\rho(V)$	§9.7
E	$\mathrm{Hq}_{\pm}$ additive error term	§2
$U_{\pm}$	$\pm$-representative non-resonant fiber set	§9

2 The main theorem, stated conditionally

The closures of §5–§8 settle the covering obstruction at the base rung, at the composite micro-cases, and at the critical cells of the higher rungs. What remains is the frontier: the $(1K, 5U)$ cell at $T = 8$ and its generalizations $(1K, T{-}3U)$ at every $T \geq 8$, the cells whose five-and-more-difference censuses are too large to enumerate and whose closure must therefore be structural. This section states the program's central result in its stated form: a single named conditional theorem, preceded by a ledger of hypotheses in which every input carries its status—proved, measured, or derived from a disclosed fitted constant. The theorem is the consolidation of the conditional statements scattered through the cascade analysis of §9; stating it once, with named hypotheses, is the point of this section.

The statement needs the higher-rung vocabulary of §8, which follows later; we give the gloss here and defer all derivations. At $N = p^2$ with $p = Tk + \epsilon$ prime, a covering family consists of *kernel* members (speeds divisible by p) and *unit* members (speeds coprime to p); a cell (aK, bU) collects the families with a kernel and b unit members. The kernel covers the fibers of a kernel bad set and leaves $|U| = (T-2)k + \epsilon - 1$ uncovered fibers $F_j \cong \mathbb{Z}_p$; on each, the b unit footprints are arithmetic progressions which must cover $\mathbb{Z}_p$ (Lemma 8.1). The *census normal form* pins the normalizer arc's difference to 1 and takes canonical differences in $[2, D]$, $D = (p-1)/2$; for the frontier cell, $m = T - 3$ arcs carry a two-value size window $\{2k, 2k+1\}$ or $\{2k+1, 2k+2\}$ and an *overlap budget* $\mathrm{ov} = \Sigma s - p \approx (T-6)k + O(1)$. A family is *census-admissible* if it admits at least one covering placement in the normal form; the *fiber cascade* is the iterated filter of §9.1, which retains the anchor placements surviving the uncovered fibers one by one and kills the family when the survivor set empties.

Theorem 2.1 (scaling closure; conditional, ±-reduced)**.** *Assume the hypotheses* NF, $\mathrm{Hq}_{\pm}$, H1g, H2r *of Table 3 (statuses as in the table;* $\mathrm{Hq}_{\pm}$ *is the ±-representative, additive-error form and subsumes* H2m *as its one-step case), together with the committed identity* EQ *of the same table. Let $T \geq 8$, let $p = Tk + \epsilon$ be prime with $k \geq 2$ and $0 \leq \epsilon < T$ (so $k = \lfloor p/T \rfloor$ is canonical; $p \geq p_0(T)$, the least such prime, $p_0(8) = 17$), and let the frontier cell $(1K, T{-}3U)$ at $N = p^2$ be given, with $m = T - 3$ unit arcs, the two-value size window of Lemma 8.1, and overlap budget $\mathrm{ov} = \Sigma s - p = (T-6)k + O(1)$. Let $U_{\pm}$ be a ±-representative non-resonant fiber set (one representative per pair $\{j, -j\}$, the reflection fiber excluded), $|U_{\pm}| = \frac{1}{2}((T-2)k + \epsilon - 1)$. Then:* (i) *every census-admissible family in the cell dies in the fiber cascade, processed over $U_{\pm}$, within*

$$C(T,k) \;=\; \frac{\log(c_1\,\mathrm{ov}^{\,m-1})}{(m-1)\,\log(p/\mathrm{ov}) - \log(c_1\Gamma)} \;+\; O(1)$$

fibers, where Γ is the uniform constant of $\mathrm{Hq}_{\pm}$ *and the $O(1)$ absorbs the additive-error floor $E/(1-r)$ of Proposition 2.2 plus the deterministic kill of the $O(1)$ remainder. The denominator is uniformly positive under* H1g*'s margin clause (which includes Γ). The depth is $O_T(\log k)$ with the explicit leading term $\log k/\log(T/(T-6))$; the $k \to \infty$ form $(m-1)\log(T/(T-6)) - \log(c_1\Gamma)$ gives the T-flat plateau $\approx 9, 9.2, 10.1$ at $T = 8, 9, 10$ (at $k = 2, 3, 3$).* No k-uniform linear bound is claimed*; the earlier "$\leq T + 3$" is withdrawn (it compared the depth against the un-reduced budget $|U|$).* (ii) *since $C(T,k) \ll |U_{\pm}| = \frac{1}{2}(T-2)k$ for every $T \geq 8$, $k \geq 2$, with a surplus growing linearly in k (conditional on the ledger: the depth bound is what (i) provides; the surplus language refers to the ±-reduced fiber budget), the cascade exhausts the budget: no census-admissible family admits a covering placement valid on all of $U_{\pm}$ (equivalently on all of U, by the reflection identification).* (iii) *the cell $(1K, T{-}3U)$ at p^2 is empty.* This is a statement about the covering obstruction at the cell only. *It does not by itself imply* LRC *at the rung:* (TAU-n) *requires a certifying pair for every V at its own pair-sum moduli, and ruling out one cell type at moduli p^2 leaves prime moduli (the rigid zone), pq and other composites, and the remaining cells untouched. The cell-emptiness result is exactly the covering obstruction at the frontier cell. The measured*

range of the calibration is $T = 8, 9, 10$ at $p \le 73$, with committed ground-truth cells at $p \le 61$ for $T \in \{7, 8\}$; the statement at primes beyond the calibration is exactly what the hypotheses purchase.

Proposition 2.2 (the $\mathrm{Hq}_{\pm}$-induction; the derivation step)**.** *Let the frontier cell and notation be as in Theorem 2.1, and assume* $\mathrm{Hq}_{\pm}$ *with constant* Γ *and additive error term* E*, and the volume law* H1g *with* $|\mathrm{Sol}(f, \mathrm{sz})| \le c_1 \,\mathrm{ov}^{\,m-1}$ *for every admissible family and size tuple. Let the cascade process the* $\pm$*-representative non-resonant fibers of* $U_{\pm}$ *(one per reflection pair, reflection fiber excluded). Write* S_i *for the survivor set after* i *processed fibers,* $A_i = \lambda_i^{-1}(\mathrm{Sol}_i - \tau_i)$ *for the* i*-th fiber's dilate constraint,* $d = m - 1$*,* $B = c_1 \,\mathrm{ov}^{\,d}$*, and* $r = \Gamma B / p^{\,d}$*. Then* $|S_0| \le B$*, each step obeys* $|S_i| \le r\,|S_{i-1}| + E$*, and consequently*

$$|S_q| \;\le\; B\, r^{\,q} + \frac{E}{1-r} \;\xrightarrow[q\to\infty]{}\; \frac{E}{1-r} \;=:\; S_\infty = O(1).$$

Once $|S_q| \le S_\infty$ *the set is a structured* $O(1)$ *remainder; a single further fiber (a generic non-resonant* λ*) moves the fixed drift off the finite survivor set, dropping it to* $\varnothing$ *(the deterministic kill). Hence the cascade is empty whenever*

$$q \;>\; \frac{\log B}{(m-1)\,\log(p/\mathrm{ov}) - \log(c_1\Gamma)} \;+\; O(1),$$

the $O(1)$ *absorbing the error floor and the deterministic kill. The denominator is uniformly positive iff* $c_1\Gamma < (p/\mathrm{ov})^{m-1}$ *with a margin uniform in* (T, k)*; in the* $k \to \infty$ *limit* $p/\mathrm{ov} \to T/(T-6)$*, and the uniform condition is the threshold clause of* H1g*.*

Proof. S_0 is the family's initial placement set, bounded by the volume law. For each i, $S_i = S_{i-1} \cap A_i$ with A_i the i-th fiber's constraint set, so $\mathrm{Hq}_{\pm}$ applied to the admissible history S_{i-1} (a $\pm$-representative fiber, so A_i is a genuine constraint, not a reflection duplicate) gives $|S_i| \le \Gamma\,|S_{i-1}|\,|\mathrm{Sol}_i|/p^d + E \le r\,|S_{i-1}| + E$ using $|\mathrm{Sol}_i| \le B$. Induction gives $|S_q| \le B\,r^q + E(1 + r + \cdots + r^{q-1}) \le B\,r^q + E/(1-r)$ (for $r < 1$, which is the positivity clause). The contraction continues until the r^q term falls below $S_\infty = E/(1-r)$, i.e. after $q = \log(B(1-r)/E)/\log(1/r)$ steps; the residual $O(1)$ set is then killed by one further fiber's drift, which has no fixed point on a nonempty finite set unless the set is a union of drift-fixed configurations—excluded for a generic non-resonant λ. The resonance tail (weak-cut fibers with $\bar\lambda \le \Lambda_0$) is excluded from $U_{\pm}$ by construction, so no N_{res} term is needed. □

The reduction and its hypotheses. An unquantified "sequential composition" input (the H0 row: $C_i \approx \bar\gamma^{\,i} |\mathrm{Sol}|^{i+1}/p^{(m-1)i}$) is not used; a recurrence for the *independent-extension model* and a bound on *unconditional* pair intersections do not compose into a bound on the *iterated, history-dependent* intersections the actual cascade performs: pairwise sparsity does not control higher-order correlations, and three sets can have acceptable pairwise overlaps while sharing a structured common core. The multi-fiber survivor inequality $\mathrm{Hq}_{\pm}$ (above) replaces both, with H2m as its one-step case. The derivation from the ledger is exactly Proposition 2.2's induction, conditional on named open inputs only.

The cascade is modelled on the $\pm$-representative fiber set $U_{\pm}$ (one representative per reflection pair; the reflection fiber excluded, by Lemma 9.6), because the fiber $-j$ imposes the same covering constraint as fiber j; an oriented-fiber model would double-count every constraint and make the survivor inequality fail on the reflection duplicate. The survivor inequality $\mathrm{Hq}_{\pm}$ carries an additive error term E; the induction contracts to an $O(1)$ floor and closes it by a deterministic kill (Prop. 2.2), and the depth is $O_T(\log k)$ (Prop. 9.5). Clause (iii) states cell emptiness and nothing more.

What is conditional and what is not. The reduction chain *(i)→(ii)→(iii)* is proved from committed machinery: the two-fiber reduction (Lemma 9.1: the cascade step *is* a dilate-intersection of census solution sets), the drift law (Lemma 8.1, machine-verified 3,528/3,528), the reflection identification (Lemma 9.6), the T-free fiber machinery, and the lossless identity of Corollary 3.3. The arithmetical content—that the per-fiber cut beats the pool within $C(T,k)$ fibers—is proved *given* the ledger's inputs, by the induction of Proposition 2.2. None of those inputs is silently established, and none is silently weakened: that is the content of Table 3. The pq side of the composite frontier is carried by the row translation (Theorem 7.1) with the same input structure; Theorem 2.1 is stated for the p^2 side only. The reduction chain (i)→(ii)→(iii) is stated over the ±-representative fiber set $U_\pm$: (iii) is cell emptiness, not LRC at the rung.

The hypothesis ledger. Each row of Table 3 names one input, its precise form, its status, and its evidence or attack route. The rows are *conditional inputs*, not evidence toward the theorem's validity: three are open hypotheses with measured support ($\mathrm{Hq}_\pm$, H1g, H2r, the first over the ±-representative fiber set with an additive error term); one is the measured one-step base case (H2m); the others are proved, machine-validated structure (NF) and a committed identity (EQ). The statuses are stated once here and are not restated with each citation.

Table 3: The hypothesis ledger of Theorem 2.1. Statuses: *OPEN* (a conditional input with measured support), *proved* (with the verification record), *measured* (base case with tested range), or *committed*. The substantive open inputs are $\mathrm{Hq}_\pm$, H1g, and H2r; every row is an input the theorem assumes, not evidence that the theorem holds.

	hypothesis (precise form)	status	evidence / route
NF	Census normal-form completeness: every covering family at an uncovered fiber is census-admissible (the normal form—normalizer difference 1, canonical differences, two-tier sizes—loses no coverings).	proved in structure, machine-validated	The footprint and arc derivations are T-free and verified 957/957 against direct $\mathbb{Z}_{p^2}$ enumeration; the $T=7$ census-restricted search equals brute force at the boundary prime $p=5$ (the 8-covering control); the $T=8$ zoo census is the measured object.
$\mathrm{Hq}_\pm$	Multi-fiber survivor inequality, ±-representative form with additive error: for every admissible cascade history S_{i-1} over the ±-representative fiber set $U_\pm$ (one representative per reflection pair $\{j,-j\}$; the reflection fiber excluded) and next such fiber i, $\lvert S_{i-1}\cap A_i\rvert \le \Gamma\,\lvert S_{i-1}\rvert\,\lvert\mathrm{Sol}_i\rvert/p^{m-1}+E$, where $A_i=\lambda_i^{-1}(\mathrm{Sol}_i-\tau_i)$, with Γ uniform over families, size tuples, dilations, translations, and prior surviving fibers, and E a (T,k)-uniform additive error (k-free, structural).	OPEN (measured benign in range)	Depths ≤ 4 at the committed cells (764-pair record, ±-reduced); ≤ 11 oriented (halved: ≤ 6 reduced) across the three calibration rungs; exit time ≤ 2 exhaustive over admissible families at $T=7$ and $T=8$-critical; the $i=1$ case is the measured H2m. Route: tangency codimension ≥ 1 (the reflection identification removes the $j=-j$ duplicate); the error term E is the tangency remainder; the iterated conditional bound is the open content.

Table 3, continued

	hypothesis (precise form)	status	evidence / route
H1g	Distinct-family volume law with uniform margin: $\lvert\mathrm{Sol}(f,\mathrm{sz})\rvert \le c_1\,\mathrm{ov}^{m-1}$ for every $\pm$-distinct family f and admissible size tuple, with $c_1\,\Gamma \le \big(T/(T-6)\big)^{m-1} e^{-\eta}$ for some $\eta > 0$ uniform in (T,k,f,sz)—the margin clause now includes the $\mathrm{Hq}_\pm$ constant Γ, so that the denominator of Proposition 2.2 is uniformly positive.	OPEN (measured, margin growing in k; chain class proved; three attack routes closed, the automaton route by theorem)	At $T = 8$: worst ratio $\lvert\mathrm{Sol}\rvert/\mathrm{ov}^4$ of 0.214/0.083/0.013 at $p = 17/19/29$ (chain-class benchmark 2.17/3.61/2.03); measured margins over the trivial box 95 × /1,207 × /3,803×. Remaining distance: the strand census (§9.5)—necessary-condition and marginal-rigidity routes measured closed, the coordinates fixed by Lemma 9.12, and the transfer-matrix/automaton route closed by Lemma 9.13 (single-anchor scope): the instrument computes the census exactly (its cyclic trace *is* the survivor count) and consumes H1g rather than proving it, every spectral quantity in the framework being exactly 0 or 1.
H2m	Mid-region dilate-sparsity—now the $i = 1$ (one-step, unconditional) case of $\mathrm{Hq}_\pm$, retained for its stronger measured record: $\lvert\mathrm{Sol}_0 \cap \lambda^{-1}(\mathrm{Sol}_j - \tau)\rvert \le \gamma_0\,\lvert\mathrm{Sol}_0\rvert\,\lvert\mathrm{Sol}_j\rvert/p^{m-1}$ for every fiber pair with $\bar\lambda > \Lambda_0$, $\gamma_0 = O(1)$ uniform.	OPEN (measured; proved for the chain class with loose constants)	764 pairs at 5 committed cells: mid-zone γ median 0 (instant kills), max 40.6. Chain class proved (Theorem 9.9) with constants $((m-1)!)^2 \cdot 4(m-1) \cdot s_{\max}$—loose by orders of magnitude. Route: the Lemma 5.16 three-distance pair-line count.
H2r	Resonance budget: the weak-cut fibers—those whose dilation ratio λ_i has small rational height, $\bar\lambda \le \Lambda_0$—are excluded from the $\pm$-representative set $U_\pm$ by construction, so they contribute no tail to the $\pm$-reduced cascade; the $\lambda = -1$ fiber is the reflection (Lemma 9.6), identified with $+1$ in $U_\pm$. $\Lambda_0 \approx 4$ is data-chosen, not proved.	OPEN (measured support; the $\lambda = -1$ clause proved)	Lemma 9.6 (the $x \mapsto -x$ involution; survival fraction exactly 1 at every committed family and cell); the small-rational dilation count is a Gauss-circle-type estimate; $\Lambda_0 \approx 4$ *is data-chosen.*
EQ	Lossless identity (NOT a cell-implies-LRC claim): the pair-sum equality $M(V) = \max_{\{u,w\}} \tau(u,w)$ is an identity, and cell emptiness at the frontier cell is a statement about the covering obstruction at that cell only.	committed theorem	The pair-collapse equality $M(V) = \max_{\{u,w\}} \tau(u,w)$ (Theorem 3.2, machine-verified with a 2,870/2,870 independent re-verification); the fibration threshold-compatibility.

The free-parameter disclosure. Standing practice for this paper, applied to every constant entering Theorem 2.1: a claim is called *derived* only when it is derived outright, and a formula containing a fitted constant is labeled as such. The constants are:

- c_1 *(H1g):* bounded by data only—≤ 0.214 at $T = 8$ over the three census primes $p = 17, 19, 29$, tightening with k (Table 13); *not proved.* The chain class, the only proved case, carries $c_1 \approx 2.0$–3.6. The threshold the constant must beat is structural, not fitted: the trivial bound realizes $c_1 = \big(T/(T-6)\big)^{m-1}$ (Proposition 9.11), so the content of H1g is the margin over ≈ 250—which every measured constant carries and no proved constant yet does.

- $\Lambda_0 \approx 4$ *(H2r):* data-chosen—the $\bar{\lambda}$ zone in which the measured per-fiber cuts at the committed cells first exceed the random model by $\sim 10\times$. In the $\pm$-reduced model the weak-cut fibers are excluded from $U_\pm$ rather than added as a tail, so Λ_0 enters only the definition of $U_\pm$, not an additive N_{res} term; it is *a data-chosen zone cutoff*, not a fitted depth constant.
- γ_0 *(H2m):* bounded by data (≤ 41 in the non-resonant zone over the committed cells); proved only for the chain class, with loose constants.
- R *(a junction-volume scale fitted at* 4.2 *in an earlier formulation): retired.* For the chain class it is replaced by the derived overlap budget $T-6$; for the thin families it is superseded by c_1 under H1g. No fitted R remains in Theorem 2.1.

The depth formula's remaining constants are structural: the leading term is built from the overlap budget $T-6$ and the modulus T, and the $\pm$-representative fiber set $U_\pm$ is defined by the reflection identification (Lemma 9.6), with Λ_0 only the data-chosen zone cutoff excluding the weak-cut fibers.

The sampled numbers, with intervals. Theorem 2.1's calibration rests on exhaustive censuses where the cells are enumerable and on disclosed samples where they are not. The $T = 8$ saturation figures of Table 10 (1.000/0.571/0.357/0.133/0.015 at $p = 17/19/23/29/31$) are exhaustive censuses and carry no sampling error. The $T = 9$ figures are stride-sampled with full decisions per sampled tuple: at $p = 29$, saturation 0.903, 95% Wilson interval $[0.889, 0.915]$ (n = 2,005), projected pool 1.39×10^5 with interval $[1.37, 1.41] \times 10^5$; at $p = 31$, saturation 0.490, $[0.468, 0.511]$ (n = 2,002), projected pool 1.18×10^5, $[1.12, 1.23] \times 10^5$. Both pools are *projected*—the falling-factorial bound times the sampled fraction, at one prime each—and the rung-monotone reading rests on two primes with $\epsilon \in \{2, 4\}$, whose modulation (0.90 against 0.49) is large and unmodeled. The $T = 10$ saturation (extrapolated ≥ 0.9 at the 40% budget from the rung-monotone pattern—two primes, unmodeled ϵ-modulation, no error control) is *not measured*. Cascade-depth figures at the committed cells are exact runs (median 2, maximum 4 over the zoo random-40); the maxima 9/11/10 are oriented-fiber counts (both j and $-j$ processed) and exceed the $\pm$-reduced formula by the reflection duplicate; the $\pm$-reduced maxima are ≤ 6 at $T = 9$ and match the formula of Table 11 to within the $O(1)$ error floor, quoted as measurements.

Scope levels, fixed once. Three different closures circulate in this paper and must not be conflated: *cells* (the $T = 7$ and $T = 8$-critical cells are closed within the tested primes, $p \leq 61$, by exact census plus the shear step); *rungs* (the frontier cells of every $T \geq 8$ are closed only conditionally, by Theorem 2.1 under its ledger); and *rigid-zone finiteness* (open; the five-prime zone of §5 is an empirical finding of the verified range). Statements below use the matching noun.

Where the derivation lives. Section 9 derives the cascade model and proves the proved pieces (Lemmas 9.1, 9.2, 9.3, 9.6, 9.10; Theorem 9.9); reports the measurements with their scopes and ground-truth records; states the two open obligations (the strand census for H1g, the sharp pair-line constants for H2m) in their reduced forms; and closes with the epistemic status. Theorem 2.1 should be read as the summary of that section, not as an independent result.

3 The pair-sum equality theorem

3.1 Setup

Fix $n \geq 2$ and speeds $V = (v_1, \ldots, v_n)$, not necessarily distinct in this section (no argument below uses distinctness; the conjecture adds it). Normalize $\gcd(V) = 1$. For $t \in \mathbb{T}$ write

$$f_p(t) = \|v_p t\|, \qquad g(t) = \min_{1 \leq p \leq n} f_p(t), \qquad M(V) = \max_{t \in \mathbb{T}} g(t).$$

The function g is continuous on the compact circle, so the maximum is attained, and $M(V) > 0$: at $t = 1/(2v_{\max})$ every f_p equals $v_p/(2v_{\max}) \leq \frac{1}{2}$, whence $g \geq v_{\min}/(2v_{\max}) > 0$.

Definition 3.1 (pairs, lattices, reduced optima)**.** For an unordered pair of indices $\{u, w\}$, $u \neq w$, the *pair sum* is $N = N_{uw} := v_u + v_w$ (≥ 3), and the *pair-sum lattice* of $\{u, w\}$ is the set of times $\{k/N : 1 \leq k \leq N-1\}$. The *reduced optimum* of the pair is

$$\tau(u, w) \;:=\; \max_{1 \leq k \leq N-1} \; \min_{p \neq w} \left\| \frac{v_p k}{N} \right\|.$$

This is the $(n-1)$-runner optimal loneliness of the effective residues $(v_p \bmod N)_{p \neq w}$ on the cyclic group $\mathbb{Z}_N$: runner w has been deleted, and by Lemma 3.4 below the deleted runner is carried by its partner u at every lattice time.

Theorem 3.2 (pair-sum equality; lossless reduction)**.** *For every V as above,*

$$M(V) \;=\; \max_{\{u,w\}} \; \tau(u, w),$$

the maximum over the $\binom{n}{2}$ unordered pairs. Moreover the maximum defining $M(V)$ is attained at a time $t^* = k^*/N$ on the pair-sum lattice of some pair $\{u, w\}$ with $1 \leq k^* \leq N-1$; and if $M(V) < \frac{1}{2}$ the binding runners at t^* can be chosen two-sided, one at residue m^* and one at residue $-m^*$ on $\mathbb{Z}_N$.

Corollary 3.3 (lossless reformulation)**.** LRC-n *is equivalent to the finite statement* (TAU-n)*: for every* gcd*-normalized speed set V there is a pair $\{u, w\}$ with*

$$(n+1) \;\cdot\; \max_{1 \leq k \leq N-1} \min_{p \neq w} \left\| \frac{v_p k}{N} \right\| \;\geq\; N = v_u + v_w.$$

The corollary is immediate: LRC-n asserts $M(V) \geq 1/(n+1)$, i.e. $\max_{\{u,w\}} \tau(u, w) \geq 1/T$ with $T = n+1$, which unfolds to exactly (TAU-n); the equivalence is an *identity* between a continuous maximization and a finite family of finite maximizations, so no information is lost in either direction. The proof of the theorem occupies the rest of the section.

3.2 The mirror identity

Lemma 3.4 (mirror / pair collapse)**.** *Let $N = v_u + v_w$. For every integer k:*

(i) $v_u k \equiv -v_w k \pmod N$*, hence* $\|v_u k/N\| = \|v_w k/N\|$*;*

(ii) $\min_p \|v_p k/N\| = \min_{p \neq w} \|v_p k/N\|$.

Proof. (i) The two speeds sum to N, so $v_u k + v_w k = Nk \equiv 0 \pmod N$; the two positions are reflections of each other through the origin, and distances to the origin coincide. (ii) The left side is a minimum over a superset of the index set on the right, so it is at most the right side. For the reverse inequality, split the full minimum as $\min(\min_{p \notin \{u,w\}} f_p,\, f_u,\, f_w)$ and replace the last two terms by f_u alone, which equals them by (i); the resulting expression is exactly the right side, since $\{p \neq w\} = \{p \notin \{u, w\}\} \cup \{u\}$. □

Part (ii) has a reading worth isolating, because the entire finitization rests on it: *at every time of the pair-sum lattice, the full n-runner minimum and the $(n-1)$-runner reduced minimum coincide.* Deleting runner w costs nothing precisely at the times when w is mirrored by u. This is the sense in which the pair-sum lattice is the natural finite skeleton of the continuous problem.

3.3 Attainment at a pair-sum time

Lemma 3.5 (attainment)**.** *There exist a pair $\{u, w\}$ and an integer k^* with $1 \le k^* \le N-1$ such that $g(k^*/N) = M(V)$. Specifically, let t^* be any global argmax, $m := M(V) = g(t^*)$, and let $B = \{p : f_p(t^*) = m\}$ be the binders. Then:*

(a) if $0 < m < \frac{1}{2}$: some binder is rising *(its residue $v_p t^* \bmod 1$ lies in $(0, \frac{1}{2})$) and some binder is* falling *(residue in $(\frac{1}{2}, 1)$); choosing u rising and w falling, $(v_u + v_w)t^*$ is an integer, so $t^* = k^*/N_{uw}$;*

(b) if $m = \frac{1}{2}$, the half case of Lemma 3.6 applies: all speeds are odd, $t^* = \frac{1}{2}$, and every pair has even N with $k^* = N/2$;

(c) $m = 0$ cannot occur.

Lemma 3.6 (the half case)**.** *If $M(V) = \frac{1}{2}$ then every speed is odd and $t^* = \frac{1}{2}$ attains it; every pair sum is even, so $t^* = (N/2)/N$ is a pair-lattice time for any pair with $N \ge 4$ (such a pair exists for $n \ge 2$ unless all speeds equal 1; in that degenerate case $M = \frac{1}{2} = t^*$ with $N = 2$, $k^* = 1$).*

Proof. (c) was noted above ($M(V) > 0$).

(a) Assume $0 < m < \frac{1}{2}$. Each binder $p \in B$ has residue exactly m (*rising*: on the arc $(0, \frac{1}{2})$ the position $x = v_p t \bmod 1$ satisfies $f_p = x$, which is increasing at slope $+v_p$) or exactly $1 - m$ (*falling*: on $(\frac{1}{2}, 1)$, $f_p = 1 - x$ with slope $-v_p$). No binder sits at a peak ($f = \frac{1}{2}$) or a valley ($f = 0$) because m is neither $\frac{1}{2}$ nor 0.

Claim: not all binders rise, and not all binders fall. Suppose all binders rise; the falling case is symmetric. For non-binders let $\delta_p = f_p(t^*) - m > 0$. All f_p are v_p-Lipschitz. Choose $s > 0$ with

$$s < \min\Big(\min_{p \notin B} \delta_p / v_p,\ \min_{p \in B} \tfrac{1}{2}(1/2 - m)/v_p \Big).$$

Two remarks make the estimates airtight. First, the non-binder bound uses only the *global* Lipschitz property $f_p(t \pm s) \ge f_p(t) - v_p|s|$, which holds across the cusps of $f_p = \|v_p t\|$—a non-binder crossing a cusp during the perturbation is immaterial, because the one-sided Lipschitz lower bound is cusp-free. Second, the binder linearity window: a rising binder's position $x_p = v_p t^* \bmod 1$ equals $m \in (0, \frac{1}{2})$, and f_p is linear on $(0, \frac{1}{2})$; the window survives the shift iff $x_p + v_p s' < \frac{1}{2}$, which the s-bound guarantees per binder ($v_j s' \le \frac{1}{2}(\frac{1}{2} - m) < \frac{1}{2} - m$)—this is also where binders with *different* slopes are handled: each binder gets its own window from the same minimum. For binders (all rising), $0 < s' \le s$ keeps the position inside $(0, \frac{1}{2})$ where f_p is linear, so $f_p(t^* + s') = m + v_p s' > m$. For non-binders, $f_p(t^* + s') \ge f_p(t^*) - v_p s' = m + (\delta_p - v_p s') > m$ by the choice of s. Hence $g(t^* + s') > m = M(V)$, contradicting maximality. If all binders fall, perturb backwards ($s' < 0$); the symmetric argument gives the same contradiction. This proves the claim.

Take $u \in B$ rising and $w \in B$ falling. Then $v_u t^* \equiv m$ and $v_w t^* \equiv -m \pmod 1$, so $(v_u + v_w)t^* \equiv 0 \pmod 1$: the integer $k^* := (v_u + v_w)t^*$ satisfies $t^* = k^*/N_{uw}$. Since $g(0) = 0 < m$, $t^* \ne 0$; representing $t^* \in (0, 1)$ gives $0 < k^* < N$, i.e. $1 \le k^* \le N - 1$.

□

Proof of Lemma 3.6. If $m = \frac{1}{2}$ then *every* runner binds (no f_p exceeds $\frac{1}{2}$), i.e. $v_p t^* \equiv \frac{1}{2} \pmod 1$ for all p. Write $t^* = a/b$ in lowest terms. Then $b \mid v_p - v_1$ for all p (as $\gcd(a, b) = 1$), and $2v_1 a = b(2j + 1)$ for some integer j, so $b \mid 2v_1$. If b were odd, $b \mid v_1$, hence $b \mid v_p$ for all p, hence $b \mid \gcd(V) = 1$ and $b = 1$—but then $t^* = 0$, excluded. So $b = 2c$ is even; then $c \mid v_1$ and $v_p \equiv v_1 \pmod{2c}$ forces $v_p \equiv 0 \pmod c$ for all p, so $c \mid \gcd(V) = 1$, $c = 1$, $b = 2$: $t^* = \frac{1}{2}$. Then $v_1/2 \equiv \frac{1}{2} \pmod 1$ forces v_1 odd, and $v_p \equiv v_1 \pmod 2$ makes every speed odd. Conversely, if all speeds are odd then $t = \frac{1}{2}$ has $f_p = \frac{1}{2}$ for every p, which proves attainment. □

Remark 3.7 (which speed convention is proved where). The equality theorem and both lemmas above are proved for possibly non-distinct speeds; nothing in this section uses distinctness. The conjecture itself is stated for distinct speeds; the reduction is the standard WLOG: collapsing a repeated speed to one copy leaves $g(t)$, and therefore M, unchanged (a duplicate runner's constraint is identical to its twin's), while the target $1/(n{+}1)$ *increases* as n drops—so the distinct-speed case is the harder one and suffices. The covering framework downstream assumes distinct speeds where it says so. The half case above is the one place where non-distinctness changes the picture (all speeds equal 1), and it is handled inside Lemma 3.6.

Remark 3.8 (what forced two-sidedness means). The perturbation argument is the rigorous form of an empirical regularity: in exhaustive scans at $n = 4, 5, 6$, every exact argmax breakpoint has binders on both sides (480/480 random sets [17]). It also shows why the folklore's *difference* denominators $|v_u - v_w|$ never have to carry the optimum: same-sided binder pairs are exactly the configurations that produce difference relations, and Lemma 3.5(a) rules out their being the only binders. The sharpening is confirmed structurally in §3.5: in 885 test sets and, under a from-scratch reimplementation, in 2,870 further sets spanning $n = 2$ to 6 and speeds to 200, no unrestricted argmax lies off the pair-sum skeleton.

3.4 Proof of the theorem

Proof of Theorem 3.2. (≥, the constructive bound). Fix any pair $\{u, w\}$ and any $1 \le k \le N - 1$. Lemma 3.4(ii) gives $\min_{p \ne w} \|v_p k/N\| = g(k/N) \le M(V)$; take the max over k to get $\tau(u, w) \le M(V)$, hence $\max_{\{u,w\}} \tau \le M(V)$.

(≤, attainment). By Lemma 3.5 there is a pair $\{u, w\}$ with $g(k^*/N) = M(V)$. Lemma 3.4(ii) again:

$$\min_{p \ne w} \left\| \frac{v_p k^*}{N} \right\| = g(k^*/N) = M(V),$$

so $\tau(u, w) \ge M(V)$. In case (b)—all speeds odd—this works for every pair at $k^* = N/2$: all runners sit at $\frac{1}{2}$, the mirror is trivially exact, and the reduced minimum equals $M(V)$.

Combining both directions, $\max_{\{u,w\}} \tau(u, w) = M(V)$, with equality attained at the pair produced by Lemma 3.5. The two-sided conclusion in the $M < \frac{1}{2}$ case is part (a) of the lemma: the rising binder u has residue m^* and the falling binder w has residue $-m^*$. □

3.5 Verification and novelty status

The theorem was verified by two deliberately non-circular computations, recorded in the artifact bundle [17]. The first, the script `V1_equality_fresh_check.py`, computes M *independently* from the folklore breakpoint set

$$D = \{2v_p\} \cup \{v_p + v_q\} \cup \{|v_p - v_q|\},$$

which is exhaustive for the vertices of the piecewise-linear lower envelope and carries no pair-sum assumption, and compares it against the pair-lattice maxima: on 885 speed sets (479 at $n \le 4$ with $v \le 12$; 393 random at $n = 5$ with $v \le 20$; 13 targeted edge cases including all-odd, tight, and the exceptional set $(2, 6, 8, 10, 11)$), equality holds exactly (885/885), every unrestricted argmax lies on a pair-sum lattice (885/885, and *zero* sets have a difference-only argmax), and the all-odd pole case is exact. The second, `EQ_independent_verify.py`, is a from-scratch reimplementation written against the theorem statement alone—a different code path, exact integer arithmetic, time domain: it maximizes g over the full candidate set of individual breakpoints $j/(2v_p)$ together with *all* pairwise crossings $c/(v_p \pm v_q)$, differences included, and compares against the pair-sum lattices. On 2,870 sets (300 to 1,200 random primitive sets at each $n = 2, \ldots, 6$, of which 1,200 lie in the paper's measurement domain $v \le 56$, plus a beyond-corpus probe at $v \le 200$), equality holds exactly (2,870/2,870), the LRC bound holds on every set, and the anchors are exact:

the regular sets $(1,\dots,n)$ at $1/(n{+}1)$, the tight sets $(1,2,3,4)$, $(1,3,4,7)$, $(1,3,4,5,9)$, and the base rung's open set $(3,5,8,13)$, whose exact loneliness is $M = 5/21$. The fresh scan adds one further point: on 962 of the 2,870 sets a *difference* time also attains the optimum (e.g. $t = \frac{1}{2}$ for $V = (1, 59)$). Difference denominators sometimes carry an argmax, but never a needed one; the sharpening, as Lemma 3.5 states it, is that sum lattices alone always suffice, and that is exactly what the equality check verifies. Cross-checks with a third method (direct grid maximization) and at $n = 5, 6$ on targeted tight sets bring the total to over 3,700 sets checked by at least two independent routes.

On novelty: two literature rounds (56 queries, the arXiv ring enumerated for 2024–2026) found the "sum or difference" denominator observation in a comment on Tao's 2015 post [16] and nothing else at the level claimed here—but the spectrum line computes exact optima on infinite families [12], so pieces may exist in another language. The position stated throughout: the technique is folklore; the sum-only sharpening, the exact equality, the (TAU-$_n$) statement, and everything downstream of them are not located in the searched literature (two rounds, 56 queries, the arXiv ring enumerated for 2024–2026); the correct prior, given how actively the spectrum line computes exact optima [12], is that pieces may exist in another language.

4 The covering framework

4.1 Bad sets and certification

Fix a pair $\{p, q\}$ with $N = v_p + v_q$ and recall $T = n + 1$, $h = \lfloor (N-1)/T \rfloor$. The *effective residues* of the pair are the $n - 1$ residues

$$\mathrm{eff}(p, q) \;=\; (v_p,\ (v_i)_{i \neq p,q}) \bmod N,$$

runner q having been deleted; since $v_q \equiv -v_p \pmod N$, its bad set coincides with that of v_p and carries no independent information (Lemma 3.4). The pair *certifies* (TAU-$_n$) iff some lattice time k has $\min_w \|wk\|_N \geq N/T$ for all effective residues w—that is, iff the *bad sets*

$$B_w \;=\; \{\, k \in \mathbb{Z}_N \;:\; T\,\|wk\|_N < N \,\} \;=\; \{\, k \in \mathbb{Z}_N \;:\; \|wk\|_N \leq h \,\}$$

of the $n - 1$ effective residues do *not* cover $\mathbb{Z}_N$. All covering questions in this paper are finite and decidable by exact bitmask arithmetic; on the corpora of §10 they are decided for every pair of every set.

Every bad set contains 0, and each B_w is antipodal-saturated ($k \in B_w \Rightarrow -k \in B_w$, as $\|-wk\|_N = \|wk\|_N$). The single most useful structural parameter of a residue is its *additive order* $\mathrm{ord}_N(w) = N/\gcd(w, N)$, because multiplication by w maps $\mathbb{Z}_N$ onto the cyclic subgroup $\langle w \rangle$ of order $j = \mathrm{ord}_N(w)$, with kernel of size N/j. Consequently B_w is a union of cosets of that kernel, and its exact size depends only on j:

Lemma 4.1 (exact size law)**.** *For every residue $w \in \mathbb{Z}_N$ with $\mathrm{ord}_N(w) = j$,*

$$|B_w| \;=\; \frac{N}{j}\left(2\left\lfloor \tfrac{j-1}{T} \right\rfloor + 1\right).$$

In particular $j \leq T$ forces $|B_w| = N/j$ and B_w is exactly *the subgroup $\frac{N}{j}\mathbb{Z}_N$; and w invertible ($j = N$) forces $|B_w| = 2h + 1$, a dilated copy of the arc $A = \{x : \|x\|_N \leq h\}$.*

Proof. Multiplication by w maps $\mathbb{Z}_N$ onto $\langle w \rangle = \{tN/j : 0 \leq t < j\}$, each image value having exactly N/j preimages (a coset of the kernel). Hence $|B_w| = \frac{N}{j}\,|A \cap \langle w \rangle|$. Now $\|tN/j\|_N = \frac{N}{j}\min(t,\, j - t)$, so membership $\|tN/j\|_N < N/T$ is equivalent to $\min(t, j-t) < j/T$. For $t \in \{1, \dots, j-1\}$ the inequality $t < j/T$ holds for exactly $\lfloor (j-1)/T \rfloor$ values of t (whether or not T divides j), and by the symmetry $t \mapsto j - t$ the same count holds on the other side; the two

ranges are disjoint because $t = j - t$ would give $\min = j/2 \geq j/T$ for $T \geq 2$. With $t = 0$ included, $|A \cap \langle w \rangle| = 2\lfloor (j-1)/T \rfloor + 1$. For $j \leq T$ this equals 1, so B_w is the single coset containing 0, i.e. the kernel subgroup $\frac{N}{j}\mathbb{Z}_N$ itself. □

Table 4: The obstruction taxonomy: five empirically distinct obstruction classes of the program's earlier experiments are five additive orders of one covering phenomenon. "World" is the idiom used in §5–10.

$\mathrm{ord}_N(w) = j$	B_w	exact size (Lem. 4.1)	world
1 ($w \equiv 0$)	all of $\mathbb{Z}_N$	N	stuck
2	$\frac{N}{2}\mathbb{Z}_N$	$N/2$	half
3	$\frac{N}{3}\mathbb{Z}_N$	$N/3$	third
$4 \leq j \leq T$	$\frac{N}{j}\mathbb{Z}_N$	N/j	polygon
$T < j < N$	union of $2\left\lfloor \frac{j-1}{T} \right\rfloor + 1$ kernel cosets	$\frac{N}{j}\left(2\left\lfloor \frac{j-1}{T} \right\rfloor + 1\right)$	kernel
$j = N$ (unit)	dilated arc $w^{-1}A$	$2h + 1$	covering

Table 4 is the framework's unification statement: what the earlier experiments had classified as five unrelated failure modes (stuck, half, third, blocked, covering) is one phenomenon—bad sets covering $\mathbb{Z}_N$—organized by the additive order of the effective residues. The polygon rows are order and nothing more; the kernel row is where sizes have slack; the unit row is where the multiplicative structure of $\mathbb{Z}_N$ enters, and it is the subject of §5.

4.2 Proved certification lemmas

The battery of §10 draws on the following lemmas, each proved uniformly in N (no per-modulus computation). Their combined soundness—every fired flag implies certification—is itself verified exhaustively in §12.

Lemma 4.2 (polygon witness)**.** *Let $j \mid N$ with $2 \leq j \leq T$, and suppose every effective residue satisfies $w \not\equiv 0 \pmod{j}$. Then $k = N/j$ certifies the pair: $\|wk\|_N \geq N/j \geq N/T$ for every effective residue w.*

Proof. Write $r = w \bmod j \in \{1, \ldots, j-1\}$ and $N = jm$ with $k = m$. Then $wm \equiv rm \pmod{jm} = N$, and $\|rm\|_N = m \cdot \min(r, j-r) \geq m = N/j$ because $1 \leq r \leq j-1$. Since $j \leq T$, $N/j \geq N/T$. □

The witness $k = N/j$ is constructive, and its regular-polygon geometry—the $n+1$ lattice points $\{0, m, 2m, \ldots, (j-1)m\}$ visited by the residues at speed multiples—explains the classical tightness of regular speed sets $\{1, \ldots, n\}$: the pair $(1, n)$ on $N = n+1$ has all residues invertible and every lattice time pinned to the polygon. In the corpora, the polygon lemma alone certifies the large majority of pairs at small V.

Lemma 4.3 (moat counting)**.** *If $\sum_w |B_w| - (n-2) < N$, the sum over the $n-1$ effective residues, then the pair certifies.*

Proof. All $n-1$ bad sets contain 0; adding them one at a time, each set after the first contributes at most $|B_w| - 1$ new elements to the union, so $|\bigcup_w B_w| \leq \sum_w |B_w| - (n-2) < N$ and the union cannot be all of $\mathbb{Z}_N$. □

With Lemma 4.1 the hypothesis is computable from the order signature alone, which is what makes the counting lemma cheap and uniform. Its weakness is equally structural: in the kernel world the sizes have slack (an order-2 residue already contributes $N/2$), so counting certifies nothing there—that slack is precisely the residual measured in §10.

Lemma 4.4 (coincidence counting)**.** *Suppose all effective residues are invertible and span p_m distinct antipodal classes. Then $\left|\bigcup_w B_w\right| \le p_m(2h+1) - (p_m - 1)$; in particular the pair certifies whenever $p_m(2h+1) - (p_m - 1) < N$. At $T = 6$: $p_m = 2$ always certifies, and $p_m = 3$ certifies iff $N \not\equiv 1 \pmod 6$.*

Proof. If $w' \equiv \pm w$ then $B_{w'} = B_w$, so the union is unchanged by merging antipodal partners; it is a union of at most p_m distinct bad sets, each of size exactly $2h+1$ by Lemma 4.1 (invertible case), all containing 0, whence the bound as in Lemma 4.3. At $T = 6$, $h = \lfloor (N-1)/6 \rfloor$: for $p_m = 2$, $4h + 1 \le \frac{2(N-1)}{3} + 1 < N$ for all $N \ge 2$; for $p_m = 3$, $6h + 1 < N$ fails exactly when $h = (N-1)/6$, i.e. $N \equiv 1 \pmod 6$. □

Lemma 4.4 is the sound replacement for a statement that fails. At $n = 4$ the statement "two effective residues coincide or are antipodal ⇒ the pair certifies" holds (verified for all $N \le 80$). At $n = 5$ it is false: the effective residues $(1, 1, 2, 3)$ on $\mathbb{Z}_7$ contain a coincidence, yet $B_1 \cup B_2 \cup B_3 = \{0, \pm 1\} \cup \{0, \pm 4\} \cup \{0, \pm 5\} = \mathbb{Z}_7$—three distinct bad sets suffice to cover because their antipodal classes $\{1, 6\}, \{3, 4\}, \{2, 5\}$ exhaust $\mathbb{Z}_7^*$. The replacement keeps the true part (coincidence shrinks the number of *distinct* bad sets) and adds the counting hypothesis that fails exactly at the falsifier: $N = 7 \equiv 1 \pmod 6$ is the excluded case of the $p_m = 3$ bound.

Lemma 4.5 (lift law)**.** *Let $M \ge 2$, $m \ge 2$, and let $w_1, \ldots, w_{n-1}$ be residues mod M. Then (w_i) admits a covering of $\mathbb{Z}_M$ by bad sets iff (mw_i) admits a covering of $\mathbb{Z}_{mM}$ by bad sets (same T).*

Proof. Let $\pi : \mathbb{Z}_{mM} \to \mathbb{Z}_M$ be reduction. For $k \in \mathbb{Z}_{mM}$, $(mw_i)k \bmod mM = m\,(w_i\,\pi(k) \bmod M)$, so $\|(mw_i)k\|_{mM} = m\,\|w_i\,\pi(k)\|_M$ and $T\|(mw_i)k\|_{mM} < mM \iff T\|w_i\pi(k)\|_M < M$. Thus $B^{(mM)}_{mw_i} = \pi^{-1}(B^{(M)}_{w_i})$, and since π is surjective, $\bigcup_i B^{(mM)}_{mw_i} = \mathbb{Z}_{mM}$ iff $\bigcup_i B^{(M)}_{w_i} = \mathbb{Z}_M$. □

The lift law explains the composite half of the capability map of §5: every covering-capable composite modulus in range is a lift of a smaller *base* modulus (at $n = 4$ the base covering moduli for sum-triples are exactly $\{6, 7, 8, 11, 13, 16\}$, of which $6, 8, 16$ are themselves composite but are *base* in the sense that they are not lifts of any smaller covering-capable modulus—a prime power p^a is its own base unless it is a proper power lift of a capable p^b, $b < a$), so the primitive obstructions live at the base moduli, and the interesting question is which moduli are base.

4.3 The class reduction at prime moduli

At a prime modulus N every nonzero residue is invertible, and the multiplicative structure enters through the *class group* $C = \mathbb{Z}_N^*/\{\pm 1\} \cong \mathbb{Z}_m$ with $m = (N-1)/2$. Write $\mathrm{cls}(x) = \{x, -x\}$ for the antipodal class of x.

Lemma 4.6 (class reduction)**.** *Let N be prime and let $w \ne 0$. Then*

$$B_w \;=\; \{0\} \;\cup\; \bigcup_{j=1}^{h} \mathrm{cls}(j\,w^{-1}).$$

Consequently, $n-1$ unit bad sets cover $\mathbb{Z}_N$ iff the class sets $X_w \cdot S_h$ cover C, where $X_w = \mathrm{cls}(w^{-1})$ and $S_h = \{\mathrm{cls}(1), \ldots, \mathrm{cls}(h)\}$.

Proof. For $k \ne 0$: $k \in B_w \iff \|wk\|_N \le h \iff wk \equiv \pm j$ for some $1 \le j \le h \iff k \equiv \pm j\,w^{-1}$. Since N is prime, every nonzero k is a unit, so $\mathbb{Z}_N \setminus \{0\}$ is the disjoint union of the m antipodal classes; 0 lies in every B_w. Thus the bad sets cover iff their nonzero parts cover every class, and the nonzero part of B_w is the translate $X_w \cdot S_h$ of the class segment S_h. □

At composite N the reduction breaks, and where it breaks is exactly where the composite landscape of §5 lives: a unit bad set reaches only elements $k \equiv \pm j\,w^{-1}$ with $1 \le j \le h$, so a non-unit element can be covered only through a $j \le h$ sharing its gcd with N. This gives the composite safety mechanism.

Proposition 4.7 (composite reachability)**.** *Let N be composite. In any covering of $\mathbb{Z}_N$ by unit bad sets, every non-unit antipodal class must be of the form* $\mathrm{cls}(j\,u)$ *with* $1 \le j \le h$, $\gcd(j, N) > 1$, *and u a unit. If some non-unit class is not of this form, no such covering exists. If N is even, no covering by unit bad sets exists at all.*

Proof. A unit bad set B_w contains exactly the classes $\mathrm{cls}(j\,w^{-1})$ with $1 \le j \le h$ (the computation of Lemma 4.6 is valid for any N as long as w is invertible). The class $\mathrm{cls}(j\,w^{-1})$ is non-unit iff $\gcd(j, N) > 1$, and as w ranges over units, $\mathrm{cls}(w^{-1})$ ranges over all unit classes; hence the non-unit classes reachable by unit bad sets are exactly the classes $\mathrm{cls}(j\,u)$ with $1 \le j \le h$, $\gcd(j, N) > 1$, u a unit, and a covering must cover every class. For even N: if all $n-1$ effective residues are units, they are all odd, hence nonzero mod 2; since $2 \mid N$ and $2 \le 2 \le T$, Lemma 4.2 at $j = 2$ exhibits the certifying time $k = N/2$, so no unit tuple covers. □

Remark 4.8 (the two-runner base)*.* The $n = 4$ run of the program also isolated a two-residue law used as a base case: for all $N \in [3, 80]$ and all residue pairs (a, b) with $a, b \not\equiv 0$, $\max_k \min(\|ak\|_N, \|bk\|_N) \ge N/5$. Equality is attained by the pentagon configuration $(1, 2)$ on $\mathbb{Z}_5$ (and only there: on $\mathbb{Z}_{5m}$ with $m \ge 2$ the left side is $mN/5 > N/5$, e.g. $3, 5, 6$ on $\mathbb{Z}_{10}, \mathbb{Z}_{15}, \mathbb{Z}_{20}$ against $N/5 = 2, 3, 4$); the $5 \mid N$ case is proved (pentagon times $k = jN/5$ when $5 \nmid$ the ratio, a kernel-coset argument otherwise), and the rest is verified per modulus. The nonvanishing hypothesis is needed ($a \equiv 0$ gives value 0). We record it because the pentagon extremal reappears as the tight case of the whole framework: $V = \{1, 2, 3, 4\}$ has $M = \tau = 1/5$ exactly.

5 The rigid zone and the covering classifications

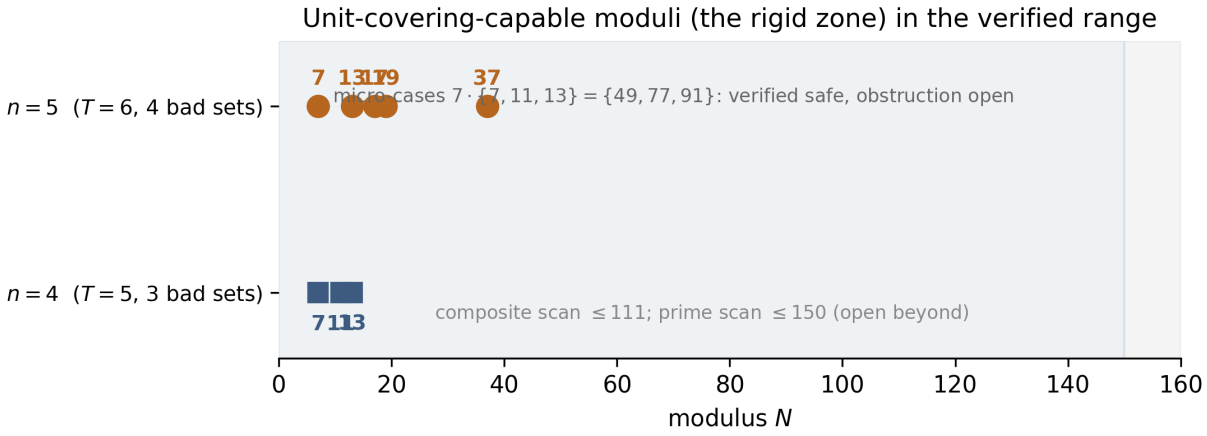
5.1 Capability and the rigid zone

Definition 5.1 (unit capability)**.** Fix the rung n (so $T = n + 1$ and $n - 1$ bad sets per pair). A modulus N is *unit-capable* if some $(n-1)$-multiset of invertible residues $w_1, \ldots, w_{n-1}$ has $\bigcup_i B_{w_i} = \mathbb{Z}_N$.

Two normalizations make capability a finite, exactly decidable property of N alone. First, common scaling: if u is a unit then $B_{uw} = \{k : \|u\,w\,k\|_N \le h\} = u^{-1}B_w$ (because $uwk = w(uk)$), so scaling all residues by u scales each bad set by u^{-1}, a bijection of $\mathbb{Z}_N$—covering is invariant, and we may normalize one residue to 1, leaving $\binom{\varphi(N)+n-3}{n-2}$ scale-normalized tuples to check. Second, the covering counts below are counts of scale-normalized covering $(n-1)$-multisets $(1, b_2, \ldots, b_{n-1})$.

Table 5: The rigid zone: unit-capable moduli in the verified range, with the number of scale-normalized covering configurations at each. The $n = 4$ column is the committed base rung; the $n = 5$ column subsumes the scan of this paper. Verified range: primes ≤ 150, composites ≤ 111 at $n = 5$; all moduli ≤ 60 at $n = 4$.

rung	unit-capable moduli	covering counts
$n = 4$ ($T = 5$, 3 bad sets)	{7, 11, 13}	4, 12, 4
$n = 5$ ($T = 6$, 4 bad sets)	{7, 13, 17, 19, 37}	20, 68, 16, 64, 32



Three features of Table 5 organize this section. First, the zones are *small and scattered*: no arithmetic progression, no obvious multiplicative family—17, 19, 37 are not lifts of anything smaller (they are prime, and prime moduli are their own base by Lemma 4.5). Second, the $n = 4$ zone does not contain the $n = 5$ zone and vice versa: 11 is capable with three bad sets but not with four—counting kills it, since four unit bad sets at $N = 11$ ($h = 1$, $|B_w| = 3$) have union of size at most $4 \cdot 3 - 3 = 9 < 11$—while 17, 19, 37 are incapable with three bad sets and capable with four. The rigid zone is a genuinely rung-dependent object. Third, within the verified range the $n = 5$ zone is *constant* as the range grows: after 37, no modulus whatsoever (prime or composite) is capable, through primes 150 and composites 111. This is the rigid-zone map; it is an empirical finding, stated with its range, and its finiteness is not proved (Conjecture 5.18 is the natural positive statement).

The rest of the section classifies the covering configurations at every capable prime, at both rungs. The classifications are exact if-and-only-if theorems; their proofs at $N \le 17$ are group-theoretic (tilings and matchings), and at 19, 37 rest on an exhaustive finite check whose verification record is in §12. Throughout, $m = (N-1)/2$, $C = \mathbb{Z}_N^*/\{\pm 1\} \cong \mathbb{Z}_m$, and by Lemma 4.6 a unit tuple covers $\mathbb{Z}_N$ iff the class sets $X_w \cdot S_h$ cover C.

5.2 The base rung: three bad sets

Theorem 5.2 (K_7; $n = 4$)**.** *Let $N = 7$ ($h = 1$, $C \cong \mathbb{Z}_3$). Three unit bad sets cover $\mathbb{Z}_7$ iff their antipodal classes $\{\pm w_i^{-1}\}$ are the three distinct classes of $\mathbb{Z}_7^*$. There are 4 scale-normalized covering pairs (b, c) at $a = 1$: $(2, 3)$, $(2, 4)$, $(3, 5)$, $(4, 5)$.*

Proof. Here $S_h = S_1$ is a single class, so B_w contributes exactly the class X_w, and three bad sets cover iff $\{X_1, X_b, X_c\} = C$. The covering pairs are read off $\mathbb{Z}_7^*$: the classes are $\{1, 6\}, \{2, 5\}, \{3, 4\}$; with $a = 1$ contributing $\{1, 6\}$, the pair (b, c) must contribute $\{2, 5\}$ and $\{3, 4\}$, i.e. $b \in \{2, 5\}$, $c \in \{3, 4\}$ with the classes of b, c distinct and completing C—enumerating gives exactly the four listed pairs. □

Theorem 5.3 (K_{11}: domino adjacency)**.** *Let $N = 11$ ($h = 2$, $C \cong \mathbb{Z}_5$, generated by $g = \mathrm{cls}(2)$). Each unit bad set covers the* domino *$\{X_w, X_w g\}$ of C. Three dominos $\{t_i, t_i + 1\}$, $t_i \in \mathbb{Z}_5$, cover C iff the complement of the start set $T = \{t_1, t_2, t_3\}$ (two vertices of C_5) contains no adjacent pair. There are 12 scale-normalized covering pairs.*

Proof. A vertex $p \in C$ is uncovered iff $p \notin T$ and $p - 1 \notin T$; since $|T| = 3$ and $|C| = 5$, covering fails iff the two-element complement contains an adjacent pair $\{p - 1, p\}$. With the scale normalization $w_1 = 1$ (so $e(X_1) = 0 \in T$) the admissible complements are $\{1, 3\}, \{1, 4\}, \{2, 4\}$—that is, the start sets $\{0, 2, 4\}, \{0, 2, 3\}, \{0, 1, 3\}$—and each is realized by four multisets $(1, b, c)$ (each of the two remaining starts is realized by two residues, b and $-b$, with the same class), giving $3 \times 4 = 12$, matching the direct scan. □

Theorem 5.4 (K_{13}: coset tiling)**.** *Let $N = 13$ ($h = 2$, $C \cong \mathbb{Z}_6$, generated by $g = \mathrm{cls}(2)$). Three unit bad sets cover $\mathbb{Z}_{13}$ iff their start classes X_w form a coset $\{X, Xg^2, Xg^4\}$ of the order-3 subgroup of C. There are 4 scale-normalized covering pairs: $(3,4)$, $(3,9)$, $(4,10)$, $(9,10)$ at $a = 1$.*

Proof. Each bad set covers the domino $\{X_w, X_w g\}$; three dominos on the 6-cycle cover iff they are pairwise disjoint (three sets of size two covering six vertices). Disjointness of $\{t, t+1\}$ and $\{s, s+1\}$ on $\mathbb{Z}_6$ is equivalent to $s - t \notin \{-1, 0, 1\}$, i.e. $s - t \in \{2, 3, 4\}$ up to sign. Three starts pairwise at such differences must be $\{t, t+2, t+4\}$: if two starts differ by 3, no third start is compatible (differing from both by $\{2, 3, 4\}$ mod 6), and the remaining possibilities are exactly the cosets of $\{0, 2, 4\}$. The four listed pairs are the realizations with $e(X_1) = 0$ fixed. □

5.3 The second rung: four bad sets

At $n = 5$ the same three group-theoretic mechanisms persist at 7 and 13 (now with four dominos: at $N = 7$, four classes cover C_3 iff they hit all three classes; at $N = 13$, four dominos cover C_6 iff the complement of the start set contains no adjacent pair), and one new mechanism appears at 17.

Theorem 5.5 (K_{17}: perfect matching)**.** *Let $N = 17$ ($h = 2$, $C \cong \mathbb{Z}_8$; $c = \mathrm{cls}(2)$ has order 4 since $2^4 \equiv -1 \pmod{17}$). Each unit bad set covers the domino $\{X_w, X_w c\}$, and dominos live inside the two cosets of the order-4 subgroup $\langle c \rangle$, each coset a 4-cycle under multiplication by c. Four unit bad sets cover $\mathbb{Z}_{17}$ iff each coset contains exactly two of the four starts and the two are non-adjacent in that 4-cycle (equivalently, differ by c^2). There are 16 scale-normalized covering 4-multisets.*

Proof. Within one 4-cycle $\{x, xc, xc^2, xc^3\}$, two dominos $\{X, Xc\}$, $\{Y, Yc\}$ cover the cycle iff they are disjoint iff $Y \notin \{X, Xc, Xc^{-1}\}$, i.e. $Y = Xc^2$. If one coset contained three starts, the other would contain one, and a single domino covers only two of its cycle's four vertices; if one coset contained all four, the other would be untouched. Hence exactly two starts per coset, at antipodal positions of the 4-cycle. The count of scale-normalized realizations is 16, matching the direct scan; the sparsity is the matching constraint. □

Example 5.6 (a covering at 17, by hand)**.** The residues $(1, 4, 6, 7)$ cover $\mathbb{Z}_{17}$:

$$B_1 \cup B_4 \cup B_6 \cup B_7 = \{0, \pm 1, \pm 2\} \cup \{0, \pm 4, \pm 9\} \cup \{0, \pm 3, \pm 6\} \cup \{0, \pm 5, \pm 10\} = \mathbb{Z}_{17},$$

the eight classes $\{\pm 1, \pm 2, \pm 4, \pm 9, \pm 3, \pm 6, \pm 5, \pm 10\}$ exhausting C_8. This is realizable as a genuine speed set: the pair $(1, 16)$ of $V = (1, 4, 6, 7, 16)$ has effective residues $(1, 4, 6, 7)$ on $\mathbb{Z}_{17}$ and is a true (non-certifying) pair. Hand-checkable anchors of this kind anchor every classification to the original problem.

At $N = 19$ and $N = 37$ the covering radius h reaches 3 and 6, the class segments S_h grow past two elements, and the tiling gloss ends: the classifications become explicit finite lists. Fix a generator $g = \mathrm{cls}(2)$ of C (at both moduli $2^m \equiv -1$, verified), write $e(\mathrm{cls}(g)^t) = t$ for the *exponent* of a class, and let $S = e(S_h) \subseteq \mathbb{Z}_m$. By Lemma 4.6, a unit 4-tuple covers $\mathbb{Z}_N$ iff

$$\bigcup_{i=1}^{4} (e(X_{w_i}) + S) = \mathbb{Z}_m.$$

Multiplying all four residues by a unit u shifts every exponent by the constant $-e(\mathrm{cls}\, u)$, so covering depends only on the *translate class* of the exponent multiset in $\mathbb{Z}_m$.

Theorem 5.7 (K_{19})**.** *Let $N = 19$: $m = 9$, $h = 3$, $S = \{0, 1, 4\}$. Four unit bad sets cover $\mathbb{Z}_{19}$ iff the exponent multiset $\{e(X_{w_i})\}$ is a translate of*

$$A = \{0, 1, 2, 7\} \quad (\textit{gaps } 1, 1, 5, 2) \qquad \textit{or} \qquad B = \{0, 2, 4, 6\} \quad (\textit{gaps } 2, 2, 2, 3),$$

the latter being the doubled interval $2 \cdot \{0,1,2,3\}$. *There are exactly two covering translate classes and* 64 *scale-normalized covering residue* 4*-multisets: each class has* 4 *translates and, per translate,* $2^3 = 8$ *sign realizations (each nonzero exponent realized by* $\pm$ *its residue), i.e.* 32 *residue* 4*-multisets per class; all have four distinct exponents.*

Theorem 5.8 (K_{37})**.** *Let* $N = 37$*:* $m = 18$*,* $h = 6$*,* $S = \{0,1,2,5,8,9\}$*. Four unit bad sets cover* $\mathbb{Z}_{37}$ *iff the exponent multiset is a translate of the single class*

$$C_0 = \{0,2,12,15\} \quad (\textit{gaps } 2,10,3,3).$$

There is one covering translate class and 32 *scale-normalized covering residue* 4*-multisets.*

Proof of Theorems 5.7 and 5.8. The reduction to translate classes is proved above. The class lists are then decided by exhaustive enumeration of all contains-0 4-multisets of $\mathbb{Z}_m$ (165 candidates at $m = 9$, 1,140 at $m = 18$—hand-checkable scale), checking $\bigcup(e_i + S) = \mathbb{Z}_m$ exactly. Correctness of the packaging was verified bidirectionally against the residue level: over *all* unit 4-multisets (5,985 at $N = 19$, 82,251 at $N = 37$), the predicate "exponent multiset is a translate of a listed class" agrees with direct bitmask covering, with zero mismatches in either direction; the residue covering counts 64 and 32 were reproduced by three independent implementations (bitmask, frozenset, exponent-space). The full verification record is in §12. □

Remark 5.9 (what the classifications say structurally). Three structural facts deserve the reader's attention, because they are the paper's central finding. First, *the exponent multiset has four distinct entries at every capable prime* $N \geq 17$: at $N = 7$ and $N = 13$ the class group is too small ($|C| = 3, 6$) to support four distinct classes, and covering there necessarily repeats a class (at $N = 7$, all 12 n=5 coverings repeat a class; at $N = 13$, 24 of 68 do); from $N = 17$ onward ($17, 19, 37$, the primes where the classification was verified with distinct exponents) every covering has four distinct classes and no exponent is repeated. This is consistent with Lemma 4.4: $19 \equiv 1 \pmod 6$ is exactly the case its $p_m = 3$ bound cannot reach, and the classification shows covering there needs the full $p_m = 4$. Second, *exponent inversion (class reflection) is a diagnostic, not a symmetry*: the map $w \mapsto w^{-1}$ sends an exponent class to its reflection, and it does *not* preserve the covering property—at $N = 19$ it carries a covering tuple to a covering tuple in 144 of 288 cases and to a non-covering tuple in the other 144 (verified). The earlier "negation" phrasing conflated two maps: the exponent negation $e \mapsto -e$ (which fixes the bad set, $B_{-w} = B_w$, since $\|-wk\| = \|wk\|$—a genuine trivial invariance that identifies w with $-w$ in the class) and the class inversion $e \mapsto -e$ on exponents (the nontrivial reflection, which is the diagnostic). Inversion is used as a diagnostic only; quotienting the census by it is not valid (the map does not preserve coverings). Third, the character of the classification changes at $h \geq 3$: K_{13} and K_{17} are group-theoretic statements (tilings, matchings) with slick proofs, while K_{19} and K_{37} are finite lists. The classification is complete but the tiling gloss does not persist; this is a change in the texture of the classification, not a gap in the method.

5.4 Composite moduli

Composites split cleanly by parity (Proposition 4.7): even composites are never unit-capable, and odd composites are governed by the reachability mechanism—a covering by unit bad sets must cover every non-unit class, but unit bad sets reach non-unit classes only through $j \leq h$ with $\gcd(j, N) > 1$. Table 6 records the outcome for every odd composite in the open part of the range (49–111): a *proved* safety whenever some non-unit class is unreachable, and a handful of *micro-cases* where all non-unit classes are reachable, the slot counts do not obstruct, and no *unit* covering exists anyway (Table 6; the qualification "unit" is load-bearing — see Remark 5.10).

The micro-cases form a pattern too clean to be coincidence:

$$49 = 7 \cdot 7, \qquad 77 = 7 \cdot 11, \qquad 91 = 7 \cdot 13,$$

Table 6: Odd composites 49–111 at $n = 5$ ($T = 6$): non-unit antipodal classes, how many are reachable by unit bad sets (via $j \le h$, $\gcd(j, N) > 1$), and capability. All entries cross-checked by an independent set-based scanner; zero capable moduli.

N	h	non-unit classes	reachable	status
$49 = 7^2$	8	3	**3**	micro-case (verified safe)
$51 = 3 \cdot 17$	8	9	8	proved safe
$55 = 5 \cdot 11$	9	7	5	proved safe
$57 = 3 \cdot 19$	9	10	9	proved safe
63	10	13	12	proved safe
$65 = 5 \cdot 13$	10	8	6	proved safe
69	11	12	11	proved safe
75	12	17	14	proved safe
$77 = 7 \cdot 11$	12	8	**8**	micro-case (verified safe)
$81 = 3^4$	13	13	12	proved safe
$85 = 5 \cdot 17$	14	10	8	proved safe
$87 = 3 \cdot 29$	14	15	14	proved safe
$91 = 7 \cdot 13$	15	9	**9**	micro-case (verified safe)
$93 = 3 \cdot 31$	15	16	15	proved safe
$95 = 5 \cdot 19$	15	11	9	proved safe
$99 = 9 \cdot 11$	16	19	18	proved safe
$105 = 3 \cdot 5 \cdot 7$	17	28	25	proved safe
$111 = 3 \cdot 37$	18	19	18	proved safe

that is, the products of 7 with the $n = 4$ rigid-zone primes $\{7, 11, 13\}$. These are precisely the odd composites in range where *every* non-unit class is reachable, so the reachability mechanism is silent, yet no unit covering exists; the obstruction is multiplicative. The pattern predicts that the next micro-cases, if any, should be sought at $7 \cdot 17 = 119$ and $7 \cdot 19 = 133$ (and $7 \cdot 37 = 259$), all outside the scanned composite range. We flag this as an open problem (§11) rather than a conjecture: three data points do not make one.

Remark 5.10 (post-submission development at the composite moduli). Work completed after this manuscript was assembled (companion note) changes the micro-case picture in three ways, and the open problem is restated accordingly in §11. First, the *fibration*: a bad set whose speed has $d = \gcd(w, N) > 1$ is the full preimage of a unit bad set at N/d with the *same* threshold. Consequently kernel-speed coverings *do* exist at the micro-cases — for example $B_7 \cup B_{14} \cup B_{21} = \mathbb{Z}_{49}$ is the transferred K_7 classification — so "no covering at composites" is false as a blanket statement; what the micro-cases exclude is unit coverings, and more generally any covering in which units are load-bearing. Second, the unit-side statement that survives is *no unit assistance*: at $N = p^2$ and $N = pq$ with $p \ge 5$, a covering family of at most four distinct bad sets with at most two unit members has its kernel subfamily covering; three unit bad sets never cover at all for $N \ge 31$, $N \not\equiv 0 \pmod 6$; and a pure-unit family covers only above rigid-zone primes — which makes the unit safety of $77 = 7 \cdot 11$ a theorem rather than a verification (its reduction demands a covering of $\mathbb{Z}_{11}$, and 11 is not in the $n = 5$ zone). Third, the residual is explicit: the cells with three unit sets and at most one kernel set, occupied at $N = 9$ (a genuine unit-assisted covering, and the reason the prime hypothesis cannot be dropped), verified empty at $p = 5$ and throughout the tested composite range.

Fourth (the p^2 closure, completing the companion note's programme): the last open cell at square moduli — one kernel set with three unit sets — is now *empty at every prime* $p \ge 5$. For unit triples whose residues are pairwise $\pm$-distinct mod p, the trichotomy programme applies: Lemma B ($p \equiv 5 \pmod 6$) is proved for $p \ge 17$ (the kills run on the contained census, Lemma T, clean from $p = 17$, and Corollaries T1–T3; a pre-T3 draft of the overlap kill had required $p \ge 23$), with $p = 11$ closed by exhaustive finite case analysis (36 viable configurations, each carrying

a positive residual-fit margin of at least 2, cross-validated by an independent brute force that reproduces the known failure counts 50 at $p = 7$ and 2 at $p = 13$ exactly, the $p = 17$ finite analysis a redundant verification); Lemma A ($p \equiv 1 \pmod 6$) is proved for $p \geq 31$ unconditionally — its one remaining input, the foot-count trichotomy in its consumed form (the one-foot statement at the two size pairs $(2k{+}2, 2k{+}1)$ and $(2k{+}1, 2k{+}1)$, viable differences confined to $\{2, (p-1)/2\}$), is discharged by Lemma 5.16 of §5.5 (the three-distance foot count, proved for every $p \geq 31$ and cross-checked against the foot census of the previous revision to $p \leq 419$) — with $p = 19$ closed the same way (614 viable configurations, margin 3), and it is *false* at $p = 7, 13$, where $\pm$-distinct coverings exist and the cells are closed by direct verification. For $\pm$-colliding units — two speeds equal up to sign mod p, a configuration the cell definition does not exclude and the trichotomy lemmas do not reach — a new argument closes the case: the two colliding bad sets are shears of one another, so their per-fiber t-space placements drift *exactly linearly* in the fiber index, while the fiber covering configurations are pinned (the placements' difference set has at most 6 elements, with covering counts 28 and 68 per collision topology, *independent of* p); a pigeonhole over the $p - |B_k|$ kernel-uncovered fibers then kills every family. The mechanism is proved in general; its computational input is stated as a named lemma with its verification range (every prime $11 \leq p \leq 113$), and the ground truth is an exhaustive enumeration of 3.3 million colliding families at $p \leq 43$ with zero coverings, the nearest families failing all but two fibers.

The closure's exact dependency structure is now recorded in §6, and it is sharper than "modulo two computational inputs": of the two named inputs, one is now *proved* — the foot-count inequality for the one-foot trichotomy (in the consumed form, clean from $p \geq 31$) is Lemma 5.16 of §5.5, an elementary three-distance argument with no group-ring content — so the closure rests on a single unbounded hypothesis, the four-family covering classification for the pinning lemma (verified to $p \leq 113$, and subsuming the shear-rigidity input because the pinned per-fiber configurations are exactly the classified families). The pq side of §7 now carries the same structure at semiprime moduli: the $(1, 1, 2)$ and three-unit cells at $N = pq$ are verified empty to $N \leq 143$ and reduce, by the row translation, to exactly the same remaining input.

5.5 Discharging Input E: the three-distance foot count

Input E in its consumed form (§6.4) is the one-foot trichotomy at the two size pairs consumed by the kills M4 and M5 of Lemma A: at $p = 6k{+}1$, an arithmetic progression $A(a, d, s)$ of critical size $s \in \{2k{+}1, 2k{+}2\}$ with at most one point outside $I_1 = [s_1, p-1]$ (where $(s_1, s) = (2k{+}2, 2k{+}1)$ for M4 and $(2k{+}1, 2k{+}1)$ for M5) must have $\|d\| \in \{2, (p-1)/2\}$. This subsection proves that statement outright for every prime $p \geq 31$ (Lemma 5.16 below) — the discharge route identified in §6.4, carried out by an elementary, self-contained analysis of the rotation $j \mapsto jd \bmod p$: the three-distance theorem of that rotation, in exactly the regime the consumers need. No group-ring input is used. Throughout, $p = 6k + 1$ is prime, $3k = (p-1)/2$, and d is a $\pm$-representative in $[2, 3k]$.

Notation. For $s \geq 2$ put $X = X_d^{(s)} = \{jd \bmod p : 0 \leq j < s\}$ (so $A(a, d, s) = a + X$), let $x_1 < \cdots < x_s$ be its sorted elements and $g_i = x_{i+1} - x_i$ for $i < s$, $g_s = p - x_s + x_1$ its cyclic gaps; the pair sums $g_i + g_{i+1}$ (indices mod s) are the spans $\mathrm{next}(x) - \mathrm{prev}(x)$ over $x \in X$, and $G_2(X) = \max_i(g_i + g_{i+1})$. Recall $\mathrm{feet}(A) = s - |A \cap I_1|$, the number of points of A in $[0, s_1)$. For $R \subseteq \mathbb{Z}_n$ with $0 \in R$, sorted $0 = \rho_1 < \cdots < \rho_t$, let $\Gamma_2(R; n)$ be the largest sum of cyclically adjacent gaps of R on $\mathbb{Z}_n$ (the last gap being $n - \rho_t$; $t = 1$: $\Gamma_2 = 2n$), and write $\gamma^{\mathrm{f}}(R) = \rho_2$ (first gap) and $\gamma^{\mathrm{w}}(R; n) = n - \rho_t$ (wrap gap).

Lemma 5.11 (feet and pair sums)**.** *For $X \subseteq \mathbb{Z}_p$ with $|X| \geq 2$ and $2 \leq L \leq p$: every cyclic interval of L consecutive residues meets X in at least two points iff $G_2(X) \leq L$. In particular,* $\min_a \mathrm{feet}(A(a, d, s)) \geq 2$ *iff* $G_2(X_d^{(s)}) \leq s_1$.

Proof. An L-interval $[u, u+L)$ contains at most one point of X iff some $x \in X$ has $\mathrm{prev}(x) < u$ and $u + L \le \mathrm{next}(x)$: for the unique point of a one-point instance this is immediate, and for an empty instance it holds with x the first point at or after $u + L$. Either way $\mathrm{span}(x) \ge L + 1$; conversely a point of span $\ge L+1$ produces the one-point interval $[\mathrm{prev}(x)+1,\ \mathrm{prev}(x)+L+1)$. For the second claim, $\#(a + X) \cap [0, s_1) = \#X \cap ([0, s_1) - a)$, and $([0, s_1) - a)$ runs through all cyclic s_1-intervals as a does. □

Lemma 5.12 (column decomposition)**.** *Let $s \in \{2k+1, 2k+2\}$, $2 \le d \le 3k$, and $W = \lfloor d(s-1)/p \rfloor \ge 1$; write $r = p \bmod d$, $\sigma = d - r$, $m_0 = \lfloor p/d \rfloor$, and $c_w = (-wp) \bmod d \in [0, d)$ for $0 \le w \le W$. Then:* (i) *$c_0, \ldots, c_W$ are distinct;* (ii) *the j with $\lfloor dj/p \rfloor = w$ form the interval $[\lceil wp/d \rceil,\ \min(\lceil (w+1)p/d \rceil - 1,\ s-1)]$, and the corresponding points $jd \bmod p = dj - wp$ are exactly $c_w, c_w + d, \ldots, c_w + (h_w - 1)d$, where $h_w = \lceil (w+1)p/d \rceil - \lceil wp/d \rceil$ for $w < W$ and $h_W = s - \lceil Wp/d \rceil$;* (iii) *for $w < W$ the run is the* full *column $\{x \in [0, p) : x \equiv c_w \pmod d\}$ — of height $F(c) = m_0 + 1$ for $c < r$ and m_0 otherwise, with top $p - d + c_{w+1}$ — while run W is the prefix of the column of c_W of height $h_W \le F(c_W)$, with top $H = d(s-1) - Wp$.*

Proof. (i) p is prime and $d < p$, so $\gcd(p, d) = 1$; and $W \le d(s-1)/p \le d(2k+2)/(6k+1) < d$, so $w \mapsto wp \bmod d$ is injective on $[0, W]$. (ii) $\lfloor dj/p \rfloor = w$ iff $wp \le dj < (w+1)p$, i.e. $j \in [\lceil wp/d \rceil, \lceil (w+1)p/d \rceil - 1]$, truncated at $s-1$; the sequence $\lfloor dj/p \rfloor$ increases in steps of 0 or 1 (as $d < p$), so every $w \le W$ occurs. The first point of run w is $d\lceil wp/d \rceil - wp \in [0, d)$ with residue $\equiv -wp \equiv c_w$, so it equals c_w; the run then steps by d within its column. (iii) For $w < W$ the last point is $d(\lceil (w+1)p/d \rceil - 1) - wp = p - d + \beta_w$, where $\beta_w = d\lceil (w+1)p/d \rceil - (w+1)p \in [0, d)$ is the canonical representative of c_{w+1}. This top lies in $[p-d, p)$ and satisfies $p - d + \beta_w \equiv r + c_{w+1} \equiv c_w \pmod d$ (since $c_{w+1} \equiv c_w + \sigma$ and $r + \sigma = d$); being congruent to c_w and in $[p-d, p)$ it is the top of the full column of c_w, whose $F(c) = \lceil (p-c)/d \rceil$ elements are $c, c+d, \ldots$ — so run w is that full column and $h_w = F(c_w)$. Run W spans $j \in [\lceil Wp/d \rceil, s-1]$, hence is the bottom prefix of height $h_W = s - \lceil Wp/d \rceil \le F(c_W)$, with top $d(s-1) - Wp$. □

Lemma 5.13 (bands and the pair-sum identity)**.** *In the setting of Lemma 5.12 with $W \ge 1$, put $C = \{c_0, \ldots, c_W\}$, $C^- = C \setminus \{c_W\}$, $h = h_W$, and $T = (C^- \cap [0, r)) \cup E$ with $E = \{c_W\}$ if $h > m_0$ and $E = \emptyset$ otherwise. Then:* (i) *for $0 \le i \le m_0$, the residues of X in the band $[id, (i+1)d)$ are exactly $R_i = C$ if $i < h$, $R_i = C^-$ if $h \le i \le m_0 - 1$, and $R_{m_0} = T$; in particular $R_0 \supseteq \cdots \supseteq R_{m_0}$, all containing 0;* (ii) *the gap sequence $(g_1, \ldots, g_s)$ is the concatenation, over $i = 0, \ldots, m_0 - 1$, of the cyclic gap block of R_i on $\mathbb{Z}_d$ (consecutive differences of R_i, ending with the wrap gap $d - \max R_i$), followed by the top block of T (consecutive differences, ending with $r - \max T$);* (iii) *the pair sums of X are the cyclic two-step gaps of these blocks — a singleton band contributing the double wrap $2d$, attained iff the band below is also a singleton; singleton bands occur only for $W = 1$ — together with the boundary pairs $\gamma^{\mathrm{w}}(R_{i-1}; d) + \gamma^{\mathrm{f}}(R_i)$ (reading $\gamma^{\mathrm{f}} = d$ for a singleton band), of which those with $i \le m_0 - 1$ are at most $\gamma^{\mathrm{w}}(R_i; d) + \gamma^{\mathrm{f}}(R_i) \le \Gamma_2(R_i; d)$ by (i), together with the two outer pairs*

$$\mathrm{span}_0 = (r - \max T) + \gamma^{\mathrm{f}}(C), \qquad \mathrm{span}^* = (d - \max R_{m_0-1}) + \begin{cases} \gamma^{\mathrm{f}}(T) & |T| \ge 2, \\ r & |T| = 1. \end{cases}$$

Consequently, for $W \ge 2$,

$$G_2(X) = \max(\Gamma_2(C; d),\ \Gamma_2(C^-; d)\ \textit{[if } h \le m_0 - 1\textit{]},\ \Gamma_2(T; r)\ \textit{[if } |T| \ge 2\textit{]},\ \mathrm{span}_0,\ \mathrm{span}^*),$$

while for $W = 1$, $G_2(X) \le 2d$.

Proof. (i) By Lemma 5.12, X is the union of the runs, run w occupying bands $0, \ldots, h_w - 1$ at residue c_w. For $i \le m_0 - 1$ every full run is present ($h_w = F(c_w) \ge m_0 > i$), and run W is present iff $h > i$; hence $R_i = C$ for $i < h$ and C^- otherwise. Band m_0 sees exactly the full runs with $F(c_w) > m_0$, i.e. $c_w < r$, plus run W iff $h > m_0$: that is T. The nesting is immediate, and

$0 = c_0 \in C^-$ (as $W \geq 1$) with $0 < r$ puts 0 in every band. (ii) Band i's points, in increasing order, are $c + id$ with $c \in R_i$ in residue order, so the within-band gaps are the residue differences; the gap from band i's last point to band $i{+}1$'s first (residue 0, since $0 \in R_{i+1}$) is $d - \max R_i$; and the final wrap, from $\max T + m_0 d$ to 0, is $r - \max T$ (as $p = m_0 d + r$). (iii) The pair sum around a point of band i is either within-block (a two-step gap of R_i; around the band's last point it is the last two gaps of the block, equally a two-step gap) or crosses the boundary below, where it is the last gap of block $i-1$ plus the first gap of block i; for $i \leq m_0 - 1$ the nesting (i) gives $\gamma^{\mathrm{w}}(R_{i-1}; d) \leq \gamma^{\mathrm{w}}(R_i; d)$, and the wrap and first gaps are cyclically adjacent, so the boundary pair is at most $\Gamma_2(R_i; d)$. A singleton band $R_i = \{0\}$ arises only when $C^- = \{0\}$, i.e. $W = 1$; its lone point has pair sum $\gamma^{\mathrm{w}}(R_{i-1}; d) + d$, which equals $2d$ exactly when $R_{i-1} = \{0\}$ as well. The top band contributes the two-step gaps of T on $[0, r)$ (only for $|T| \geq 2$; around T's last point the pair is the last within-gap plus $r - \max T$, again a two-step gap on $\mathbb{Z}_r$), the boundary pair span^* around $m_0 d$, and — around 0 — the outer pair span_0. For $W \geq 2$ no band is a singleton ($|C^-| = W \geq 2$), so every pair sum is at most the maximum of the five displayed quantities; conversely each of the five is realized as a pair sum of X ($\Gamma_2(C)$ in band 0; $\Gamma_2(C^-)$ in a C^--band, which exists exactly when $h \leq m_0 - 1$; $\Gamma_2(T; r)$ in the top band; span_0 around 0; span^* around $m_0 d$). For $W = 1$: $\Gamma_2(C; d) = d$, the singleton-band terms are at most $2d$, $\mathrm{span}_0 \leq d$, and $\mathrm{span}^* < 2d$: if $|T| = 1$ then $R_{m_0-1} \in \{C, C^-\}$ gives $(d - \sigma) + r = 2r$ resp. $d + r$, both $< 2d$; if $|T| = 2$ then $h = m_0 + 1$, $R_{m_0-1} = C$, and $(d - \sigma) + \gamma^{\mathrm{f}}(T) < (d - \sigma) + r = d$. Hence $G_2 \leq 2d$. □

Lemma 5.14 (wrap count)**.** *For $k \geq 2$ and $s = 2k{+}1$: $W \geq d - 2k + 1$ for $3 \leq d \leq 3k - 2$, $W = d - 2k$ at $d = 3k - 1$, and $W = d - 2k - 1$ at $d = 3k$. For $s = 2k{+}2$: $W \geq d - 2k + 1$ for $3 \leq d \leq 3k - 2$, and $W = d - 2k$ at $d \in \{3k - 1, 3k\}$.*

Proof. Since $d - 2k + 1$ is an integer, $W = \lfloor d(s-1)/p \rfloor \geq d - 2k + 1$ iff $d(s-1) \geq (d - 2k + 1)p$, a quadratic inequality in d. For $s = 2k{+}1$ it reads $d(4k{+}1) \leq 12k^2 - 4k - 1$, i.e. $d \leq 3k - 1 - \frac{3k}{4k+1} < 3k - 1$, so it holds for $d \leq 3k - 2$ — for $d \leq 2k$ trivially, since then $d - 2k + 1 \leq 1 \leq W$ (with $W = 0 \geq 4 - 2k$ at $d = 3$, and $W \geq 1$ for $d \geq 4$). The endpoint values are direct: $W = \lfloor 2k(3k-1)/(6k+1) \rfloor = k - 1 = d - 2k$ and $W = \lfloor 6k^2/(6k+1) \rfloor = k - 1$. For $s = 2k + 2$ the inequality reads $4kd \leq 12k^2 - 4k - 1$, i.e. $d \leq 3k - 1 - \frac{1}{4k}$, again $d \leq 3k - 2$ (for $d \leq 2k$ trivially: $d - 2k + 1 \leq 1 \leq W$, $W \geq 1$ for $d \geq 3$), with endpoints $W = \lfloor (2k+1)(3k-1)/(6k+1) \rfloor = k - 1$ and $W = \lfloor 3k(2k+1)/(6k+1) \rfloor = k$. □

Lemma 5.15 (auxiliary bounds)**.** *In the setting of Lemmas 5.12–5.13:* (a) *for $R \subseteq \mathbb{Z}_n$ with $0 \in R$ and $2 \leq |R| \leq n$, $\Gamma_2(R; n) \leq n - |R| + 2$;* (b) *if $d \leq 3k - 1$ then $r \leq 2k - 1$;* (c) $\max C \leq \max T + \sigma$; (d) $\mathrm{span}_0 \leq \Gamma_2(C; d)$; (e) *if $W \geq 2$ then* $\mathrm{span}^* \leq d - W + 2$.

Proof. (a) Let $t = |R|$ and let the cyclic gaps be $\gamma_1, \ldots, \gamma_t$ (summing to n, each ≥ 1). Every two-step gap is $\gamma_i + \gamma_{i\pm1} \leq \gamma_{\max} + \gamma'$, where γ' is the largest of the other $t - 1$ gaps and hence at most $(n - \gamma_{\max})/(t-1)$; the map $\gamma \mapsto \gamma + (n - \gamma)/(t-1)$ is increasing (constant when $t = 2$), and $\gamma_{\max} \leq n - t + 1$ by the pigeonhole principle, so $\Gamma_2 \leq (n - t + 1) + 1 = n - t + 2$. (b) If $d \leq 2k$ then $r \leq d - 1$; if $2k + 1 \leq d \leq 3k - 1$ then $2d \leq 6k - 2 < p$ and $3d \geq 6k + 3 > p$, so $m_0 = 2$ and $r = p - 2d \leq 6k + 1 - 4k - 2 = 2k - 1$. (c) The orbit moves by $c_{w+1} = c_w + \sigma$ when $c_w < r$ (no wrap: $c_w + \sigma < r + \sigma = d$) and by $c_{w+1} = c_w - r$ when $c_w \geq r$; hence any orbit point $x \geq \sigma$ is $c_w + \sigma$ with $c_w = x - \sigma \in [0, r)$. If moreover $x \neq c_W$ then $c_w \in C^- \cap [0, r) \subseteq T$, so $x \leq \max T + \sigma$; for $x = c_W \geq \sigma$ the same argument uses $c_{W-1} = c_W - \sigma \in C^- \cap [0, r) \subseteq T$ ($W \geq 1$, so c_{W-1} exists). An orbit point x with $r \leq x < \sigma$ (possible only when $\sigma > r$) satisfies $x < \sigma \leq \max T + \sigma$; a point of $C^- \cap [0, r)$ is at most $\max T$; and $c_W < \sigma$, $x = 0$ are immediate. So every $x \in C$ has $x \leq \max T + \sigma$. (d) By (c), $r - \max T = (d - \sigma) - \max T \leq d - \max C = \gamma^{\mathrm{w}}(C; d)$; the wrap and first gaps are cyclically adjacent, so $\mathrm{span}_0 \leq \gamma^{\mathrm{w}}(C; d) + \gamma^{\mathrm{f}}(C) \leq \Gamma_2(C; d)$. (e) Reflect the orbit: $C \setminus \{0\} = \{d - m : m \in M\}$ with $M = \{wr \bmod d : 1 \leq w \leq W\}$, and $C^- \setminus \{0\} = \{d - m : m \in M^-\}$ with $M^- = \{wr \bmod d : 1 \leq w \leq W - 1\}$; thus $d - \max C = \min M =: \mu \leq \min M^- =: \mu^-$, so $d - \max R_{m_0-1}$ is μ or μ^-, and the elements of

$T \setminus \{0\}$ are reflections $d-m$ of multiples $m \in (d-r,\, d)$. If $|T| = 1$: no multiple of M^- lies in $(d-r, d)$ (its reflection would lie in $C^- \cap (0, r) \subseteq T \setminus \{0\}$), so the $W-1$ distinct integers of M^- lie in $[1, d-r]$ and $\mu^- \le (d-r)-(W-2) = d-W+2-\sigma$, whence $\mathrm{span}^* \le \mu^- + r \le d-W+2$. If $|T| \ge 2$, let $t = \gamma^{\mathrm{f}}(T) = \min(T \setminus \{0\})$. When $t \in C^-$: no multiple of M^- lies above $m_t = d-t$ (its reflection would lie in $C^- \cap (0, t)$, below t), so $M^- \subseteq [\mu^-, m_t]$ gives $m_t \ge \mu^- + W - 2$ and $\mathrm{span}^* \le \mu^- + (d - m_t) \le d - W + 2$. When $t = c_W$ (which forces $h > m_0$ and $c_W < r$, i.e. $m_W = d - c_W \in (d-r, d)$, and $R_{m_0-1} = C$): no multiple at all lies above m_W (a smaller-index one would reflect into $C^- \cap (0, c_W)$ below t, and index W is m_W itself), so $M \subseteq [\mu, m_W]$ gives $m_W \ge \mu + W - 1$ and $\mathrm{span}^* = \mu + (d - m_W) \le d - W + 1$. □

Lemma 5.16 (three-distance foot count; Input E discharged)**.** *Let $p = 6k+1$ be prime with $k \ge 5$, let $d \in [3, 3k-1]$, and let $s \in \{2k{+}1, 2k{+}2\}$. Then $G_2(X_d^{(s)}) \le 2k+1$. Consequently, at both consumer pairs $(s_1, s) = (2k{+}2, 2k{+}1)$ and $(2k{+}1, 2k{+}1)$, $\min_a \mathrm{feet}(A(a, d, s)) \ge 2$ for every $d \in [3, 3k-1]$: the one-foot viable set is contained in $\{2, (p-1)/2\}$, which is Input E in its consumed form, and Lemma A is proved for $p \ge 31$.*

Proof. Case $W = 0$. Only $d = 3$, $s = 2k{+}1$: $X = \{0, 3, \ldots, 6k\}$ has gaps $3, \ldots, 3, 1$, pair sums 6 and 4; $G_2 = 6 \le 2k+1$. *Case $W = 1$.* Then $p \le d(s-1) < 2p$, which at $s = 2k{+}1$ gives $d \in \{4, 5, 6\}$ and at $s = 2k{+}2$ gives $d \in \{3, 4, 5\}$; by Lemma 5.13(iii), $G_2 \le 2d$. If $d \le 5$ then $G_2 \le 10 \le 2k{+}1$ (as $k \ge 5$). If $d = 6$ (so $s = 2k{+}1$): here $r = 1$, $m_0 = k$, and $h = s - F(0) = k$, so no band is a singleton, $T = \{0\}$, $\mathrm{span}^* = 2r = 2$, and $G_2 \le \max(d, \mathrm{span}_0 \le d, \mathrm{span}^*) = 6 \le 2k{+}1$. *Case $W \ge 2$, $d \le 3k-2$.* Lemma 5.14 gives $W \ge d - 2k + 1$, and no band is a singleton (that would force $C^- = \{0\}$, i.e. $W = 1$). Lemma 5.13(iii) and Lemma 5.15 bound each candidate: $\Gamma_2(C; d) \le d - W + 1 \le 2k$ (Lemma 5.15(a) with $|C| = W + 1$); $\Gamma_2(C^-; d) \le d - W + 2 \le 2k + 1$ (same, $|C^-| = W \ge 2$; only if it occurs); $\Gamma_2(T; r) \le r - |T| + 2 \le r \le 2k - 1$ (Lemma 5.15(a),(b); only if $|T| \ge 2$); $\mathrm{span}_0 \le \Gamma_2(C; d) \le 2k$ (Lemma 5.15(d)); $\mathrm{span}^* \le d - W + 2 \le 2k + 1$ (Lemma 5.15(e)). Hence $G_2 \le 2k + 1$. *Case $d = 3k - 1$.* Here $r = 3$, $\sigma = 3k - 4$, $W = k - 1$ (Lemma 5.14), and, since $p \equiv 3 \pmod d$, $c_w = (-3w) \bmod d$; as $3w \le 3k - 3 < d$ for $w \le k - 1$, $C = \{0\} \cup \{d - 3w : 1 \le w \le k{-}1\} = \{0, 2, 5, \ldots, 3k{-}4\}$, an arithmetic progression with difference 3 together with 0, with cyclic gaps $2, 3, \ldots, 3$ and $\Gamma_2(C; d) = 6$. Full columns have $F(0) = 3$ and $F(c) = 2$ for $c \ge 3$, so $\sum_{c \in C^-} F(c) = 3 + 2(k-2) = 2k - 1$ and $h = s - (2k-1) \in \{2, 3\}$: in either case $h \ge m_0 = 2$, so C^- never occurs as a band set; $T = \{0\}$ (at $s = 2k{+}1$) resp. $T = \{0, 2\}$ (at $s = 2k{+}2$); $\Gamma_2(T; 3) \le 3$; $\mathrm{span}_0 \le 3 + 2 = 5$; $\mathrm{span}^* = (d - \max C) + r = 3 + 3 = 6$. So $G_2 = 6 \le 2k + 1$. Finally, the primes $6k+1$ with $k \le 4$ are $7, 13, 19$ (25 is not prime); at $p = 7$ the range $[3, 3k-1]$ is empty, and $13, 19$ are Lemma A's finite-cased primes, so $k \ge 5$ is exactly $p \ge 31$. The consequence on viable sets follows from Lemma 5.11 together with the endpoint values recorded in Remark 5.17. □

Remark 5.17 (endpoints and sharpness). The two excluded differences carry exact values. First, $G_2(X_2^{(2k+1)}) = 2k{+}3$: the progression $\{0, 2, \ldots, 4k\}$ has gaps $2, \ldots, 2, 2k{+}1$, and the two outer pair sums are $2k + 3$. Second, $G_2(X_{3k}^{(s)}) = 2k{+}2$ for both s: at $s = 2k{+}1$ through $\Gamma_2(C; d)$ with $C = \{0\} \cup \{2k{+}1, \ldots, 3k{-}1\}$; at $s = 2k{+}2$ through $\Gamma_2(C^-; d)$ with $C^- = \{0\} \cup \{2k{+}1, \ldots, 3k{-}1\}$, which occurs since then $h = 1$ — in both cases the dead residue run $[1, 2k]$ plus one cluster gap. Meanwhile $G_2(X_{3k-1}^{(2k+1)}) = 6$ (the case $d = 3k - 1$ above). These are exactly the one-foot exceptions $\{2, (p{-}1)/2\}$ of the consumed form. The clean threshold is sharp: at $d = 5$ every $p \ge 31$ has $G_2 = 2d = 10$ (the two-column case with two consecutive singleton bands: $m_0 - h = 2\lfloor (6k+1)/5 \rfloor - 2k \ge 2$ for $k \ge 4$), and $10 \le 2k + 1$ first holds at $k = 5$, i.e. $p = 31$; at $p = 19$ the dirty differences are $d \in \{4, 5\}$, each with $G_2 = 8$ ($2d = 8$ resp. $\mathrm{span}^* = 2r = 8$), at $p = 13$ they are $d \in \{3, 4, 5\}$ with $G_2 \in \{6, 8, 10\}$ — matching the census of §6.4 exactly. For large p the maximum of G_2 over $d \in [3, 3k-1]$ grows like $\frac{2}{3}(2k+1)$ (attained near $d = 2k$, where $r = 1$ and the orbit is the cluster $\{0\} \cup \{d - W, \ldots, d - 1\}$), so the bound is generous asymptotically; its binding cases are the two-column ones at small d.

Machine verification. The full architecture of this subsection is machine-checked by `lemmaE_master.py` (output `out_lemmaE.json`): the column decomposition, the band identity, the block identity for the gap sequence, and the pair-sum identity of Lemma 5.13 — as an exact equality, not merely a bound — hold at every (p, d, s) with $p \leq 500$ (45 primes, 10,410 triples); the bounds of Lemma 5.15 are checked on every occurring set; Lemma 5.16 is verified directly to $p \leq 2000$ (136,806 evaluations) and, through the verified identity, to $p \leq 4000$ (367,986 further checks, cross-checked against direct evaluation along the way); the endpoint and dirty-prime values of Remark 5.17 are exact; and Lemma 5.11's equivalence reproduces all 76 consumed-pair viable sets of the foot census (§6.4) verbatim at every prime $p \leq 419$. Two structural facts were confirmed exactly and are recorded for future use: the class orbit C is a rotation segment (the three-distance structure), and its low part $C \cap [0, r)$ is itself a rotation segment on $\mathbb{Z}_r$ with step $\sigma \bmod r$ — the Euclidean recursion $d \mapsto r$, which explains the uniformity of the analysis in d but which the proof above does not need.

With Lemma 5.16, Input E is discharged: the kills M4 and M5 of Lemma A receive their one-foot input unconditionally at every prime $p \geq 31$, and the p^2 and pq closures of §5.4 and §7 now rest on a single unbounded input, Input S. Their dependency statements are updated accordingly.

5.6 The non-porting, and the successor conjecture

At $n = 4$ the program's earlier work isolated a clean statement: no invertible *triple* of bad sets covers $\mathbb{Z}_N$ for any prime $N \geq 17$ (verified for all unit triples at every modulus $N \leq 60$, with hand proofs at 17 and 19). This *no-covering conjecture* does *not* port to $n = 5$: 17, 19, and 37 are capable with four bad sets (Example 5.6). What generalizes is not the conjecture but the framework—the covering language, the class reduction, and the classification program—which at $n = 5$ produced a different zone and a complete list of its covering configurations. The failure mode is instructive: the obstruction at $n = 4$ was never "three sets cannot cover" as a counting matter (three unit bad sets at a large prime have union at most $3(2h + 1) - 2 \approx \frac{3}{5}N$), it was the multiplicative tiling constraint; adding a fourth set releases exactly the primes whose class segments S_h are large enough to tile with four translates.

The natural successor statement at $n = 5$ is:

Conjecture 5.18 (successor no-covering)**.** *For every prime $N \geq 41$, four unit bad sets never cover $\mathbb{Z}_N$ (equivalently: the $n = 5$ rigid zone is exactly $\{7, 13, 17, 19, 37\}$).*

Verified within the scan range ($N \leq 150$, prime; composites ≤ 111; capability cross-checked by an independent set-based implementation at every danger modulus). Note that counting never obstructs here: for $N \geq 41$ one has $4(2h + 1) - 3 \geq N$, so the conjecture, if true, is a genuinely multiplicative statement about four translates of S_h in $\mathbb{Z}_{(N-1)/2}$—the same genre of statement as the $n = 4$ conjecture, one rung up. We do not attack it in this paper; the verified range is stated, the conjecture is open, and the finiteness of the rigid zone in general is open.

6 The group-ring translation

The covering lemmas of the p^2 programme (Lemmas A and B of the companion notes) are statements about three arithmetic progressions covering $\mathbb{Z}_p$. This section re-encodes them *exactly* as a statement about one sparse identity in the group ring $\mathbb{Z}[\mathbb{Z}_p] \cong \mathbb{Z}[z]/(z^p - 1)$: the obstruction becomes the coefficient-sign pattern of a quotient of a 24-term element by an 8-term element. The translation is proved (Theorem 6.1), not merely verified, and it is unconditional — the identity holds for *every* configuration, and covering is exactly a positivity condition on its solution. The machine verification then establishes, over an independent code path, a census law: at every prime tested, the critical coverings fall into exactly four difference families with placement

counts *independent of p* (§6.3). This is the same pinning phenomenon that the shear-rigidity argument measured fiberwise, now visible as a statement about the identity itself, and it converts the programme's computational inputs from vague debt into precisely stated, named lemmas with identified discharge paths (§6.4) — one of which, the foot-count lemma, is now proved (Lemma 5.16, §5.5).

The motive is the classical line connecting covering systems to roots of unity: the Mirsky–Newman–Davenport–Rado theorem — in an exact covering system of $\mathbb{Z}$ the largest modulus repeats — is proved by evaluating a generating function at roots of unity and reading off pole structure [18]. Lemmas A and B are the cyclic, three-term analogue, and the group ring is where that analogy becomes literal.

6.1 The identity and the translation theorem

Fix a prime p and a *configuration*: three arithmetic progressions $A_i = \{a_i + jd_i \bmod p : 0 \le j < s_i\}$ in $\mathbb{Z}_p$ with $d_i \in \mathbb{Z}_p^*$, $s_i \ge 1$. Write $\mathbf{1}_i$ for the indicator of A_i and define the *signed excess*

$$\varepsilon(x) \;:=\; \sum_{i=1}^{3} \mathbf{1}_i(x) - 1 \;\in\; \{-1, 0, 1, 2\}, \qquad E(z) \;:=\; \sum_{x \in \mathbb{Z}_p} \varepsilon(x)\, z^x.$$

Work in the group ring $R_p := \mathbb{Z}[z]/(z^p - 1)$, and put

$$Q(z) \;:=\; \sum_{i=1}^{3} z^{a_i} \left(1 - z^{d_i s_i}\right) \prod_{j \ne i} (1 - z^{d_j}),$$

$$P(z) \;:=\; \prod_{i=1}^{3} (1 - z^{d_i}).$$

Both are sparse: Q has at most $8 \times 3 = 24$ terms with coefficients ± 1 (each summand is the inclusion–exclusion of the box $a_i + \{0, d_i s_i\} + \{0, d_j\} + \{0, d_k\}$), and P has at most 8 terms. Let $\overline{\Phi}_p := \sum_{x \in \mathbb{Z}_p} z^x$ denote the all-ones element of R_p (the reduction of the cyclotomic polynomial Φ_p).

Theorem 6.1 (group-ring translation)**.** (i) *For* every *configuration,*

$$Q \;=\; E \cdot P \qquad \text{in } R_p.$$

(ii) *The configuration covers $\mathbb{Z}_p$ if and only if $\varepsilon \ge 0$ pointwise. Consequently, writing* $\mathrm{ov} := \sum_i s_i - p$*, if* $\mathrm{ov} = 0$ *then covering is equivalent to $E = 0$; if* $\mathrm{ov} = 1$ *to $E = z^e$ for some e; and if* $\mathrm{ov} = 2$ *to $E \in \{z^e + z^f,\ 2z^e\}$.* (iii) *The kernel of multiplication by P on R_p is exactly $\mathbb{Z} \cdot \overline{\Phi}_p$, and E is the* unique *$X \in R_p$ with $Q = XP$ and $X(1) = \mathrm{ov}$.*

Proof. (i) Let $\omega = e^{2\pi i/p}$ and evaluate at $z = \omega^\ell$, $\ell \in \mathbb{Z}_p$. For $\ell \ne 0$, since $d_i \in \mathbb{Z}_p^*$,

$$\widehat{\mathbf{1}}_{A_i}(\ell) = \sum_{j < s_i} \omega^{\ell(a_i + jd_i)} = \omega^{\ell a_i} \frac{1 - \omega^{\ell d_i s_i}}{1 - \omega^{\ell d_i}},$$

a geometric sum with nonzero denominator. Summing the function identity $\sum_i \mathbf{1}_i = 1 + \varepsilon$ over $x \in \mathbb{Z}_p$ against $\omega^{\ell x}$ kills the constant 1 at $\ell \ne 0$, giving $\sum_i \widehat{\mathbf{1}}_i(\ell) = \widehat{\varepsilon}(\ell)$; multiplying by $P(\omega^\ell) = \prod_j (1 - \omega^{\ell d_j}) \ne 0$ yields $Q(\omega^\ell) = \widehat{\varepsilon}(\ell) P(\omega^\ell)$, i.e. the polynomial $Q - E \cdot P$ vanishes at every nontrivial p-th root of unity. At $\ell = 0$ both sides vanish: $Q(1) = 0$ because each summand contains the factor $1 - z^{d_i s_i}$ at $z = 1$, and $E(1)P(1) = \mathrm{ov} \cdot 0 = 0$. So $Q - E \cdot P$ vanishes at all p roots of $z^p - 1$; a polynomial with integer coefficients divisible by $z^p - 1$ over $\mathbb{Q}$ (the roots are distinct) is divisible over $\mathbb{Z}$ (monic divisor, Gauss's lemma), and exponent reduction modulo p

changes a polynomial by multiples of $z^p - 1$. Hence $Q = E \cdot P$ in R_p, with no hypothesis beyond $d_i \neq 0$.

(ii) Covering means $\min_x \sum_i \mathbf{1}_i(x) \geq 1$, i.e. $\varepsilon \geq 0$ pointwise; and $\sum_x \varepsilon(x) = \sum_i s_i - p = \mathrm{ov}$ always. Pointwise $0 \leq \varepsilon \leq 2$ with total mass 0, 1, 2 leaves exactly the listed profiles.

(iii) Let $X \in \ker(\text{mult. by } P)$. Then $X(\omega^\ell)P(\omega^\ell) = 0$ for all ℓ, and $P(\omega^\ell) \neq 0$ for $\ell \neq 0$, so the coefficient vector $(X_x)_{x \in \mathbb{Z}_p}$ satisfies $\sum_x X_x \omega^{\ell x} = 0$ for all $\ell \in \mathbb{Z}_p^*$. By character orthogonality on $\mathbb{Z}_p$, (X_x) lies in the span of the all-ones vector; integrality gives $X = t\,\overline{\Phi}_p$ with $t \in \mathbb{Z}$. Conversely $(1 - z^d)\overline{\Phi}_p \equiv 0$ in R_p for every d (multiplication by z^d permutes the all-ones vector), so $\mathbb{Z}\overline{\Phi}_p \subseteq \ker$. Finally, if $Q = XP = EP$ then $X - E = t\overline{\Phi}_p$ and $X(1) - E(1) = t\,p$; the mass condition $X(1) = \mathrm{ov} = E(1)$ forces $t = 0$. □

Part (i) is stronger than the conditional identity one obtains by assuming a covering: the 24-term element Q is *always* divisible by the 8-term element P in R_p, and the quotient is always the signed excess. Individual summands of Q need not be divisible by P — the divisibility is a cancellation among the three boxes, and that cancellation is the arithmetic content of the covering problem. Part (iii) says the covering question is a genuine linear-algebra sign question about a uniquely determined object:

the configuration covers $\mathbb{Z}_p$ iff the unique right-mass solution X of $Q = XP$ has nonnegative coefficients.

Corollary 6.2. *Lemma B of the companion notes is equivalent to: for every normalized critical configuration at $p = 6k + 5$ with pairwise $\pm$-distinct differences, the right-mass solution E of $Q = XP$ has a negative coefficient; Lemma A at $p = 6k + 1$ likewise. The lemmas are sign conditions on the quotient $Q/P \in R_p$.*

Proposition 6.3 (normal form)**.** *The statement $Q = EP$, the covering property, the sizes, the overlap budget, and pairwise $\pm$-distinctness are preserved by* (a) *translation $a_i \mapsto a_i + t$, under which $Q \mapsto z^t Q$, $E \mapsto z^t E$, $P \mapsto P$; and* (b) *scaling $(a_i, d_i) \mapsto (\lambda a_i, \lambda d_i)$, $\lambda \in \mathbb{Z}_p^*$, under which all three transform by the ring automorphism $z \mapsto z^\lambda$ of R_p. Hence every configuration is equivalent to one with $a_1 = 0$, $d_1 = 1$, and the census below may be run in that normal form; reflection $z \mapsto z^{-1}$ permutes the solution set.*

Proof. Translation: ε transforms to $x \mapsto \varepsilon(x - t)$, so $E \mapsto z^t E$, and each summand of Q gains the factor z^t. Scaling: $z \mapsto z^\lambda$ is a ring automorphism because λ is coprime to p, and it carries the triple $(z^{a_i}, z^{d_i s_i}, z^{d_j})$ to the scaled configuration's data since s_i is unchanged and $\lambda(d_i s_i) = (\lambda d_i) s_i$. Covering and $\pm$-distinctness are properties of the bijections $x \mapsto x + t$ resp. $x \mapsto \lambda x$ on $\mathbb{Z}_p$. Scaling by d_1^{-1} and translating by $-a_1$ produce the normal form. □

6.2 Machine verification of the translation

Theorem 6.1 was verified by an implementation independent of the prior harness: Q, P, E are built directly from their definitions as integer arrays over $\mathbb{Z}_p$ (the 24 terms by explicit inclusion–exclusion over the three boxes), the product $E \cdot P$ by circular convolution, and the covering property by point counts. The census search uses a bit-mask residual-fit enumeration (for each (d_2, a_2) surviving the inclusion–exclusion prune, the completing placements a_3 are obtained from the minimal cyclic arc containing the scaled residual), with a full brute-force cross-check at $p = 7$ (all placements, no prune, no fit).

Two soundness items from this run are recorded in §12. First, the first implementation of the fit test had an off-by-one error in the minimal-arc length; it was caught immediately by the assertion that every enumerated covering has $E \geq 0$, and after the fix all counts agreed with the brute force. Second, the run forced a reconciliation of the prior failure counts: the "50 at $p = 7$ and 2 at $p = 13$" of the earlier harnesses are *witness-tuple* counts — one per (ordered sizes, d_2, d_3, a_2) with d_2, d_3 over positive $\pm$-reps and at least one completing a_3, the shared convention

Table 7: Verification of the translation theorem. "Identity" is the check $Q = E \cdot P$ in R_p; "sharp form" includes configurations that do not cover, where E has negative coefficients.

check	scope	outcome	count
identity, Lemma A exceptions	$p = 7, 13$ (all 52)	all hold; E-profiles exactly $\{z^e\}$, $\{z^e{+}z^f\}$, $\{2z^e\}$ as classified	52
identity, census coverings	every covering of §6.3 (census $11 \le p \le 61$ plus boundary $p = 5, 7, 13$)	all hold; $E \ge 0$	12,256
identity, sharp form	3,000 random configurations per prime, $p \in \{11, 13, 17, 19, 23, 29, 31, 37\}$, covering or not	all hold (signed E)	24,000
sign $\Leftrightarrow$ covering	same sample	agreement	24,000
kernel $= \mathbb{Z}\overline{\Phi}_p$	same sample, $X = E \pm \overline{\Phi}_p$	$Q = XP$ holds	24,000
$\pm$-distinct census	$p \in \{11, \ldots, 59\} \cup \{19, \ldots, 61\}$, exhaustive	zero coverings	—
brute-force ground truth	$p = 7$, all placements	272 normalized configurations = 68 set-triples, matching the fast enumerator	—

of the earlier search and its brute-force cross-check — while the true solution counts are 272 normalized configurations (68 set-triples) at $p = 7$ and 8 (2 set-triples) at $p = 13$, confirmed by the no-prune brute force. The content of the earlier finding (Lemma A is false at 7 and 13; the cells are closed there by direct verification) is unchanged, and the identity check ran on the *full* solution set, not on witnesses.

6.3 The covering census: four families, pinned placements

The census drops the $\pm$-distinctness hypothesis and enumerates *all* critical coverings in the normal form $a_1 = 0$, $d_1 = 1$: all ordered size triples from the class-appropriate critical window with $\sum s_i - p$ in the class-appropriate set (Lemma B: sizes $\{2k{+}1, 2k{+}2\}$, ov $\in \{0, 1\}$; Lemma A: sizes $\{2k, 2k{+}1, 2k{+}2\}$, ov $\in \{0, 1, 2\}$), over *all* $(d_2, d_3) \in (\mathbb{Z}_p^*)^2$. Write $\|d\| := \min(d, -d \bmod p)$ for the $\pm$-representative and $q := (p-1)/2$.

Proposition 6.4 (census record, machine)**.** *At every prime $p \equiv 5 \pmod 6$ with $11 \le p \le 59$, and at $p = 5$, the census returns exactly 144 normalized coverings, distributed by the $\pm$-rep pair $(\|d_2\|, \|d_3\|)$ as*

$$(1,1) : 48, \qquad (1,q) : 32, \qquad (2,2) : 32, \qquad (q,1) : 32,$$

independently of p. At every prime $p \equiv 1 \pmod 6$ with $19 \le p \le 61$ it returns exactly 1368, distributed as

$$(1,1) : 504, \qquad (1,q) : 288, \qquad (2,2) : 288, \qquad (q,1) : 288,$$

again independently of p. No other family occurs at any of these primes, and no $\pm$-distinct covering occurs (0 at every prime in range, re-deriving Lemmas A and B at those primes by an independent code path). At the boundary, $p = 5$ already shows the four families (with $q = 2$); $p = 7$ shows the $k = 1$ collapse (nine families, including 272 $\pm$-distinct configurations — the 68 set-triples of the Lemma A failure); $p = 13$ shows the four canonical families with the same counts as at general p, plus small exceptional families (among them the 8 $\pm$-distinct instances of the two Lemma A failures), totaling 1520.

Every one of the 12,256 census coverings satisfies the identity of Theorem 6.1 with $E \ge 0$ (Table 7). The four families are structural, not merely counted: they are, in order, the three-interval family, the parity family, and the two two-block families (interval plus two-block, with the

Table 8: Per-multiset family counts, constant in p (Lemma A side, $19 \le p \le 61$; the Lemma B side gives $24/16/16/16 = 72$ for each of its two multisets). "ov" is $\sum s_i - p$.

multiset	ov	$(1,1)$	$(1,q)$	$(2,2)$	$(q,1)$	total
$\{2k, 2k, 2k{+}1\}$	0	24	16	16	16	72
$\{2k, 2k, 2k{+}2\}$	1	72	32	32	32	168
$\{2k, 2k{+}1, 2k{+}1\}$	1	72	48	48	48	216
$\{2k, 2k{+}1, 2k{+}2\}$	2	288	160	160	160	768
$\{2k{+}1, 2k{+}1, 2k{+}1\}$	2	48	32	32	32	144

two-block on either index). Their placement counts per ordered size triple are also p-independent — the $(1,1)$ family has 8 per triple at ov $= 0$ and 24 at ov $= 1$ — and this is the identity-level form of the pinning that the shear-rigidity argument measured fiberwise. The p-independence has a structural explanation, and it is worth isolating, because it is the mechanism a proof of Conjecture 6.7 will exploit.

Proposition 6.5 (why the counts are p-independent)**.** *Fix a family and a multiset of Table 8, and $k \ge 1$. Every census configuration of the parity family $(2,2)$ with at most one foot per AP has the* integer-parity parametrization*: each of A_2, A_3 meets $I_1 = [s_1, p-1]$ in a run of consecutive elements of one integer parity class of I_1, does not wrap, and carries its feet as the one-point extensions of that run at either end; consequently the configurations are enumerated by a choice set — parity assignment (≤ 2), orientation of each AP (2 each), spill slot of each AP (≤ 3 each) — of cardinality bounded independently of k, and the per-multiset counts are the constants of Table 8. Written out:* 16 *for both B-side multisets,* 16 *($M1$),* 48 *($M2$),* 32 *($M3$),* 32 *($M5$); the $M4$ count* 160 *adds the wrapped spill pair $\{0,2\}$ available at* ov $= 2$*. The $(1,1)$ family is Proposition 6.6; the two-block families obey the same junction arithmetic (verified; the $s_1 = 2k{+}1$ slice is written out in §7.3).*

Proof. Let $A = \{a + 2j\}$ ($d = \pm 2$) have $|A \setminus I_1| \le 1$ and meet I_1. The step-2 cycle visits $\ldots, p{-}3, p{-}1, 1, 3, \ldots$ (or shifted by one): a wrap passes from $p-1$ (or $p-2$) to 0 (or 1). If the segment wrapped strictly inside, the post-wrap head would contribute at least two points of $A_1 = [0, s_1 - 1]$ once it re-enters I_1 — impossible at one foot unless the wrap is at an endpoint of the segment; hence the I_1-part is the pre-wrap tail: consecutive integers of one parity, and the single foot (if any) is the adjacent run extension below s_1 or above $p{-}1$ (after reduction). Covering forces $A_2 \cup A_3 \supseteq I_1$, and since each parity class of I_1 can only be covered by the AP of that parity, one AP carries the even run, the other the odd run, each extended by its spill slots within the foot budget ov. The count is then a product of {assignment} $\times$ {orientations} $\times$ {spill slots}, all bounded by constants; evaluating it per multiset (with $|E(I_1)|$, $|O(I_1)|$ determined by s_1 alone) gives the listed constants, each independent of k. E.g. the B-side multiset $\{2k{+}2, 2k{+}2, 2k{+}1\}$: at $s_1 = 2k{+}1$ the parities have $2k{+}2$ each, both APs are exact runs (2 orientations each, 2 assignments: 8); at $s_1 = 2k{+}2$ the even run has $2k{+}2$, the odd $2k{+}1$, the sizes match forced assignments (2 ordered triples $\times 2^2$ parametrizations: 8); total 16. The structural claim was verified on every census $(2,2)$ solution at six primes (32 resp. 288 per prime, 100% conformity, spills $\le$ ov in A_1). □

This is the answer to "why p-independent": the critical configurations live in a finite, k-independent choice space of interval-arithmetic patterns; the families are its connected components, and the placement counts are its component sizes. The same mechanism governs the row census of the pq side (§7.3), where the pinned placement sets have 24 resp. 312 elements at every row modulus — which is why the covering counts 28/68 of the shear-rigidity input are p-independent (§6.4).

Proposition 6.6 (the interval family)**.** *In the normal form, with $d_2, d_3 \in \{\pm 1\}$ and critical sizes with* ov $\in \{0, 1\}$ *(for $k \geq 1$, so that all sizes are ≥ 2; the degenerate prime $p = 5$ is verified directly), the coverings are exactly the consecutive arrangements: at* ov $= 0$*, the two cyclic orders of the consecutive tiling of $I_1 = [s_1, p-1]$ by A_2, A_3; at* ov $= 1$*, the six arrangements obtained from them by placing the single double point at one of three positions — the interior junction, the left spill into A_1 at $s_1 - 1$, or the wrap at 0 — each in two orders. Per ordered size triple this gives 2 resp. 6 set-arrangements, each realized by 2^2 parametrizations (independent orientation choices ± 1 for A_2 and A_3), i.e. 8 resp. 24 configurations, matching the census exactly.*

Proof. With $d_i = \pm 1$ each A_i is a cyclic interval. Let $q = p - s_1 = |I_1|$. Since $A_1 = [0, s_1 - 1]$, covering is equivalent to $A_2 \cup A_3 \supseteq I_1$. At ov $= 0$ we have $s_2 + s_3 = q$, so $A_2 \cup A_3 = I_1$ exactly; both intervals lie in I_1, and two intervals whose union is the arc I_1 are its consecutive blocks, in one of the two orders. At ov $= 1$ we have $s_2 + s_3 = q + 1$, so $|A_2 \cap A_3| = 1 - (A_2 \cup A_3 \smallsetminus I_1)$ counts the single double point (a triple point would contribute 2 to the mass, exceeding ov $= 1$). If the double point is interior ($\in I_1$) then $A_2 \cup A_3 = I_1$ and the two intervals are consecutive blocks sharing their junction point. If it lies in A_1, then $A_2 \cap A_3 = \varnothing$, and the one point of $(A_2 \cup A_3) \cap A_1$ belongs to exactly one of the intervals: an interval of size ≥ 2 containing a point $< s_1$ and points of I_1 must contain the intervening block of A_1, so it either starts at $s_1 - 1$ (left spill) or wraps through 0, in which case it contains $[0, v]$ for some v and $v = 0$ is forced. Each case occurs in the two orders of (A_2, A_3), giving $3 \times 2 = 6$ set-arrangements. The parametrization count is immediate. □

The other three families have the same character. The $(2, 2)$ family at ov $= 0$ is the parity split: the surviving structural pair (E, O) of the Lemma B proof — the two $d = \pm 2$ progressions partition I_1 into its even and odd classes, which is why $\pm$-distinctness is exactly what kills it. The $(1, q)$ and $(q, 1)$ families are the two-block families: one of A_2, A_3 is the two-block set of difference q (the proved two-block structure of the trichotomy: blocks of $\lceil s/2 \rceil$ and $\lfloor s/2 \rfloor$ consecutive integers whose right endpoints differ by exactly q), and the partner is an interval. The full placement windows of these three families are verified but not written out as propositions; they are the remaining content of the support lemma below.

6.4 The support lemma and the computational inputs

The census law suggests the all-p statement that the group-ring programme targets.

Conjecture 6.7 (support lemma: the covering classification)**.** *Let $p = 6k + 5$, $k \geq 0$ (resp. $p = 6k + 1$, $k \geq 3$) be prime. Every critical covering configuration at p in the normal form satisfies*

$$(\|d_2\|, \|d_3\|) \in \{(1, 1), (1, q), (2, 2), (q, 1)\}, \qquad q = (p-1)/2,$$

and conversely each family is realized at every such prime with the placement counts of Table 8, independent of p. Equivalently: (i) *no pairwise $\pm$-distinct critical configuration covers $\mathbb{Z}_p$ (Lemmas B and A), and* (ii) *every critical covering has a $\pm$-collision, the collision patterns being exactly the four listed.*

Its status, part by part. Clause (i) is *proved* for $p \equiv 5 \pmod 6$, $p \geq 17$ (Lemma B; see below), and for $p \equiv 1 \pmod 6$, $p \geq 31$ (Lemma A, unconditionally: its foot-count input is discharged by Lemma 5.16 of §5.5); verified to $p \leq 251$ (prior harness) and $p \leq 419$ (the foot census), and re-derived independently at every prime in $5 \leq p \leq 61$. The $(1, 1)$ family is proved (Proposition 6.6); the $(2, 2)$ family is proved (Proposition 6.5); the two-block placement structure is proved. What is *not* proved is the exhaustiveness of the four families at general p (clause (ii) and the two-block placement counts), verified by the fresh census to $p \leq 61$ on both classes (and at 5, 7, 13) and by the prior relaxed census to $p \leq 250$. The $\varepsilon = 2$ counts of the interval family are verified but, unlike Proposition 6.6, not written out.

The computational inputs, precisely. The p^2 closure — the $(1K, 3U)$ cell is empty at every prime $p \geq 5$ — rested, at the previous revision, on two named inputs beyond the proved lemmas. Both were *unbounded* hypotheses: each asserts something for all primes in a mechanism range and is verified only to a finite bound, so neither can be discharged by a bounded search, however large; each requires a lemma, and each had one identified. The first of the two (Input E below) is now *proved* — Lemma 5.16 of §5.5 — leaving Input S as the closure's single unbounded input.

Input E (the foot-count trichotomy, stated in its consumed form) — discharged. The kills M4 and M5 of Lemma A consume exactly the *one-foot* case, at exactly two size pairs: M4 at $(s_1, s) = (2k{+}2,\ 2k{+}1)$, M5 at $(2k{+}1,\ 2k{+}1)$. The consumed statement is therefore: *at $p = 6k + 1$, any critical-size AP with at most one point outside $I_1 = [s_1, p - 1]$ at these two pairs has $\|d\| \in \{2, (p{-}1)/2\}$* — at the M4 pair the viable set is in fact exactly $\{2\}$, at the M5 pair exactly $\{2, (p{-}1)/2\}$, matching the kills' branching. A complete census at every prime $p \leq 419$ gives the exact boundaries: the consumed form is clean at every prime $p \equiv 1 \pmod 6$ with $p \geq 31$ (dirty at 13 and 19). *The input is now discharged*: Lemma 5.16 of §5.5 (the three-distance foot count) proves the consumed statement outright at every prime $p \geq 31$, by an elementary column decomposition of the dilation $j \mapsto jd \bmod p$ with no group-ring content, machine-verified against the census (76/76 viable-set agreement at $p \leq 419$). Consequently *Lemma A is proved for $p \geq 31$*, with $p = 19$ closed by the finite case analysis and $7, 13$ by direct verification; the all-pairs one-foot version is clean for $p \geq 31$ on the $\epsilon = 1$ class and $p \geq 41$ on the $\epsilon = 5$ class; the two-foot version (clean only from $p \geq 61$ resp. 47) is consumed by *no* current proof — the earlier "$p \geq 43$" phrasing bundled the two-foot statement with the pair-restricted one, and a previous revision separated them. Lemma B, finally, consumes *no* foot-count input: its kills run on the contained census (Lemma T, clean for $p \equiv 5 \pmod 6$, $p \geq 17$) and Corollaries T1–T3, so *Lemma B is proved for $p \geq 17$* (the earlier $p \geq 23$ reflected a pre-T3 draft of the β-kill; at $p = 17$ the contained forms at $(s_1, s) = (6, 6)$ are exactly E and T with $|E \cap T| = 4 > 1$, as the proof requires). The earlier discharge sketch — a wrap of a middle difference $d \in [3, s_1]$ landing in A_1, the re-entry climb contributing at least $\approx s \cdot s_1/p - 2$ points — is superseded by the exact column analysis of §5.5, which proves the bound outright rather than heuristically.

Input S (the pinning lemma of the shear-rigidity argument). Statement: in the $\pm$-colliding closure, the per-fiber covering configurations of a colliding family are pinned — the relative-offset difference sets have at most 6 elements, with covering counts 28 and 68 per collision topology, independent of p. Consumer: the shear-rigidity theorem for $\pm$-colliding units — hence the closure for colliding unit triples at $p \geq 11$. Status: verified at every prime $11 \leq p \leq 113$; exhaustive ground truth 3.3 million families at $p \leq 43$ with zero coverings; mechanism (exact linear per-fiber drift against pinned configurations) proved in general. Discharge: the collision side of Conjecture 6.7. The per-fiber covering configurations are exactly critical coverings of the fiber's t-space ($\cong \mathbb{Z}_p$) whose three differences carry the colliding units' $\pm$-collision; the support lemma classifies these into the four families with p-independent placement sets, and the pinning tables are then a read-off of that classification — which is why the counts are p-independent: the families are. The pq side measures the pinning directly and confirms it: every row configuration satisfying the fiber condition is a census solution after normalization (Proposition 7.3), with at most 6 surviving rows per family against fiber arcs of length $\frac{2}{3}q$.

In this form the technical debt is finite and explicit: the prime-side foot-count lemma (Input E, one-foot at two size pairs) is now *proved* (Lemma 5.16, §5.5); what remains is the support lemma's exhaustiveness clause and the shear input — each with a statement, a consumer, a verification range, and a proof strategy; the group-ring translation is the machine that proves them, and the pq side of §7 adds no further input. The entire programme rests on a single unbounded input, Input S.

Finally, the defect statistics. Define the distance-to-solvability of a non-covering $\pm$-distinct critical configuration as $\|Q\|_0$ (the number of nonzero coefficients of Q in R_p) at ov $= 0$, and $\min_e \|Q - z^e P\|_0$ at ov $= 1$ — the least ℓ_0-distance from the identity's data to a covering

profile. Over 3,000 random configurations per prime the sampled minima grow: $5, 5, 6, 7, 9$ at $p = 11, 17, 23, 29, 41$ (medians $8, 11, 13, 14, 17$), with the minimizers concentrated in the overlap-1 multiset $\{2k{+}2\}^3$ — the case of the Lemma B proof that needed Corollaries T2/T3 rather than pure parity. The obstruction thus *strengthens* with p: this is the cyclotomic analogue of the shear-rigidity near-miss margin (families failing all but two fibers), and it is the quantity a stability argument would bound below.

6.5 Outlook: cyclotomic tools and the pq side

The translation puts the problem in the room of the Lam–Leung–de Bruijn–Schoenberg theory of polynomials with few terms divisible by cyclotomic polynomials [19, 20]. The prime-modulus case of that theory is elementary and worth recording, since it is the tool any proof of Conjecture 6.7 will consume on the nonnegative side:

Lemma 6.8 (prime-modulus Lam–Leung structure)**.** *Let $F \in \mathbb{Z}[z]$ have nonnegative coefficients and satisfy $\Phi_p \mid F$. Then, reducing exponents modulo p, all p residue classes carry equal mass: $F \equiv (F(1)/p)\,\overline{\Phi}_p$ in R_p.*

Proof. $F = \Phi_p G$ gives $p \mid F(1)$, and $F(\omega^\ell) = 0$ for $\ell \neq 0$ says the reduced coefficient vector is annihilated by all nontrivial characters of $\mathbb{Z}_p$; by orthogonality it is constant. □

The gap between this and our problem is exactly the sign structure: Q is a ± 1 combination, not a nonnegative one, so Lemma 6.8 does not apply directly, and the covering question is a *sign-mixed* sparse divisibility problem — the quotient $E = Q/P$ must be shown to have a negative coefficient in the $\pm$-distinct regime, and to be nonnegative with the exact four-family profile in the colliding regime. The specific structure available beyond general sparsity is that Q is built from factors $(1 - z^c)$: these are real-rooted at $z = 1$, and their coefficient sequences are log-concave; whether that structure propagates through the alternating combination Q in the way log-concavity propagates through products — the mechanism by which stability methods turn counting statements into theorems [21] — is the precise question the stability route must answer. Conjecture 6.7 is calibrated for it: the four families are its tightness examples, and the defect margins of §6.4 are its stability margins.

At composite modulus the translation persists with new structure: at $N = pq$ the group ring is $\mathbb{Z}[\mathbb{Z}_p \times \mathbb{Z}_q]$, the tensor product of the prime-modulus rings, the identity tensors over the CRT components, and the kernel of Lemma 6.8 enlarges — the cyclotomic field of Φ_{pq} replaces $\mathbb{Q}(\omega_p)$, and with it the Lam–Leung theory acquires its genuine (nontrivial) content [20]. This is not a black-box transfer of the prime side, and it is now carried out: §7 proves the row translation (Theorem 7.1), verifies the two-unit and three-unit cells at ten semiprime moduli, and reduces their closure to the same input structure, now a single unbounded input after the discharge of Input E — the tensor structure is the reason the pq side inherits, rather than re-derives, the prime-side obstruction theory.

7 The pq side: the tensor translation

The group-ring translation of §6 was stated and verified at prime modulus, where the three covering objects are APs of $\mathbb{Z}_p$. This section completes the structural picture at semiprime modulus: at $N = pq$ the CRT identification $\mathbb{Z}_{pq} \cong \mathbb{Z}_p \times \mathbb{Z}_q$ turns the two-unit and three-unit cells into *fiber-wise* three-AP problems of exactly the prime-modulus form, with one member fixed by the kernel side and the other members moving affinely along the fiber index. The tensor structure that the outlook of §6.5 anticipated is real, it is proved (Theorem 7.1), and it is machine-verified in every component. The outcome is the structural completion the programme wanted: *the pq cells introduce no new computational input* — they close modulo exactly the two lemmas already named for the p^2 closure (Corollary 7.4).

7.1 Setup: the CRT picture and the closed-form constants

Fix $N = pq$ with $5 \le p < q$ primes, $c = \lfloor (N-1)/6 \rfloor$, and work in the CRT coordinates $(x, y) = (k \bmod p,\ k \bmod q)$. Define

$$\rho_p := \lfloor c/q \rfloor = \lfloor p/6 \rfloor\,, \qquad \rho_q := \lfloor c/p \rfloor = \lfloor q/6 \rfloor\,,$$

the first equality by the nested-floor identity. A kernel-p member (speed pa, $a \in \mathbb{Z}_q^*$) has $\|pa\,k\|_N = p\,\|ak \bmod q\|_q \le c$ iff $\|ak \bmod q\|_q \le \rho_q$: it covers exactly the rows of

$$Y_a = \{y \in \mathbb{Z}_q : \|ay\|_q \le \rho_q\},$$

an AP of $2\rho_q + 1$ rows (the preimage of the symmetric interval under multiplication by a); symmetrically a kernel-q member (speed qb, $b \in \mathbb{Z}_p^*$) covers exactly the columns of $X_b = \{x \in \mathbb{Z}_p : \|bx\|_p \le \rho_p\}$, an AP of $2\rho_p + 1$ columns. The uncovered region of a $(1, 1, 2)$ family — one kernel of each type, two units — is the rectangle $X_b^c \times Y_a^c$ with

$$|X_b^c| = p - 2\rho_p - 1, \qquad |Y_a^c| = q - 2\rho_q - 1 \ \tfrac{2}{3}q - \mathrm{O}(1).$$

A unit $u \in \mathbb{Z}_{pq}^*$ has bad set $B_u = u^{-1}[-c, c]$, and its trace on a row is governed by one integer interval. Write $k = y + qj$ $(j \in \mathbb{Z}_p)$ for the row y and $\sigma = \sigma_u(y) := uy \bmod q \in [0, q)$; then $(x, y) \in B_u$ iff $x \equiv u^{-1}t \pmod p$ for some $t \in [-c, c]$ with $t \equiv \sigma \pmod q$, i.e. $t = \sigma + qj'$ with

$$j' \in J_u(y) := \big[\lceil (-c - \sigma)/q \rceil,\ \lfloor (c - \sigma)/q \rfloor\big], \tag{7.1}$$

an interval of $n_u(y) := |J_u(y)| = \lfloor (c - \sigma)/q \rfloor + \lfloor (c + \sigma)/q \rfloor + 1$ consecutive integers. This is the pq form of the fiber footprint $T_u(j)$ of the p^2 analysis.

7.2 The row translation

Theorem 7.1 (row translation at $N = pq$)**.** *With the notation above, writing $\alpha_u := u^{-1} \bmod p$:*

(i) *the trace of B_u on the row y is the AP*

$$f_u(y) = \{\alpha_u(\sigma + qj') \bmod p :\ j' \in J_u(y)\},$$

of step $\alpha_u q \bmod p$ and size $n_u(y)$;

(ii) *the size profile is class-appropriate: writing $p = 6k + \epsilon$ with $\epsilon \in \{1, 5\}$,*

$$n_u(y) \in \begin{cases} \{2k+1,\ 2k+2\} & \epsilon = 5, \\ \{2k,\ 2k+1\} & \epsilon = 1, \end{cases}$$

independently of q;

(iii) *the $(1, 1, 2)$ family covers $\mathbb{Z}_{pq}$ if and only if, for every row $y \in Y_a^c$,*

$$X_b^c \ \subseteq\ f_{u_1}(y) \cup f_{u_2}(y),$$

i.e. iff $(f_{u_1}(y),\ f_{u_2}(y),\ X_b)$ is a three-AP covering of $\mathbb{Z}_p$ whose third member is the fixed *AP X_b; at $\epsilon = 5$ this is the exact partition of the multiset $\{2k{+}2, 2k{+}2, 2k{+}1\}$, and at $\epsilon = 1$ a critical covering with* ov $\in \{0, 1, 2\}$ *and sizes in $\{2k, 2k{+}1\}$ — in both cases precisely the census scope of §6.3 with $s_1 = |X_b| = 2k + 1$;*

(iv) *the start $a_u(y) = \alpha_u(\sigma_u(y) + q\, j_{\min}(y))$ of the row AP moves* piecewise-affinely *in y: $a_u(y{+}1) - a_u(y) \in \{\alpha_u u,\ \alpha_u(u - q)\}$ in $\mathbb{Z}_p$; the $(1K, 3U)$ cells obey the same translation with three footprints and no fixed member (the whole row must be covered), i.e. the Lemma A/B problem itself.*

Proof. (i) is the computation above (7.1). (ii) Write $c = \rho_p q + c'$ with $c' = c \bmod q$. Since $p = 6m + \epsilon$ and $r := (\epsilon q) \bmod 6 \in \{1, 5\}$, $c = mq + (\epsilon q - r)/6$, so $c' = (\epsilon q - r)/6$, which is $> q/2$ exactly when $\epsilon = 5$ (for $q \geq 3$) and $< q/2$ exactly when $\epsilon = 1$. Now

$$n_u(y) = 2\rho_p + \lfloor (c' - \sigma)/q \rfloor + \lfloor (c' + \sigma)/q \rfloor + 1,$$

where the two floors take values in $\{-1, 0\}$ and $\{0, 1\}$: the maximum $2\rho_p + 2$ occurs iff $q - c' \leq \sigma \leq c'$ (nonempty iff $c' > q/2$, i.e. $\epsilon = 5$), the minimum $2\rho_p$ occurs iff $c' < \sigma < q - c'$ (nonempty iff $c' < q/2$, i.e. $\epsilon = 1$), and $2\rho_p + 1$ always occurs. With $\rho_p = \lfloor p/6 \rfloor = k$ this is the stated profile; the dichotomy depends on c', hence on $p \bmod 6$ and q, but the *window* $\{2k{+}1, 2k{+}2\}$ resp. $\{2k, 2k{+}1\}$ depends only on p. (iii) The uncovered region is exactly $X_b^c \times Y_a^c$; covering it is equivalent to the row inclusions. At $\epsilon = 5$, $|X_b^c| = p - 2k - 1 = 4k + 4 = 2(2k + 2)$ while $n_{u_i}(y) \leq 2k + 2$, so both footprints must have the maximal size and $f_{u_1}(y) \sqcup f_{u_2}(y) = X_b^c$ exactly: together with X_b (disjoint from X_b^c) this is a pairwise-disjoint partition of $\mathbb{Z}_p$ with sizes $(2k + 2, 2k + 2, 2k + 1)$, the B-side critical multiset with the third member fixed. At $\epsilon = 1$, $|X_b^c| = 4k$ and $n_{u_i}(y) \leq 2k + 1$, so $n_1 + n_2 + |X_b| - p \leq 2$ with all sizes in the A-side window: a critical covering with $s_1 = 2k + 1$ fixed. (iv) $\sigma_u(y{+}1) - \sigma_u(y) \in \{u \bmod q,\ u \bmod q - q\}$ and $j_{\min}(y{+}1) - j_{\min}(y) \in \{0, -1\}$, giving the two rates; the $(1K, 3U)$ statement is the same row decomposition with the kernel covering only rows. □

Part (iii) is the tensor translation: the pq cell condition *is* the prime-modulus census condition, fiber by fiber, with the kernel side supplying the fixed member and the unit side supplying the moving members. Part (ii) is what makes it sharp — the row-size profile is forced by $c \bmod q$ to sit in the class-appropriate critical window, so no out-of-scope size ever appears on a fiber. Part (iv) is the shear structure: the covering configurations of the rows are pinned (below), and their carriers drift at rates determined by the units alone.

7.3 The row census and the pinning

Theorem 7.1(iii) reduces the row condition to the census of §6.3 *restricted to* $s_1 = |X_b| = 2k + 1$. That restricted census is again four families with p-independent counts.

Proposition 7.2 (row census, machine)**.** *At every prime $p \equiv 5 \pmod 6$ with $11 \leq p \leq 59$, the census of critical coverings in the normal form with $s_1 = 2k + 1$ (sizes $(2k{+}2, 2k{+}2)$, $\mathrm{ov} = 0$) returns exactly 24 configurations, distributed by the $\pm$-rep pair $(\|d_2\|, \|d_3\|)$ as*

$$(1, 1) : 8, \qquad (2, 2) : 8, \qquad (1, q) : 4, \qquad (q, 1) : 4,$$

with $q = (p - 1)/2$. At every prime $p \equiv 1 \pmod 6$ with $19 \leq p \leq 61$, the census with $s_1 = 2k + 1$ and sizes in $\{2k, 2k + 1\}$ ($\mathrm{ov} \in \{0, 1, 2\}$) returns exactly 312, distributed as

$$(1, 1) : 104, \qquad (2, 2) : 72, \qquad (1, q) : 68, \qquad (q, 1) : 68.$$

All counts are independent of p, and no other family occurs.

The 24 of the $\epsilon = 5$ side is completely proved by the parametrization mechanism of Proposition 6.5: the $(1, 1)$ configurations are the two cyclic orders of the consecutive tiling of $I_1 = [2k + 1, p - 1]$ by two intervals of $2k + 2$, in 2^2 parametrizations (8); the $(2, 2)$ configurations are the even/odd split of I_1 in both assignments and orientations (8); and the $(1, q)$, $(q, 1)$ configurations are the *wrapped* two-block: the two-block of step $-q$ with first block $[2k{+}1, 3k{+}1]$ has its second block at $[a - q, a - q + k] \equiv [5k + 4, 6k + 4]$ through the wrap at 0, and its complement in I_1 is the single interval $[3k + 2, 5k + 3]$ of size $2k + 2$ — one set-arrangement in 2^2 parametrizations (4 each). The unwrapped two-block placements $a \in \{3k + 1, 3k + 2\}$ fail by exactly one exposed endpoint ($\{2k + 1\} \cup [3k{+}3, 5k{+}3]$ resp. $[3k{+}2, 5k{+}2] \cup \{6k{+}4\}$): the

Table 9: Verification of the pq translation and the cells. "Families" counts (a, b, u_1, u_2) resp. (a, u_1, u_2, u_3) tuples over $\pm$-canonical speeds; margins are the minimal numbers of uncovered points over all families.

check	scope	outcome	count
constants	8 moduli 35–143	$\rho. = \lfloor \cdot/6 \rfloor$; ϵ=5 partition equality $\lvert X^c \rvert = 2(2k+2)$; profile dichotomy (7.1)–(ii)	8
footprint formula	all units × all rows, 8 moduli	exact set equality $f_u(y) = \{\alpha_u(\sigma + qj')\}$	4,394
row census (Prop. 7.2)	$\epsilon = 5$: $p \le 59$; $\epsilon = 1$: $p \le 61$	24 resp. 312, counts constant, four families only	14
$(1, 1, 2)$ cell	$N \in \{35, 55, 65, 77, 85, 91, 95, 115, 119, 133, 143\}$, all $(a, b, u_1 < u_2)$	0 coverings; margins 4–18	144,138
$(1K, 3U)$ cells, both orientations	same N, all unit triples	0 coverings; margins 2–10	989,000
pinning (Prop. 7.3)	8 moduli, all families	37,672/37,672 rows are census solutions; max good rows ≤ 6	—
T-10 reproduction	$N = 77, 91$	$(1, 1, 2)$ empty, as committed	2
parity/two-block structure	every census $(2, 2)$ solution, 6 primes	one integer parity class of I_1, $\le$ ov spills	1,248

near-miss that the shear argument exploits. This is the CRT-compatible form of the four families: on each fiber, the classified objects are exactly the census objects, with the fixed member playing the normalized role.

The pinning statement is the machine result that connects this to Input S. Across the eight moduli $N \in \{35, 55, 77, 91, 95, 115, 119, 143\}$, all 132,485 (a, b, u_1, u_2) families were scanned row by row:

Proposition 7.3 (pinning at $N = pq$, machine)**.** *Every row configuration* $(f_{u_1}(y), f_{u_2}(y), X_b)$ *satisfying the row condition of Theorem* 7.1(iii) — 37,672 *of them across the eight moduli — is, after the normalization* $x \mapsto bx + \rho_p$, exactly *a census solution of Proposition 7.2 at the row modulus (*100%*; both the family label and the placement sets coincide). For a fixed family* (a, b, u_1, u_2) *the number of rows satisfying the condition is at most* 6*.*

This is the identity-level form of the shear mechanism at $N = pq$: a covering would need the row condition on *every* row of the arc Y_a^c (of size $q - 2\lfloor q/6 \rfloor - 1 \ \frac{2}{3}q \ \ge 8$ for $q \ge 11$), but the pinned placement sets have p-independent size (24 resp. 312 elements, by Propositions 7.2 and 6.5), and the carriers drift at the two rates of Theorem 7.1(iv). Either the relative drift is nonzero, and the normalized placement pair exits the pinned set within a bounded number of rows (at most 6 in every measured case, with the failure rows carrying 1–2 uncovered points each — the margins below), or the relative drift vanishes, which forces a multiplicative relation between u_1, u_2 modulo p — the collapse case, where the family degenerates to a common scaling and the kernel side must cover. The dichotomy is the same as at p^2, with the same two consumers.

7.4 Machine verification and the cell record

All checks pass (Table 9); the harness is an independent code path (formula-based footprints cross-checked against the direct definition, bit-masked cell enumeration, census by residual fit with the $p = 7$ brute-force validation of §6.2 carried over). The margin structure mirrors the prime side: the nearest $(1, 1, 2)$ families sit in the interval and parity families with a handful of good rows (the drift kills the rest, each failing row missing 1–2 points of X_b^c), and the nearest

$(1K, 3U)$ families fail entire rows by 2–10 points — the cyclotomic analogue of the shear near-miss at p^2 (families failing all but two fibers).

Corollary 7.4 (the pq closure's dependencies)**.** *The emptiness of the* $(1, 1, 2)$ *and* $(1K, 3U)$ *cells at* $N = pq$ *(*$5 \le p < q$*), verified here for* $N \le 143$*, is modulo exactly the computational inputs of the* p^2 *closure, and no new unbounded statement appears on the* pq *side:* (a) *the support lemma (Conjecture 6.7) at the row and column moduli — which classifies the per-fiber configurations into the four families with the pinned placement sets of Propositions 7.2–7.3 — and* (b) *the shear-rigidity argument applied along the row arc (Theorem* 7.1(iv)*), i.e. Input S in its fiberwise form. The* p^2 *closure's foot-count input (Input E) reaches the* pq *side only through the prime-side trichotomy inside* (a) *and is now discharged (Lemma 5.16); both closures rest on the single unbounded input S.*

Proof sketch. By Theorem 7.1(iii) a covering family yields a critical three-AP covering on every row of Y_a^c; by the support lemma each is one of the four families in the normalization of the fixed member X_b, with placement in the pinned set; the affine drift (iv) moves the normalized placements at rates $\lambda_i \in \{\alpha_{u_i} u_i, \alpha_{u_i}(u_i - q)\}$, and the pinned set is finite and p-independent while $|Y_a^c|$ grows with q; a nonvanishing relative drift exits the set (Proposition 7.3 measures at most 6 surviving rows), and a vanishing one collapses the units to a common scaling modulo the fiber modulus, whereupon the reduced problem is the prime-modulus cell, closed by Lemmas A/B. The column side exchanges the roles of p and q and of rows and columns. □

The reformulation is thus structurally complete on the semiprime side: every cell of the two-unit and three-unit counting at $N = pq$ is either closed by the proved fiber-cap theorems (Theorems 2–3 of the companion notes, with the $(1, 1, 2)$ cell's row-slack equality $p - 2\rho_p - 1 = 2(2\rho_p + 2)$ at $p \equiv 5 \pmod 6$ explaining precisely why that cell is the sole survivor of the counting), or reduced by Theorem 7.1 to the prime-modulus programme with the same named inputs — of which the foot-count lemma is now proved (Lemma 5.16), leaving a single unbounded input. What remains outside the classification is the rigid-zone lift question, unchanged.

8 The higher rungs: $T = 7$ and $T = 8$

The closures of §5–§7 live at the rung $T = 6$ ($n = 5$): four bad sets per pair, the frontier cell $(1K, 3U)$, and the fiber problem is the three-AP census classified by Conjecture 6.7. Open Problem 9 asked whether the method survives a third rung. This section answers affirmatively, with measurements: the machinery, the classification, and the shear mechanism all port, and at the *critical* cell of $T = 8$ the census is sharper than at $T = 6$ — exactly eleven families with placement counts *constant in* p. The higher rungs are also where the pinned sets grow, and the shear step must be re-established as a growth comparison; we measure that growth and the drift-exit mechanism that keeps it harmless.

8.1 The rung map and the general-T fiber machinery

At rung $T = n + 1$ the composite modulus is $N = p^2$ with $c = \lfloor (N - 1)/T \rfloor$ and bad sets $B_w = \{x : \|wx\|_N \le c\}$. Writing $p = Tk + \epsilon$ ($\epsilon = p \bmod T$), a kernel member pa covers exactly the fibers of $a^{-1}B_k$, where $B_k = \{r : \min(r, p - r) \le k\}$ has $2k + 1$ elements, so the uncovered set has $|U| = (T - 2)k + \epsilon - 1$. Cells are families of $n - 1 = T - 2$ bad sets; the frontier cells are

$$T = 7 : \ (1K, 4U) \text{ (overlap } k + O(1)), \qquad T = 8 : \ (2K, 4U) \text{ (overlap } O(1)) \text{ and } (1K, 5U),$$

$(2K, 3U)$ at $T = 7$ and $(3K, 3U)$ at $T = 8$ being dead by size ($3(2k + 2) < p$ for k past a small boundary). The $T = 8$ cell $(2K, 4U)$ is *critical* in the $T = 6$ sense — four footprints of size $2k + 1 \approx p/4$ sum to $p + O(1)$ — which is why its census mirrors (and sharpens) the base rung's.

The fiber machinery is T-free in form and is stated once.

Lemma 8.1 (footprints, the three-tier arc, and the drift at rung T)**.** *Fix rung T, $p = Tk+\epsilon$ prime, and a unit u with residue $r = u \bmod p$ and lift m ($u = r+pm$). On the fiber $F_j = \{j+pt\}$, writing $uj = a_1 p + a_0$, the trace $B_u \cap F_j$ in the t-coordinate is an AP with difference $\bar{u} = u^{-1} \bmod p$, size*

$$L(a_0) \in \{2k,\ 2k+1,\ 2k+2\}$$

given by the three-tier arc law*: with $M = k\epsilon + \delta_0$, $\delta_0 = \lfloor(\epsilon^2-1)/T\rfloor$, one has $L = 2k+1$ on the ball B_k* and *on two shallow off-bands, $L = 2k+2$ on the deep band $a_0 \in (p-1-M,\ M]$ when $2M \geq p-1$, and $L = 2k$ on the deep band $a_0 \in (M,\ p-1-M]$ when $2M \leq p-2$. At $\epsilon \equiv \pm 1 \pmod T$ the deep band is the whole off-ball (the phenomenon of §5); at intermediate ϵ a shallow band of width $|M-k|$ carries the on-size. Consequently the occurring footprint sizes are exactly two values, $\{2k, 2k+1\}$ for $\epsilon < T/2$ and $\{2k+1, 2k+2\}$ for $\epsilon > T/2$. The starts obey the* drift law

$$\sigma_u(j) \ = \ \sigma_r^0(j) - \bar{u}\, m\, j \pmod p, \qquad \sigma_r^0(j) = \bar{u}\,(S_2(a_0) - \lfloor rj/p \rfloor),$$

so the lift acts by per-fiber t-translation linear in j; for a $\pm$-colliding pair $w = u+pm$ the relative drift $\sigma_w(j) - \sigma_u(j) = -\bar{u}\, m\, j$ is exactly linear, injective in j, and vanishes only when $B_w = B_u$.

Proof. The footprint computation is the one of §5.4 verbatim with 6 replaced by T: $x \in B_u$ iff $a_0 + p\,s$ (with $s = (a_1 + ut) \bmod p$) lands in $[-c, c]$ modulo p^2, and the good s form the cyclic interval $[S_2, p-1] \cup [0, S_1]$ with $S_1 = k + \lfloor (M - a_0)/p \rfloor$, $S_2 = p - k - \lfloor (M + a_0)/p \rfloor$, which gives the three-tier law by the case split on a_0 relative to M and $p-1-M$. The drift law follows from $\lfloor (r+pm)j/p \rfloor = \lfloor rj/p \rfloor + mj$ and the a_0-dependence of the arc being r-only. Machine-verified: the law reproduces the committed $T = 6$ tables boundary for boundary ($p \leq 37$, both classes); it holds for every a_0 at 19 primes of $T = 7/T = 8$ (all residue classes); the footprint formula matches direct Z_{p^2} enumeration (957/957); the drift identity holds in 3,528/3,528 checks at $T \in \{7, 8\}$. □

8.2 The critical census at $T = 8$: eleven families, constant counts

The $(2K, 4U)$ fiber problem — four APs with sizes in the two-value window covering $\mathbb{Z}_p$, normal form $(0, 1, s_1)$ plus three moving APs with canonical differences — is the exact analogue of the Conjecture 6.7 census one rung up, and it is *finite*.

Proposition 8.2 (the 4-difference partner representatives; machine)**.** *At every prime $p = 8k+\epsilon$ tested — seventeen primes $p \in \{19, 29, 31, 41, 43, 47, 59, 61, 67, 71, 73, 79, 89, 97, 103, 137\}$ plus the boundary $p = 17$, covering all four odd ϵ classes with k up to 17 — the census of four-AP coverings in the normal form returns* exactly eleven difference families*:*

$$(1,1,1), \quad \textit{the permutations of } (1,1,t),\ (1,2,2),\ (1,q,q), \quad (3,3,3),$$

where $q = (p-1)/2$ is the antipode and $t = \lfloor (p+1)/3 \rfloor$ (equivalently $t \equiv \pm 3^{-1} \bmod p$, $3t \in \{p-1, p+1\}$). The placement counts are independent of p and depend only on the class and the ϵ-class of p:

ϵ *class*	*families*	*placements per family*	*total*
$\epsilon \in \{3, 7\}$	$(1,1,1)$*:* 48*; each other:* 24	288	
$\epsilon \in \{1, 5\}$	*three classes:* 528, 232, 216	2,784	

No pairwise $\pm$-distinct covering occurs at any $k \geq 2$ ($\epsilon \in \{3, 5, 7\}$) or $k \geq 5$ ($\epsilon = 1$); the $k = 2$ boundary carries twelve $\pm$-distinct configurations, all enumerated. At $\epsilon = 5$, $k = 3$ only, twelve further families with six placements each occur; they are gone by $k = 7$.

The five mechanisms are hand-derived, and each count is the junction arithmetic of a placement-rigid structure — the Corollary T1 phenomenon two rungs up. **(M1)** *Four intervals* $(1,1,1)$: the partition multiset $\{2k+1,(2k+2)^3\}$ gives 4 ordered size tuples $\times 3! = 6$ cyclic arrangements $= 24$ exact tilings; the one-overlap multiset $(2k+2)^4$ gives $3! \times 4 = 24$; total 48. **(M2)** *Two intervals + the parity split* $(1,2,2)$: the intervals tile an arc of length $4k+3$; the complementary arc of length $4k+4$ splits into its two parity classes, each *being* a 2-AP of size $2k+2$ — the $(2,2)$ mechanism of §6.3, one rung up, with an interval pair in place of the single interval. **(M3)** *Three intervals + the three-block sampler* $(1,1,t)$: since $t \equiv \pm 3^{-1}$, the t-AP decomposes into three interleaved strands — three short blocks spaced $\approx p/3$ apart — the wrapped two-block one block up; the three intervals tile the complement of the three holes. This is why the *third-point* replaces the antipode in the double-interval family: three intervals leave three holes. **(M4)** *Intervals + antipodal two-blocks* $(1,q,q)$: the q-AP is the wrapped two-block of §6.3 verbatim; the interval plugs its gap structure. **(M5)** *The thickened arc* $(3,3,3)$: three 3-APs with starts spaced $1,2$ have union a contiguous arc of length $3s-2 = 6k+4$; the normalized interval covers the complementary arc of length $2k+3$. Every count is p-free because each configuration is determined by its junction pattern.

8.3 The $T = 7$ census: the core, the tail, and the growing pinned sets

At $T = 7$ the frontier cell $(1K, 4U)$ carries structural linear slack (the overlap window is $[k-\epsilon+4,\ k-\epsilon+8]$), and the census is a *core–tail* object rather than a finite list.

Proposition 8.3 (the $T = 7$ census; machine)**.** *At eighteen primes $p = 7k+\epsilon$, all six ϵ classes, k up to 11: (i)* no pairwise $\pm$-distinct four-AP covering occurs *at any $k \geq 4$ ($\epsilon \leq 3$) resp. $k \geq 2$ ($\epsilon \geq 4$) — the clause-(i) analogue of Conjecture 6.7 one rung up; the boundary dirt at $k \leq 3$ is finite and enumerated. (ii) The same eleven families as Proposition 8.2 — with $t = \lfloor (p+1)/3 \rfloor$, $q = (p-1)/2$ — are the top eleven by configuration count at every prime and carry $\approx 90\%$ of all configurations at $k \geq 6$. (iii) The core's placement counts grow with the overlap window (the $(1,1,1)$ count is $528, 2160, 5712, 11952, 21648, 35568$ at $k = 2, \ldots, 11$: quadratic growth, the junction freedom under the slack; the count depends only on the window, whence exact coincidences between primes with equal windows). (iv) Beyond the core, the family count falls from 55–183 at $k \leq 4$ to 43–91 at $k \geq 5$ (decaying with k: $183 \to 43$ over the tested range), with rational-anchor differences ($\{4,5,6,12,14,18,28,33\}$ at $p = 71$) and counts a factor 15–50 below the core.*

This is the measured form of the "growing pinned sets": the solution sets grow $\approx k^2$ (two-dimensional families), and the $T = 6$ pigeonhole $|\Delta| \leq 6 < |U|$ must be replaced by a growth comparison. The replacement is supplied by the projections and the exit time.

8.4 The shear step at the higher rungs

Proposition 8.4 (projections, exit, and the empty cells; machine)**.** (i) *The one-coordinate projections of the census solution sets (starts and relative offsets) have maximum size*

$$11,\ 12,\ 18,\ 24,\ 24,\ 30,\ 36,\ 36,\ 42,\ 42,\ 42 \quad \text{at } k = 1, \ldots, 11,$$

that is, saturating at 42, *while $|U| = 5k+\epsilon-1$ grows linearly ($20 \to 60$ over the same range); the pigeonhole $|U| >$ maxproj holds at every $k \geq 5$ except the single case (p=29, ϵ=1, k=4), where $20 < 24$.* (ii) *The exit time of an affine drift line from a census solution family is at most 2 consecutive fibers, over all drift vectors fixed by the first fiber (top-ten families at $p \in \{17, 29, 43\}$; all admissible families in the ground-truth runs).* (iii) *Ground truth, exhaustive over census-admissible families with the bad-set distinctness filter: the $(1K, 4U)$ cell at $T = 7$ is* empty *at every prime $11 \leq p \leq 61$ (all six ϵ classes; best exit 2); the $(2K, 4U)$ cell at $T = 8$ is* empty *at every prime $17 \leq p \leq 43$ tested (best exit ≤ 1; at $p = 29, 41$ no admissible family even passes the*

per-fiber size profile). Controls: brute force at $p \in \{5, 7, 11\}$ *(all unit triples, distinct bad sets, no census restriction) finds* 262,350 *triples and* 0 *coverings at* $p = 11$*, and* 8 *coverings at the degenerate* $p = 5$ *(*$k = 0$*) — exactly the* 8 *found by the restricted search;* 0 *at the* $\epsilon = 0$ *prime* $p = 7$*.*

The closure architecture is the $T = 6$ one, rung-shifted: the census forces a $\pm$-collision (Propositions 8.2–8.3); the colliding pair's relative offset drifts exactly linearly and injectively (Lemma 8.1); the relative-offset projections are bounded — measured at ≤ 42, with the bound's mechanism the span-exact junction arithmetic of M1–M5, the same analysis that proves Corollary T1 — and $|U|$ is linear in k; the drift line exits the solution family within two fibers. The degenerate $p = 5$ ($k = 0$) is the boundary phenomenon, recorded as at $T = 6$.

The $(1K, 5U)$ cell named below is the object of the conditional closing attack of §9 (Theorem 2.1); the boundary recorded here is the measured boundary there.

8.5 The boundary of the method

Three things do not transfer, and are recorded here. *First*, the $(1K, 5U)$ cell at $T = 8$: the five-difference census at $k = 2$ has 8,168 families and 32.5M configurations, 840 of them $\pm$-distinct — the collision-only law fails at the fifth difference, the census restriction does not confine this cell, and its closure (sampled: best run 2 over 20,000 random families at $p \in \{17, 19\}$; heuristic: solution families of dimension 3 in a 4-coordinate relative space still meet a generic affine line boundedly) is the rung's open piece. *Second*, the $T = 7$ tail: the family count decays ($183 \to 43$) but is not proved to die; only the $T = 8$-critical census is a finite exact list. *Third*, the pinning lemma — projections ≤ 42 — is measured with the mechanisms identified but not yet proved; the span-exactness proof per family is the recorded obligation, the direct analogue of the $T = 6$ pinning-size lemma (Input S's fiberwise form).

For the programme's single unbounded input the meaning is this: the four-family classification of Conjecture 6.7 is not a $T = 6$ accident. One rung up the same anchors occur (1, 2, q) with a new pair (3, t); the counts stay p-independent; the exhaustiveness clause survives; and the shear mechanism that consumes the classification is T-free in its linear part. The two-page computation that would discharge Input S now has its higher-rung shape: prove the eleven-family exhaustiveness at $T = 8$-critical first (a finite, exactly verified statement), then the projection bound, and the $T = 6$ statement — the base rung of the same phenomenon — follows by the same mechanism analysis at one lower rung.

9 The scaling-closure lemma: derivation and calibration

This section is the derivation record behind Theorem 2.1. Its order is the order of the reduction: the cascade model and the dilate-intersection step (§9.1); the pool law, its saturation tables, and the status of the 840 identity (§9.2); the cut law—spacing, k-cancellation, and the conditional comparison that fixes the depth formula's two ends (§9.3); the two-regime structure of the dilate cuts, with the reflection fiber and the chain class proved (§9.4); the volume law at $T = 8$ with its threshold form, the renormalization lemma, the character system, the reduction of the remaining obligation to the strand census, and the transfer-matrix route that closed the third route by theorem (§9.5); and the epistemic status (§9.7). Throughout, a claim is labeled *proved* (with its verification record), *measured* (with its scope, sample size, and, where sampled, an interval), or *fit* (with the data it is fit to); extrapolations beyond the tested range are labeled *projected*; and no claim is made about the amount of work needed to close the open obligations.

9.1 The model: the cascade as a dilate-intersection

Fix the rung $T \geq 8$, the prime $p = Tk + \epsilon$, and the composite modulus $N = p^2$. The frontier cell $(1K, mU)$, $m = T - 3$, consists of the covering families with one kernel member and m

unit members; by Lemma 8.1 the kernel covers the fibers of a kernel bad set B_k of size $2k+1$, leaving $|U| = (T-2)k + \epsilon - 1$ uncovered fibers $F_j \cong \mathbb{Z}_p$, and on each fiber the m unit footprints are arithmetic progressions whose sizes lie in the two-value window and whose starts obey the drift law. In the census normal form—the normalizer arc pinned with difference 1 at the origin, the remaining differences canonical in $[2, D]$, $D = (p-1)/2$—the placement on fiber j is the relative-start tuple

$$c(j) \;=\; c^0(j) - \mu\, j \;\in\; (\mathbb{Z}_p)^{m-1},$$

where $c^0(j)$ is the residue-determined part (a function of the unit residues and the three-tier size assignment $\mathrm{sz}(j)$) and $\mu \in (\mathbb{Z}_p)^{m-1}$ is the free drift, one coordinate per moving arc, determined by the lifts. The fiber is covered if and only if $c(j) \in \mathrm{Sol}(\mathrm{sz}(j))$, the census solution set of the size tuple and the family's difference tuple; this is exactly the committed ground-truth check. The *fiber cascade* processes the uncovered fibers one by one, retaining the anchor placements that survive each; the family dies when the survivor set empties, and the *cascade depth* is the number of fibers needed.

Lemma 9.1 (two-fiber reduction)**.** *Fix an anchor fiber $j_0 \in U$ and an anchor solution $a \in \mathrm{Sol}(\mathrm{sz}(j_0))$. The anchor determines the drift, $\mu = (c^0(j_0) - a)\, j_0^{-1}$. For a second fiber j with $\lambda := j\, j_0^{-1} \bmod p$, the anchor solution a survives fiber j if and only if*

$$\lambda\, a + \tau \;\in\; \mathrm{Sol}(\mathrm{sz}(j)), \qquad \tau = c^0(j) - \lambda\, c^0(j_0) \;\in\; (\mathbb{Z}_p)^{m-1}.$$

Hence the two-fiber candidate set is the dilate-intersection $\mathrm{Sol}_0 \cap \lambda^{-1}(\mathrm{Sol}_j - \tau)$, and the cascade after i processed fibers is the iterated intersection $\bigcap_{\ell \le i} \lambda_\ell^{-1}(\mathrm{Sol}_\ell - \tau_\ell)$.

Proof. Substitute $\mu = (c^0(j_0) - a)\, j_0^{-1}$ into the covering condition $c^0(j) - \mu\, j \in \mathrm{Sol}(\mathrm{sz}(j))$ and rearrange; all maps are affine over $\mathbb{Z}_p$. □

The cascade step is thus exactly a *dilate-intersection of census solution sets*: the covering constraints across fibers are affine conditions on the anchor placement, with dilation ratios $\lambda = j/j_0$ given by the fiber set. This is the precise form of the picture under which the anchored-core analysis was organized. Write $\bar{\lambda} = \min(\lambda, p - \lambda)$ for the cyclic representative. The dilate $\nu \mapsto \lambda\nu$ maps the junction range into itself strongly when λ is a *small rational* ($\bar{\lambda} \le \Lambda_0$, or $\lambda \equiv a/b$ with $a, b \le O(L)$ for the junction length L): these are the *resonant* fibers, where the cut is weak. The remaining fibers impose genuine spacing (Lemma 9.3). At the committed zoo prime ($p = 17$, $k = 2$) the resonance zone $\bar{\lambda} \le 4$ covers half the dilations—the small-p extreme; its fraction vanishes as p grows. The per-pair *cut quality* is measured by

$$\gamma \;=\; \frac{\#(\mathrm{Sol}_0 \cap \lambda^{-1}(\mathrm{Sol}_j - \tau))}{|\mathrm{Sol}_0|\,|\mathrm{Sol}_j|\,/p^{m-1}},$$

the survivor count against the random-placement baseline; $\gamma = 0$ is an *instant kill* (the actual intersection is smaller than the random model), $\gamma \approx 1$ is no cut at all, and $\gamma \gg 1$ marks a resonant pair whose solution sets are too small for the random baseline to be meaningful.

The ±-reduced convention. The cascade is modelled on a ±-representative fiber set $U_\pm$: one representative per reflection pair $\{j, -j\}$, the reflection fiber excluded. This is forced by Lemma 9.6 (the bad sets are symmetric under $x \mapsto -x$, so fiber $-j$ imposes exactly the same covering condition as fiber j); processing both would double-count every constraint, halve the effective fiber budget, and make the survivor inequality $\mathrm{Hq}_\pm$ fail on the reflection duplicate. The measured depths of Table 11 are oriented-fiber counts (both j and $-j$), exceeding the ±-reduced formula by this duplicate; the theorem's budget is $|U_\pm| = \frac{1}{2}|U|$.

9.2 The pool law

Lemma 9.2 (pool law)**.** *The family pool after the ±-collision reduction is bounded by the distinct canonical difference tuples:*

$$P(T,p) \;\le\; (D-1)(D-2)\cdots(D-(T-4)) \;\le\; \big((p-1)/2\big)^{T-4}, \qquad D=(p-1)/2,$$

with difference count $T-4$ (the normalizer's difference is pinned to 1).

Proof. The census key of a ±-distinct family is an ordered tuple of distinct canonical differences in $[2,D] \subset [1,D]$; the falling factorial counts them. □

The law that is *not* a theorem is the saturation identity $P=(D-1)_{T-4}$. The committed zoo point ($T=8$, $(1K,5U)$, $p=17$, $k=2$) records exactly 840 ±-distinct families and $840 = 7{\cdot}6{\cdot}5{\cdot}4$—the bound saturated, every distinct tuple admitting a covering placement. Table 10 is the verdict at the demanded primes, and it is unambiguous: *the identity is a local accident of the slackiest prime, and the pool law is rung-monotone.*

Table 10: Pool saturation at the frontier cells. The $T=8$ rows are exhaustive censuses (two independent decision procedures, cross-checked against a third brute force; §9.6). The $T=9$ rows are stride-sampled with full decisions per sampled tuple: saturation carries a 95% Wilson interval with the stated n; the projected pools (bound × sampled fraction, one prime each) are *projected*, not established.

rung	p	(k,ϵ)	bound $(D-1)_{T-4}$	admissible	saturation
$T=8$	17	$(2,1)$	840	840	1.000
$T=8$	19	$(2,3)$	1,680	960	0.571
$T=8$	23	$(2,7)$	5,040	1,800	0.357
$T=8$	29	$(3,5)$	17,160	2,280	0.133
$T=8$	31	$(3,7)$	24,024	360	0.015
$T=9$	29	$(3,2)$	154,440	—	0.903 $[0.889, 0.915]$
$T=9$	31	$(3,4)$	240,240	—	0.490 $[0.468, 0.511]$

Three readings. *First*, the saturation mechanism: a covering by arcs with pairwise-distinct differences must waste overlap—the chain mechanisms of the repeated-difference core all exploit pairing (the interval tiling needs $d=1$ twice, the parity mechanism needs $d=d'$, the antipodal needs the q-pairs), so saturation requires the overlap budget to absorb the *distinctness penalty*. The budget-versus-penalty race is lost at $T=7$ (the distinct pool is empty for $k\ge 2$: the committed 18-prime census records 24/24 at $p=11$ and 36/60 at $p=13$ at $k=1$, and zero at every $k\ge 2$ prime), marginal at $T=8$ (it holds only at $p=17$, the prime with the largest overlap window, $\mathrm{ov}\in[3,8]$; it collapses as the window tightens and as k grows), and won with margin at $T\ge 9$ (at $T=9$ the extra arc's budget $\mathrm{ov}\approx (T-6)k+O(1)$ covers the penalty for most tuples: 90.3% at the slackier $\epsilon=2$ window, 49.0% at the tighter $\epsilon=4$). *Second*, the projected $T=9$ pools: 1.39×10^5 (interval $[1.37,1.41]\times 10^5$) at $p=29$ and 1.18×10^5 ($[1.12,1.23]\times 10^5$) at $p=31$—the measured $\sim 10^5$ pool scale at $T=9$ is (approximately) the distinct-tuple pool, and the $T=10$ reading ($\sim 10^8$ at $k\ge 4$, the falling factorial times the anchor multiplicity) is consistent with the rung-monotone law; the $T=10$ saturation itself (predicted ≥ 0.9 at the 40% budget) is *not measured*, and at $k\ge 4$ the pools exceed the cascade's reach in any case. *Third*, the set-form: admissibility is invariant under permuting the differences (0 order-conflicts over the 6,720 checked tuples at $p=19$ and $p=23$, forced by the size-assignment symmetry of the census), so the pool is $\#\mathrm{sets}\times(m-1)!$ with $\#\mathrm{sets}\le\binom{D-1}{m-1}$, and the saturation statement becomes "all $\binom{D-1}{m-1}$ sets admissible"—true only at the slackiest (rung, prime) pairs, asymptotically in T. The absolute admissible count at $T=8$ grows across the table ($840\to 960\to 1{,}800\to 2{,}280$) but collapses at $p=31$ (360): the pool is real but ϵ-modulated, never a smooth polynomial in p.

9.3 The cut law: spacing, k-cancellation, and the comparison

Lemma 9.3 (dilate-count spacing bound)**.** *Let p be prime, $\lambda \in \mathbb{Z}_p^*$, $0 \le L < p$, $t \in \mathbb{Z}_p$. Then*

$$N(\lambda, t) \;:=\; \#\{\nu \in \{0, \dots, L\} : (\lambda\nu + t \bmod p) \in \{0, \dots, L\}\} \;\le\; \lfloor L/b_{\min} \rfloor + 1,$$

where $b_{\min}(\lambda, L) := \min\{b \ge 1 : \|\lambda b\|_p \le L\}$ is the first return of the λ-walk to the L-ball ($b_{\min} = L + 1$ if no such $b \le L$ exists, giving $N \le 1$).

Proof. Let $\nu_1 < \nu_2$ be two solutions. Then $\delta = \nu_2 - \nu_1 \in [1, L]$ satisfies $\lambda\delta \equiv (\lambda\nu_2 + t) - (\lambda\nu_1 + t)$ with both right-hand residues in $[0, L]$, so $\|\lambda\delta\|_p \le L$; hence $\delta \ge b_{\min}$. The solutions are therefore $b_{\min}$-separated in ν, and at most $\lfloor L/b_{\min} \rfloor + 1$ of them fit in $[0, L]$. □

The corollary is the resonant/non-resonant split. The survival fraction per coordinate is $\sigma(\lambda) := \max_t N(\lambda, t)/(L+1) \le 1/b_{\min} + 1/(L+1)$: for an integer dilate $\bar{\lambda} \in [2, L]$ one has $b_{\min} = 1$ and $N \approx (L+1)/\bar{\lambda}$, so the cut is weak for $\bar{\lambda} \le 2$ and real for $\bar{\lambda} \ge 3$; for a small rational $\lambda \equiv a/b$ with $a, b \le O(L)$ one has $b_{\min} \le b$ and the cut degrades by the resonance factor, the number of such dilations among the fibers being $\le \sum_{b \le B}(2L+1)$ for strength B—the resonance budget; and otherwise (no good small-rational representative —the bulk of the fibers at large p) $b_{\min} \gtrsim p/(2L+1)$, so $\sigma(\lambda) \lesssim (2L+1)/p + 1/(L+1)$, and with $L = Rk$, $p = Tk$: $\sigma \lesssim 2R/T$. One regime caveat must be recorded: the box form requires $L = Rk < p/2$, i.e. $R < T/2$—true for the thin majority of zoo families ($R \approx 4.8$ at $T = 8$) but *false* for the fat top families (R up to ≈ 10.5, k-free). For those, the cut is carried not by the spacing lemma but by the solution sets' own sparsity (their volume is far below p^{m-1}; the measured instant kills), which is exactly H2m's content.

Proposition 9.4 (k-cancellation)**.** *Assume the volume law in either proved or hypothesized form—for the chain class, $|\mathrm{Sol}| \approx \mathrm{ov}^{m-1}$ with $\mathrm{ov} \approx (T-6)k$ (Lemma 9.8); for general families under H1g, $|\mathrm{Sol}| \le c_1\,\mathrm{ov}^{m-1}$. Then the per-fiber cut ratio at the frontier cell is k-free:*

$$\text{cut} \;\approx\; \frac{p^{m-1}}{|\mathrm{Sol}|} \;\gtrsim\; \Big(\frac{T}{T-6}\Big)^{m-1} / (\textit{order and tie factors}).$$

Proof. The junction volume and the modulus scale together: $p^{m-1} = (Tk)^{m-1}$ against $|\mathrm{Sol}| \approx ((T-6)k)^{m-1}$, so the k^{m-1} in the denominator cancels the k^{m-1} in the numerator and the ratio is a function of T alone, up to the order factors (the $(m-1)!$ orders and the tie bookkeeping), which are k-free by the census rigidity. □

The earlier form of this proposition was stated with a fitted junction scale R ($|\mathrm{Sol}| \le (Rk)^{m-2}$, $R \approx 4.2$); the chain-class analysis of §9.4 falsified that exponent for the dominant family (the correct chain exponent is $m-1$, not $m-2$) and *derived* the volume constant from the overlap budget $T-6$. No fitted R is used; under H1g the thin families' constant is c_1, bounded by data. Either form leaves the cut k-free—the cancellation is robust to the correction.

Proposition 9.5 (comparison; the $\pm$-reduced cascade, evaluated under the calibrated constants)**.** Derivation, not a cascade theorem*: the model below is the calibration of Theorem 2.1's clause (i) under Proposition 2.2's $\pm$-reduced induction ($Hq_\pm$ with additive error term E, quantified over the $\pm$-representative non-resonant fiber set $U_\pm$). Write $d = m - 1 = T - 4$, $B = c_1\mathrm{ov}^d$, and $r = \Gamma\,(p/\mathrm{ov})^{-d}$ (the per-fiber contraction factor of Prop. 2.2). The survivor count after q processed fibers obeys $|S_q| \le B\,r^q + \frac{E}{1-r}$, so the cascade reaches the $O(1)$ floor $|S| \le E/(1-r)$ within*

$$q \;=\; \frac{\log(c_1\,\mathrm{ov}^{m-1})}{(m-1)\log(p/\mathrm{ov}) - \log(c_1\Gamma)} \;+\; O(1)$$

fibers (the leading term; the $O(1)$ absorbs the error-term floor), after which a single further fiber kills the structured $O(1)$ remainder. The $\pm$-representative fiber budget is $|U_\pm| =$

$\frac{1}{2}((T-2)k+\epsilon-1)$. *Consequently:* (a) *(depth is $O_T(\log k)$, never linear) the leading term is* $\dfrac{(m-1)\log \mathrm{ov}+O(1)}{(m-1)\log(p/\mathrm{ov})-\log(c_1\Gamma)}=\dfrac{\log k}{\log(T/(T-6))}(1+o(1))$*, which grows like* $\log k$ *with a T-dependent slope and has* no *k-uniform linear bound;* (b) *(T-flat plateau at practical k) the pool-route term stays in* $[1.1, 2.9]$ *for* $k \le 10$, $8 \le T \le 20$*, so the typical depth is* $\approx$ 2–3 *plus the* $O(1)$ *floor;* (c) *(the "$\le T+3$" form is not used) the quantity* $(T-4)+N_{\mathrm{res}}$ *is* not *an upper bound on the depth and is not k-uniform; the explicit $O_T(\log k)$ formula above is the depth bound;* (d) *(closure margin against the $\pm$-budget) for every* $T \ge 8$, $k \ge 2$ *the contraction count q satisfies* $q \ll |U_\pm| = \frac{1}{2}((T-2)k+\epsilon-1)$*, with surplus growing linearly in k; equivalently the per-fiber factor r may be as large as* $(T/(T-6))^{m-1}$ *(the threshold value) and the cascade still closes within the $\pm$-budget. The factor* $\frac{1}{2}$ *in* $|U_\pm|$ *is the cost of the reflection identification (Lemma 9.6: fibers $\pm j$ are one constraint).*

Table 11: The depth calibration at the three rungs. Measured: the committed cascade record (maxima over $k \le 9/8/3$; medians 2/3/5). Formula: Proposition 9.5 evaluated with derived constants ($c_1 = 1$, $\Gamma = 1$) at the top-of-window overlap; the window shifts the leading term by ± 1.5 at $T = 8$. The "worst case" column is *withdrawn*: the earlier $\le T+3$ reading was an artifact of comparing the depth to the un-reduced fiber budget $|U|$, and the $O_T(\log k)$ law of Prop. 9.5 admits no k-uniform linear bound. The $T = 9$ measured maximum (11) exceeds the leading-term formula (9.2) because the measurement counts oriented fibers (both j and $-j$) while the derivation is $\pm$-reduced; the discrepancy is the reflection duplicate, not a mechanism failure, and it is resolved by the $\pm$-reduced model.

T	k	measured max (oriented)	$\pm$-reduced formula
8	2	9	8.5
9	3	11	9.2
10	3	10	10.1

The calibration carries its provenance explicitly: the per-fiber cut ratios ($\approx$ 120/960/2,160), the medians, and the maxima of Table 11 are the program's three-rung cascade record from the $T = 8/9/10$ measurement runs; those runs predate the present artifact bundle, whose committed cascade scripts cover the committed cells only (depths median 2, maximum 4; §9.6). The three-rung record is therefore *calibration*, not proof input, and the derivation of the model rests on the committed machinery (Lemmas 9.1 and 8.1) alone. The measured maxima are oriented-fiber counts and exceed the $\pm$-reduced formula by the reflection duplicate (Lemma 9.6); three rungs cannot distinguish a real growth of $\bar{\gamma}(T)$ from this duplicate or from the junction fatness, but the $\pm$-reduced model removes the duplicate as a source of discrepancy.

9.4 The two-regime structure: the measured γ, the reflection fiber, and the chain class

The zone-resolved measurement of the cut quality γ over the committed cells (764 admissible fiber pairs; five cells; canonical signs; the identical seed reproducing the zoo random-40) splits into two regimes, and the split is sharp at every cell.

The reading is threefold. *(i)* The non-resonant zone satisfies H2m with enormous margin: the median pair is an instant kill—the actual intersection is *smaller* than the random model, which is the lattice sparsity—and the worst mid-zone excess over all five cells is $\gamma = 40.6$ (a single pair at $T = 7$, $p = 29$); a uniform $\gamma_0 \approx 40$ covers every committed mid-zone pair, with 90th percentiles in 14–422. *(ii)* The γ tail is entirely a $\bar{\lambda} \le 4$ phenomenon. *(iii)* The tail's dominant component is proved structure, not noise: at $T = 8$-critical every small-zone $\gamma \in [205, 819]$ is a $\lambda = -1$ fiber with survival fraction exactly 1 (the random baseline is tiny because the solution sets there are nearly rigid), and the extreme zoo values ($\le 2.09 \times 10^4$) occur at pairs with $|\mathrm{Sol}_j| \le 4$—nearly-rigid size

Table 12: Zone-resolved cut quality γ (survivors against the random baseline) over the committed cells: pair counts, median, and maximum per zone. Mid zone: $\bar{\lambda} > 4$ and $\bar{\lambda} < p - 4$; small zone: $\bar{\lambda} \leq 4$. The tail (up to 2.09×10^4) lives entirely in the small zone; its dominant component at $T = 8$-critical is the $\lambda = -1$ reflection fiber of Lemma 9.6 (survival fraction exactly 1).

	mid zone ($\bar{\lambda} > 4$)			small zone ($\bar{\lambda} \leq 4$)		
cell	n	median	max	n	median	max
$T = 7$, $p = 17$, top-10	40	0.0	0.0	68	0.0	204.7
$T = 7$, $p = 29$, top-10	120	0.0	40.6	70	0.0	508.1
$T = 8$-crit., $p = 17$, top-10	4	0.0	0.0	11	409.4	818.8
$T = 8$ zoo, $p = 17$, top-10	40	0.0	10.7	70	0.0	193.3
$T = 8$ zoo, $p = 17$, random-40	128	0.0	17.6	213	0.0	$2.09{\times}10^4$

tuples where one or two survivors outweigh a minuscule baseline. Cascade depths re-confirmed on the same scope: medians 2 everywhere, maximum 4 (zoo random-40).

Lemma 9.6 (the free reflection fiber)**.** *The map $x \mapsto -x$ on $\mathbb{Z}_{p^2}$ preserves every bad set $B_w = \{x : \|wx\|_N \leq c\}$ and maps the fiber F_j onto F_{p-j}. Hence, for every unit system, the covering condition at fiber j is equivalent to the reflected covering condition at fiber $p - j$; in particular, for every anchor solution $a \in \mathrm{Sol}_0$ with drift μ, the reflected placement covers fiber $-j_0$ with the same drift: the fiber with dilation $\lambda \equiv -1$ has survival fraction exactly 1 and is never a cut.*

Proof. $\|w(-x)\|_N = \|wx\|_N$, so $B_w = -B_w$. For $x = j + pt \in F_j$, $-x \equiv (p-j) + p(p-1-t) \in F_{p-j}$: the involution $t \mapsto p - 1 - t$ carries the footprint $B_u \cap F_j$ onto $B_u \cap F_{p-j}$. Reflections carry arcs to arcs (an AP with canonical difference d reflects to an AP with the same canonical difference), sizes transfer (the three-tier arc law is even in a_0), and the whole fiber covering reflects to a fiber covering; the anchor's drift is a property of the composite configuration, which reflects with it. □

Measured confirmation: survival fraction exactly 1 at $\bar{\lambda} = 1$ in *every* family at *every* committed cell (840/840 for the zoo chain family $(1, 1, 1, 1)$; 6/6, 12/12, 24/24 at $T = 8$-critical). The resonance budget H2r is then derived modulo the one fitted constant: the free fiber contributes the 1, the $\bar{\lambda} \leq \Lambda_0$ dilations among the fiber set contribute $\leq 2\Lambda_0|U|/(p-1) \approx 2\Lambda_0(T-2)/T$, and $\Lambda_0 \approx 4$ is where the measured cuts turn on—data-chosen, and disclosed as such.

The case class whose structure is exactly known is the all-interval chain family $(1, \ldots, 1)$—the number-one zoo family by placement mass, with 123,240 solutions at $p = 17$ over all 32 size tuples. For it, the dilate-sparsity reduces to lattice-box counting between junction boxes, with the load-bearing invariant the overlap-sum identity.

Lemma 9.7 (overlap-sum identity)**.** *For the all-interval family and any size tuple s with $\Sigma s = p + \mathrm{ov}$: every covering placement, written in its sorted cyclic order σ, has the overlap vector $o_i = s_{\sigma(i)} - (a_{\sigma(i+1)} - a_{\sigma(i)})$ with $\sum_i o_i = \mathrm{ov}$ exactly, $o_i \leq s_{\sigma(i)}$, and every inter-start gap $\leq s_{\max}$ (the reach bound: the point before the next start is covered by some arc starting at or before the current one, whose reach is $\leq s_{\max}$ past it).*

Lemma 9.8 (chain-box parameterization)**.** $\mathrm{Sol}(s)$ *is contained in the union over the $(m-1)!$ cyclic orders σ of the affine-unimodular images of the boxes $\{o \in \mathbb{Z}^m : \sum o = \mathrm{ov},\ s_{\sigma(i)} - s_{\max} \leq o_i \leq s_{\sigma(i)}\}$; on the $o \geq 0$ stratum the parameterization is exact up to start ties, with count $\sum_\sigma \#\{o \geq 0 : \sum o = \mathrm{ov},\ o_i \leq s_{\sigma(i)}\}$. Consequently $|\mathrm{Sol}| \approx \mathrm{ov}^{m-1}$—the volume law with exponent $m - 1 = T - 4$ and constant derived from the overlap budget,* $\mathrm{ov} \approx (T-6)k$.

Both lemmas are machine-checked on the full solution set: 0 violations of either bound in all 123,240 solutions; the pure-chain ($o \geq 0$) stratum is 120,960 of them (98.2%), the nested

excess 2,280 (1.8%); the count 131,520 of (order, o) pairs against 123,240 placements is the tie/degenerate factor; the top size tuple $(5,5,5,5,5)$ gives 10,080 pairs against the committed maximum 8,880.

Theorem 9.9 (chain-class dilate reduction)**.** *For the all-interval family, the two-fiber survivor set at dilation λ is contained in*

$$\bigcup_{\sigma,\sigma'} \{o \in \mathrm{Box}_\sigma : \ \lambda\, \tilde{U}_{\sigma,\sigma'}\, o + \tilde{v} \in \mathrm{Box}'_{\sigma'} \pmod p\},$$

where $\tilde{U} = A_{\sigma'}^{-1} A_\sigma$ is integer-unimodular (A_σ the lower-triangular chain matrix) and the boxes have sides $\le 2s_{\max}$. Substituting $u = \tilde{U}o$ (whose box has sides $\le 2(m-1)\, s_{\max}$, the normalizer arc pinned), each term is bounded by Lemma 9.3 applied per coordinate:

$$\mathrm{survivors}(\lambda) \ \le \ ((m-1)!)^2 \Big(1 + \frac{4(m-1)\, s_{\max} + 1}{b_{\min}(\lambda,\, 2s_{\max})}\Big)^m.$$

Consequently the chain-class per-fiber cut is k-free and polynomial in $b_{\min}$: it exceeds any fixed γ once $b_{\min}(\lambda, 2s_{\max}) \ge B(T)$, and the fibers failing this are $\le B(4s_{\max}+1)$ in number—a vanishing fraction of U as k grows. Pulling the σ'-sum condition back through the dilation gives one affine congruence $\lambda\,(w_{\sigma,\sigma'}^\top o) \equiv c(\sigma,\sigma')$ per order pair; for matched sizes it degenerates to the o-independent congruence $\lambda\,\mathrm{ov} \equiv c$—either every anchor candidate satisfies it or none does: the instant-kill dichotomy, measured at $\bar\lambda \in \{2,4\}$ (survival 0) against the partial resonance at $\bar\lambda = 3$ (survival 120/840) in the zoo chain pair data.

Proof. Step 1 (the parameterization): with $a_1 = 0$ the sorted cyclic order σ of the starts has $\sigma(1) = 1$; the gaps $g_t = x_{t+1} - x_t$ partition $[0,p)$, so $\sum g_t = p$ and the overlap vector $o_t = s_{\sigma(t)} - g_t$ sums to ov (Lemma 9.7). For the reach bound: the point $y = x_{t+1} - 1$ is covered by some arc $[x_i, x_i + s_i)$ with $x_i \le x_t$ (no start lies in (x_t, x_{t+1})), whence $x_{t+1} \le x_i + s_i \le x_t + s_{\max}$. Conversely, given (σ, o) in the box with $\sum o = \mathrm{ov}$, the chain recursion $x_{t+1} = x_t + s_{\sigma(t)} - o_t$ from $x_1 = 0$ reconstructs the starts by a lower-triangular integer map A_σ with determinant ± 1. *Step 2* (the dilate pullback): a surviving pair (a, a') with $a \in \mathrm{Sol}_0(s)$ and $a' = \lambda a + \tau \in \mathrm{Sol}_j(s')$ writes $a = A_\sigma o + b_\sigma$ and $a' = A_{\sigma'} o' + b_{\sigma'}$, so $o' = \lambda \tilde{U} o + \tilde{v}$ with $\tilde{U} = A_{\sigma'}^{-1} A_\sigma \in \mathrm{GL}_m(\mathbb{Z})$. *Step 3* (the box count): the substitution $u = \tilde{U}o$ is a bijection of $\mathbb{Z}^m$ carrying the coordinate box (sides $\le s_{\max}$) into the box of sides $\le 2m\, s_{\max}$; dropping the sum-slice (enlarging), the count factors per coordinate and each factor obeys Lemma 9.3: $\le |I''|/b_{\min}(\lambda, s'_{\max}) + 1$. Multiplying over the m coordinates and the $((m-1)!)^2$ order pairs gives the displayed bound. *Step 4* (the sum congruence): for actual solutions Lemma 9.7 gives $\sum o' = \mathrm{ov}'$ as integers; pulling through the affine expression, $\lambda\,(w^\top o) \equiv \mathrm{ov}' - 1^\top \tilde{v} \pmod p$ with $w = \tilde{U}^\top 1$, a necessary congruence per order pair; o-independent for matched sizes. □

What the theorem does and does not give. It gives the exact structural account of the measured H2 phenomenology for the dominant family: the instant kills (the sum-congruence failure), the confinement of the resonance to small $\bar\lambda$ (the $b_{\min}$ regime), the k-freeness of the cut, and the volume law that fixes the depth formula's leading constant. It does *not* give the sharp uniform $\gamma_0 = O(1)$ with clean constants—the order factors $((m-1)!)^2$, $4(m-1)\, s_{\max}$ are loose by orders of magnitude at $T = 8$, where the measured mid-zone cut is an instant kill and the bound gives ≈ 1; the route to sharpness is to count *on the slice* (keep $\sum o = \mathrm{ov}$ in the lattice count—it is load-bearing; dropping it is what costs the constants), a geometry-of-numbers lemma in the genre of successive minima. And it does not cover the distinct-difference families, whose junction structure is not a chain: that is H1g, the subject of the next subsection.

9.5 The volume law at $T = 8$: the census, renormalization, and the strand census

H1g is the volume law with uniform margin, $|\mathrm{Sol}(f, \mathrm{sz})| \le c_1 \,\mathrm{ov}^{m-1}$ for the $\pm$-distinct families, with $c_1 \,\Gamma \le \big(T/(T-6)\big)^{m-1} e^{-\eta}$ for an $\eta > 0$ uniform in (T, k, f, sz)—the margin clause that keeps Proposition 2.2's denominator positive (the $\mathrm{Hq}_{\pm}$ constant Γ is now included). The bare-c_1 form below is the measured content; the margin clause is the open content. It was attacked at the smallest rung where the pattern is visible, $T = 8$, by full census where the cell admits it: $p = 17$ ($k = 2$, $\epsilon = 1$; all 840 families, all 32 size tuples, ov $\in [3, 8]$; per-family totals median 2,981, 90th percentile 9,633; per-(family, size) maximum 876 at $(5, 5, 5, 5, 5)$, ov $= 8$); $p = 19$ ($k = 2$, $\epsilon = 3$; all 1,680 families; admissible 960, exactly the pool saturation of Table 10 reproduced by an independent code path; totals median 102, 90th percentile 645, maximum 801; per-size maximum 108 at ov $= 6$); and $p = 29$ ($k = 3$, $\epsilon = 5$; *targeted*: every cascade-containing set, the doubling-path quadruples, the anchors, and a 120-set random sample—138 of the 715 difference sets, 710 families of the 17,160 total, a scope disclosed and not claimed beyond; 20/138 sets admissible, consistent with the full-scope 0.133; per-size maximum 186 at $(8, 8, 8, 8, 8)$, ov $= 11$). Every decision procedure was validated against an independent direct brute force (plain p^4 enumeration, no cascade, no pruning) before any number was cited: 9/9 keys match exactly across the three primes, including the chain controls 8,880/4,680/29,760 and zero keys.

Lemma 9.10 (renormalization invariance)**.** *Let $D = \{1\} \cup S$ be the difference multiset of a census family (S a $\pm$-distinct $(m-1)$-set), and for $\delta \in D$ let* $\mathrm{renorm}_\delta(S) = \{\mathrm{canon}(d\,\delta^{-1} \bmod p) : d \in D,\ d \ne \delta\}$, $\mathrm{canon}(x) = \min(x, p - x)$. *Then the total covering count summed over all size assignments is equal for S and* $\mathrm{renorm}_\delta(S)$*:* $\sum_{\mathrm{sz}} |\mathrm{Sol}(S, \mathrm{sz})| = \sum_{\mathrm{sz}} |\mathrm{Sol}(\mathrm{renorm}_\delta(S), \mathrm{sz})|$.

Proof. The covering condition is invariant under the global dilation $\varphi(x) = \delta^{-1}x$ of $\mathbb{Z}_p$ (a bijection carrying an arc $a + d\,[0, s)$ onto the arc $\delta^{-1}a + (d\,\delta^{-1})[0, s)$). Apply φ to a covering placement of the family S with size assignment $(s_d)_{d \in D}$: the image is a covering placement of the family with difference multiset $\delta^{-1}D$, in which the arc that had difference δ now has difference 1—it is the normalizer of the transformed system. Re-anchor at that arc's start (covering is translation-invariant) and canonicalize each difference $d \mapsto \mathrm{canon}(d)$ (an arc with difference $-d'$ is the reflection, $x \mapsto -x$ composed with a translation, of an arc with difference d'; reflections preserve coverings). The result is a census-normal-form covering placement of $\mathrm{renorm}_\delta(S)$ with the sizes carried along. Summing over all size tuples removes the bookkeeping dependence on which arc received which size (the admissible size-tuple set is permutation-closed), and the map is a bijection at each size assignment. □

Verified exactly: at $p = 17$ the 35 difference sets partition into 7 renormalization orbits (each of size 5, one per choice of $\delta \in D$), and the 7 volume classes are precisely these orbits; at $p = 19$ the 70 sets form 14 orbits (8 positive, 6 zero), and the 9 volume classes are exactly the unions of orbits. The volume is therefore a function of the dilation class of the difference multiset alone—not of the family key. This cuts the family space 5-to-1 in general and identifies the correct equivalence for any future classification: at $T = 8$ the census object is not the $\binom{D-2}{m-1}$ sets but their renormalization orbits. It is a dilation-invariance statement in the spirit of the class reduction at prime moduli (Lemma 4.6) and the census law of Conjecture 6.7: it identifies the equivalence; it does not count. For the cascade the consequence is load-bearing: the two-fiber cut at dilation λ compares Sol_0 with a dilate of Sol_j, and Lemma 9.10 says the volumes involved are dilation-class invariants—the cut law's input volumes are orbit functions, and H1g needs constants uniform over orbits, not over keys.

The reading of Table 13 is fourfold. *(i)* The uniform bound with $c_1 = 1$ holds at every measured prime, with margin growing in k (and tightening with the ϵ window at fixed k). *(ii)* The distinct class is uniformly *thinner* than the chain class at equal ov, by a factor growing from 10× to 160×. *(iii)* The direction of risk for H1g is therefore not a fat counterexample family but

Table 13: The volume law at $T = 8$: the worst $\pm$-distinct renormalization orbit against the chain benchmark, at the top size tuple. Normalized: the maximum divided by ov^4. The uniform bound $|\mathrm{Sol}| \le \mathrm{ov}^4$ (that is, $c_1 = 1$) holds at every census prime with margin $4.7\times/12\times/79\times$, *growing in* k; the distinct class sits $10\times/43\times/160\times$ below the chain class at equal ov. The $p = 29$ top orbit is identified within the targeted scope; a family outside it would have to exceed ov^4 outright to change the law's direction, and the admissibility collapse argues against it.

p	top distinct orbit	max @ ov	normalized	chain @ tuple	ratio
17	$\{1,2,3,4,8\}$	876 @ 8	0.214	8,880	0.099
19	$\{1,2,4,5,8\}$	108 @ 6	0.083	4,680	0.023
29	$\{1,2,4,7,8\}$	186 @ 11	0.013	29,760	0.0063

proof coverage—the opposite of an obstruction. *(iv)* The caveats: three primes, two k-values, both ϵ classes at $k = 2$ but only one at $k = 3$; the $p = 29$ top orbit rests on the targeted scope; the measurements fix the law's direction and margin, not its proof.

Proposition 9.11 (the threshold form of H1g)**.** *The per-coordinate bound* $|\mathrm{Sol}(f,\mathrm{sz})| \le p^{m-1}$ *is trivial at every cell of the measured family (the moving-start space is* $\mathbb{Z}_p^{m-1}$*), and* $p/\mathrm{ov} \to T/(T-6)$ *because* $\mathrm{ov} = (T-6)k + O(1)$ *against* $p = Tk + \epsilon$*, so the trivial bound realizes*

$$c_1^{\mathrm{triv}} = \big(T/(T-6) + o(1)\big)^{m-1}$$

in the normalization of the volume law—exactly the value at which the depth denominator $(m-1)\log(T/(T-6)) - \log c_1$ *of Theorem 2.1 vanishes. At* $m - 1 = T - 4$ *the threshold is* 256*,* 243*,* 244.1 *at* $T = 8/9/10$*: near-constant across rungs, so neither the cascade's task nor the volume law's softens at higher* T*.*

Proof. The first inequality is the per-coordinate count: a solution is a choice of the $m-1$ moving starts, each in $\mathbb{Z}_p$. The second is the size window of Lemma 8.1: the $m = T-3$ arcs of the two-value window give $\mathrm{ov} = \Sigma s - p = (T-6)k + O(1)$ against $p = Tk + \epsilon$. Dividing, $|\mathrm{Sol}| \le (p/\mathrm{ov})^{m-1}\mathrm{ov}^{m-1}$, which is the volume law with $c_1 = (p/\mathrm{ov})^{m-1}$; substituting this c_1 in the depth formula of Theorem 2.1 zeroes its denominator, so the trivial bound purchases no cascade depth at all. □

The threshold form restates what H1g must purchase: not the free bound $|\mathrm{Sol}| \le p^{m-1}$ but the *margin* over it, growing in k. Measured, the margins below the trivial box are $95\times/1{,}207\times/3{,}803\times$ at $p = 17/19/29$ (top distinct family at its top size tuple: $17^4/876$, $19^4/108$, $29^4/186$—the maxima of Table 13); the committed margins over ov^4 ($4.7\times/12\times/79\times$) are the same fact in volume-law units. The form also unifies the program's constant-quality obligations under one numerical criterion: every volume constant must beat $\big(T/(T-6)\big)^{T-4} \approx 250$. Every measured constant clears it—the chain class's 2.0–3.6 by roughly two orders of magnitude, the distinct class by the factors above—and no proved constant yet does (the chain constants of Theorem 9.9 are loose by orders of magnitude). The sharp-constants question of §9.4 and H1g are one criterion stated twice.

The mechanism of the top class was probed on the full solution sets. Every solution needs all five arcs (essential-arc pattern $(1,1,1,1,1)$ in 876/876): no spectator redundancy. The overlap-budget identity $\sum_{i<j} O_{ij} = \mathrm{ov} + \sum_x \binom{o(x)}{2}$ holds exactly (0 violations), with pairwise overlaps small (mean 0.5–1.5, maximum 2–3) and under the spacing caps with room: no pair carries the covering; the arrangement is collective. Coordinate projections of the solution set at the probed families sit far from full ($[16,17,14,12]$ for the top family; $[9,9,7,8]$ for the minimum class)—per-family statistics that initially read as a $T = 7$-style pinning law persisting at the zoo cell; the exhaustive key-level check of the character-system paragraph below shows that reading is an artifact, not a law. And the block geometry exhibits the mechanism: the interval covers

$[0, s_1)$; the $d = 2$ arc covers every other point of a window of span $2s - 1$; the $d = 4$ arc every fourth point of a span $4(s - 1)$; the $d = 8$ arc is the antipodal two/three-block sampler; the circle closes by interleaving these against the interval and each other. The top class is the *binary cascade* $\{1, 2, 4, 8\}$ plus a fifth difference—at $p = 17$ the orbit swallows every fifth value $\{3, 5, 6, 7\}$; at $p = 19$ the fifth is 5; at $p = 29$ it is 7 (the observed pattern "fifth $= \epsilon + 2$" holds at the three primes measured; three points, flagged as such, not a law). At $p = 29$, where 2 is a primitive root, the canonicalized doubling orbit visits every difference and only cascade-containing quadruples survive among the path quadruples: resonance structure is necessary for admissibility at $k = 3$, consistent with the pool collapse.

The character system: the corrected coordinates. The projection reading above was tested as a law and is false in both naive forms. Call the *character marginals* of a solution set the projection sizes along the coordinate functionals of the moving-start space. The two natural invariance readings—(V′) the multiset of the four singleton marginals is a renormalization-orbit invariant, and (V″) the same for the six pairwise-difference marginals—are both false, each caught by machine check (V′: profile multisets differ across orbit members at both full primes; V″: 0/32 σ-matched size tuples agree at $p = 19$). The mechanism was pinned by an explicit construction of the renormalization bijection (at $p = 19$, $S = (2, 4, 5, 8)$, $\delta = 5$, with image $S' = (3, 4, 6, 8)$): the image of each moving slot is δ^{-1} times a source slot coordinate *minus the anchor coordinate* a_δ, plus a size-dependent constant, and the new normalizer is $\delta^{-1}a_\delta$—every moving slot is coupled to the anchor. Consequently the image's slot singletons are the source's anchor pairs, and the image's pairwise differences (in which the anchor cancels) are the source's non-anchor singletons and pairs: neither subsystem is closed, and the union is.

Lemma 9.12 (the extended character system)**.** *Let $X = \{e_i\} \cup \{e_i - e_j : i < j\}$ on the moving-start space $\mathbb{Z}_p^{m-1}$, so $|X| = (m - 1) + \binom{m-1}{2}$ (ten functionals at $T = 8$). Under the renormalization bijection of Lemma 9.10, each functional of X maps to a nonzero scalar multiple of a functional of X plus a size-dependent constant. Consequently the multiset of marginal sizes $\{|\pi_\chi(\mathrm{Sol}(S, \mathrm{sz}))| : \chi \in X\}$, aggregated over the permutation-closed set of size tuples, is a renormalization-orbit invariant.*

Proof. The closure statement is the anchor-mixing computation just recorded, verified symbolically and then end-to-end at $p = 19$ for $S = (2, 4, 5, 8)$: all 801 images of $\mathrm{Sol}(S, \mathrm{sz})$ land in $\mathrm{Sol}(S', \mathrm{sz} \circ \sigma)$, injectively, at every size tuple, with per-key counts matching under the explicit σ (0 mismatches)—which also confirms the per-assignment-bijection step of Lemma 9.10's proof at a stronger level than the committed record (totals were verified; assignments now are). The invariance follows: each character's marginal size is carried to another character's marginal size up to the size permutation, so the aggregate multiset is constant along the orbit. Verified on the full top orbits at both primes—$p = 17$: the orbit of $(2, 3, 4, 8)$, five members, 9,633 solutions per set; $p = 19$: the orbit of $(2, 4, 5, 8)$, five members, 801 per set—with the aggregate ten-functional multisets identical across members at both primes; and at $k = 3$, the two admissible $p = 29$ probe sets $(2, 4, 7, 8)$ and $(3, 4, 8, 13)$ form one orbit ($\delta = 7$) with equal totals 1,311. □

Two consequences. First, the coordinate system for the census is now fixed: any class-level (orbit-level) law must be stated on X—not on absolute starts, not on pairwise phases alone—so the symmetry analysis of the census object is complete (Lemma 9.10 fixes the equivalence, and Lemma 9.12 the coordinates in which a law can be stated). Second, the *marginal route into the strand census is measured dead* at the zoo cell. Across 1,678 admissible keys (exhaustive at $p = 17$ and $p = 19$, targeted at $p = 29$; every key inside the committed census scope matches it, 1,654/1,654, 0 mismatches), the projection envelope over all keys and all ten characters is p itself at both $k = 2$ primes and 27/29 at $k = 3$ (fullness fractions 1.000/1.000/0.931); a sufficient marginal law on X— $|\pi_\chi| \le \gamma_0 \,\mathrm{ov}$ with $\gamma_0^{m-1} < (T/(T - 6))^{m-1}$—would imply the volume law, since Sol is contained in the product of its singleton marginals, but it requires the fullness fraction

to fall below $\gamma_0(T-6)/T$ asymptotically (below 75% already at $\gamma_0 = 3$), and no downward trend is in evidence at the three committed primes. The top-family profiles are non-monotone in (k, ϵ) ($[12, 17, 17, 16] \to [4, 4, 9, 8] \to [9, 7, 10, 27]$; the free character 27 at $p = 29$ is the $d = 8$ antipodal sampler of the block geometry above): family- and window-dependent statistics, not a law. The only invariant sparsity is *joint*: the median density of Sol inside its own marginal box is 1.5%/7.4%/6.9% at the three primes—the pointwise no-holes condition along I, which is what the strand language below encodes.

The strand census: where H1g now lives. The reduction runs in three steps, each elementary. *Step 1* (the interval spine): the complement of the normalizer interval is the interval $I = [s_1, p)$ of length $\ell = p - s_1 \approx (T-2)k$; every covering of $\mathbb{Z}_p$ by the interval plus the $m-1$ moving APs restricts to a covering of I by the I-restrictions of the moving APs (plus the two boundary points at the wrap), and the restriction map is injective up to the boundary bookkeeping, so an upper bound on the I-restricted count bounds |Sol|. The moving arcs' total size is $(m-1) \cdot 2k \approx (T-3) \cdot 2k$ against $\ell \approx (T-2)k$: a near-exact cover with slack ov. *Step 2* (the strands): the I-restriction of an AP with step d is a union of *strands*—maximal runs of consecutive t whose points $a + td$ stay inside I; within a strand the points sit d apart. Covering consecutive points of I requires strands of different steps to interleave; a pair (d, d') interleaves only at compatible phases, and the phase-compatibility condition per pair is a three-distance-type statement about $\lambda = d'/d$—exactly the Lemma 5.16 genre, on the pair lines $td - ud' \equiv \delta$ that already govern the pairwise overlaps. *Step 3* (the arrangement form): every covering induces a cyclic *arrangement*—the maximal runs of I and, for each run, the subset of strands carrying it, with the handoffs between runs subject to the pair-phase conditions—and the volume decomposes as $|\mathrm{Sol}| \le \sum_{\text{arrangements}}$(phase freedom of the arrangement), with boundary corrections.

What is missing for a proof is the uniform count: bound the number of arrangements (the handoff patterns) and the phase freedom per arrangement, uniformly over the resonance classes of the difference set—equivalently, by Lemma 9.10, over the renormalization orbits. The chain class is the degenerate resonance class where all steps coincide and the strand picture collapses to the telescoping of Lemmas 9.7–9.8; the binary cascade is the pure ratio-2 class; a general orbit mixes ratio classes. The three families of necessary conditions already available—the overlap budget, the pair spacing caps, and the last-arc fit count—do *not* discriminate: random placements typically satisfy the budget (the expected total pairwise overlap ≈ 12 exceeds ov at $p = 17$). The discriminating structure is the pointwise no-holes condition along I, which is what the strand language encodes. The reduction is to a named, bounded combinatorial object, the *strand census*; the injectivity of the restriction map and the boundary corrections are machine-consistent at the census cells but are not proved in general, and the strand census is therefore a *plausible* reduction of H1g, not an established equivalence. The work is class-by-class (per resonance type), and uniformity across classes is itself a lemma—the precise shape of the remaining distance.

The transfer-matrix route: the third route, closed by theorem. The third candidate route—a transfer-matrix or automaton model of the joint density under the arc-covering constraint—was examined as a proof attempt under a pre-committed success criterion: a bound on the whole-cascade survival fraction, survivors $\le \rho^{\mathrm{depth}} \cdot$ initial mass, with $\rho \le 1 - \epsilon$ for some $\epsilon > 0$ independent of k. Two specifications were fixed in advance, as the route's design demanded. *The state space*: the covering constraint at fiber j is membership $c(j) \in \mathrm{Sol}(\mathrm{sz}(j))$—a function of the placement state alone; the cascade is non-consuming (the alive-set semantics is a pure intersection), so the constraint is Markovian in the natural state $(\mathbb{Z}_p)^{m-1}$, and the theorem below is invariant under state augmentation in any case, so no resolution of this specification could change the verdict. *The comparison target*: the whole-cascade fraction, with both factors' k-dependence acknowledged—the initial mass is $|\mathrm{Sol}_0|$, and the per-step ratios inherit the k-scaling of the forbidden sets.

Lemma 9.13 (the monomial dichotomy of the cascade, single-anchor scope)**.** *Fix the rung, the prime, and any sequence of fibers $j_1, \dots, j_D$ from the uncovered set with solution sets $\mathrm{Sol}_t = \mathrm{Sol}(\mathrm{sz}(j_t))$, and let σ_t be the affine bijection between consecutive fiber placements of Lemma 9.1. Then:* (i) *every constrained transition $P_{\mathrm{Sol}_{t+1}}\,\sigma_t\,P_{\mathrm{Sol}_t}$ is a partial permutation matrix (at most one nonzero entry, equal to* 1*, per row and per column), and so is every product along the chain; its spectrum is* $\{0\}$ *(the directed paths) together with the roots of unity of the surviving cycle lengths.* (ii) *The affine maps compose to the identity around any closed fiber loop (each re-expresses the fixed drift μ; the dilation ratios telescope), so the cyclic transfer product is a diagonal* 0/1 *matrix whose trace is the number of placements surviving the loop—the cascade's own survivor count—and whose spectral radius is* 1 *if that count is positive and* 0 *otherwise.* (iii) *Consequently every constrained transfer matrix obtainable in the framework—open chain, cyclic, homogenized single pair, or state-augmented—has spectral radius exactly* 0 *or* 1*: never an interior value, at any k.*

Proof. (i) σ_t is a bijection; composing with coordinate projections leaves at most one nonzero per row and per column, and partial permutations are closed under products. The digraph of a partial permutation has in- and out-degree at most 1, hence decomposes into directed paths and cycles; paths give nilpotent blocks, and a cycle of length ℓ gives a cyclic permutation block whose spectrum is the ℓ-th roots of unity. *(ii)* Each σ is the change of drift coordinates $c(j') = (j'/j)\,c(j) + c^0(j') - (j'/j)\,c^0(j)$ (Lemma 9.1); composing around a loop telescopes the dilation ratios to 1 and the translations to 0, and deletion only zeroes diagonal entries. *(iii)* Augmentation preserves determinism (the register update is a function of the state), so the augmented constrained matrix is again a partial permutation, and parts (i)–(ii) apply verbatim. □

Scope. The lemma is true but narrow: it closes only the *single-anchor* automaton—the operator that tracks one anchor placement along the drift. It does *not* close the general strand census, whose natural transfer operator scans I position by position with local coverage states and *branches* (a state records which strands are active at the current position, and several are), so its constrained operator is not a partial permutation and may have interior spectral radius—as the 2×2 example above ($\rho \approx 1.122$) shows. The lemma therefore does *not* establish that "the toolkit is provably not the right shape" in general; it establishes the weaker, correct statement that *this* single-anchor transfer operator consumes H1g rather than proving it, every spectral quantity in its framework being 0 or 1. The general strand-census automaton is the open route; the lemma's scope is the single-anchor special case, and the abstract and §2 state the general version only as the open obligation.

The machine record runs at three levels. *Generic*: 1,500/1,500 random constrained systems (random bijections with random deletion sets) satisfy the structural dichotomy—the partial-permutation premise, and acyclic if and only if nilpotent—and 12,000 exact trace fingerprints $\mathrm{tr}(Q^t) = \sum_{\ell \mid t} \ell\, c_\ell$ (over 300 systems, $t \le 40$, integer-exact) match the cycle inventory with zero mismatches. *Committed objects*: six committed families (the zoo chain and top-distinct orbits at $p = 17$ and $p = 19$; the two committed admissible $p = 29$ sets), 66 homogenized pair systems—the route's own model, iterating a single pair map $x \mapsto \lambda x + \tau$ under the deletion set Sol_j—of which exactly the 6 reflection pairs ($\bar{\lambda} = 1$) have $\rho = 1$ and all 60 others have $\rho = 0$; the loop-composition identity of part (ii) holds algebraically and numerically (500 random points per family); the automaton's per-step survivor counts equal the committed μ-space cascade convention at every step (the chain family's depth 2 reproduces the committed zoo record); and the end-to-end ground truth—$[\sigma(x) \in \mathrm{Sol}_j]$ if and only if the direct fiber-covering test accepts $\sigma(x)$—holds on 2,800/2,800 sampled checks, with the committed volume margins reproduced exactly (top $|\mathrm{Sol}| = 876/108/186$, margins $95\times$ / $1{,}207\times$ / $3{,}803\times$ over the trivial box). *The reflection fiber in automaton language*: the two-fiber survival fraction at $\lambda = -1$ is exactly 1 at all six families across all three primes, and the homogenized reflection sub-shift closes the whole

of Sol_j into 2-cycles—survival flat forever: Lemma 9.6 as a statement about the instrument itself, the deletion at the free fiber being empty.

What the instrument measures at the committed cells is equally sharp. The iterated-pair survival counts collapse as $[840, 120, 0]$, $[224, 1, 0]$, $[4{,}680, 240, 0]$—piecewise extinction within three iterates, the intermediate values set by maximal lattice runs of solution points along the drift orbits (runs of length at most 2 everywhere; extinction confirmed at every horizon to 12)—the three-distance genre of Lemma 5.16, not geometric decay with an interior rate. And the k-scaling that the obligations need (the margins $95\times$ / $1{,}207\times$ / $3{,}803\times$) lives in $|\mathrm{Sol}|/p^{m-1}$, in the census counts, and in no spectral quantity: the spectral outputs are binary at every prime, with no k-trend because there is nothing to trend.

Nor does any reformulation of the instrument escape the dichotomy. A coarse-grained or weighted automaton (states as classes, class multiplicities as weights) can carry an interior spectral radius numerically, but for non-negative matrices the radius dominates every directed cycle's geometric mean, so $\rho \le 1-\epsilon$ would force uniform decay of the weight products along drift orbits—and the weights are census counts, so this is the margin itself entering as an input rather than an output; the converse fails besides (the 2×2 witness $\left(\begin{smallmatrix} 0 & 0.9 \\ 0.9 & 0.4 \end{smallmatrix}\right)$: every directed cycle has geometric mean ≤ 0.9 yet $\rho \approx 1.122$—branching resurrects mass), so a second external input, a branching control, would be needed as well. Averaged or randomized operators collapse to the independent-extension baseline, whose deficit is the γ-correlation—the cascade model's own deepest unproved input. Every door leads through an external uniform-in-k estimate of exactly the margin type; none generates one. In automaton language the whole-cascade survival fraction is the product of the consecutive conditional ratios $r_t = |S_t|/|S_{t-1}|$, each bounded by $\gamma_0\,|\mathrm{Sol}_{j_t}|/p^{m-1}$ under H2m with the thinness of $|\mathrm{Sol}|$ supplied by H1g: under its own model the route's output is Proposition 9.5's recursion, and the uniform deficit it owes is the hypothesis stack it was commissioned to prove. The instrument is the census in different notation.

The verdict against the pre-committed trichotomy of outcomes: the success criterion ($\rho \le 1-\epsilon$) is unattainable by theorem—there is no interior spectral radius to bound, in any state formulation, at any k; and the failure mode is not $\rho \to 1$ as $k \to \infty$ but $\rho \in \{0, 1\}$ as an identity of the lossless dynamics. The type-mismatch reading is thereby upgraded from observation to mechanism *for the single-anchor operator*: bijections preserve measure exactly (the monomial structure), deletion filters without dissipating (partial permutations carry spectrum in $\{0\}$ together with the unit circle), and the loop composition is the identity (the cyclic instrument measures alive-or-dead, not decay). Margins require dissipation; losslessness forbids generated dissipation *for this operator*. The reformulation is bounded by its own fidelity *at the single-anchor level*. The scope caveat is load-bearing: the general strand-census automaton branches and is not covered by the lemma, so the stronger claims "the toolkit is provably not the right shape" and "bounded by its own fidelity" in general are *not* established here and are withdrawn from the abstract and §2; what is established is the single-anchor dichotomy. Lemma 9.13 is the program's third k-invariance theorem—after the spacing lemma's k-cancellation and the character system's closed coordinates—and, like them, a closure-type statement: the single-anchor toolkit's signature holds even for the instrument built to break it.

9.6 The verification record of this section

Every number above is either an exhaustive census with two independent decision procedures, or a sample with its size and interval, or a proved statement with its machine check; the records, in the order of appearance: the pool censuses (the pruned-cascade and the flattened-popcount procedures agree exactly at $p = 17/19/23$; an independent pure-enumeration brute force agrees on 21/21 sampled tuples across $p = 17/19/23/31$, covering both admissible and inadmissible branches; the $T = 9$ stride samples carry full decisions, and the $m = 6$ code path shares the cascade code brute-force-verified at four $m = 5$ primes, with the external consistency check the agreement of the projected pools with the independently measured pool scale); the zone-

resolved γ table (764 pairs, five committed cells, the identical seed reproducing the committed family selections); the chain-class machine checks (123,240 solutions, 0 violations of the sum identity or the reach bound; the count reconciliations 131,520 against 123,240 and 10,080 against 8,880); the reflection fiber (survival exactly 1 at every $\bar{\lambda} = 1$ pair, every family, every cell); the renormalization-orbit law (machine-checked set-by-set at $p = 17$ and $p = 19$); the volume census and its ground truth (9/9 brute-force keys; the admissibility reproductions 840/840 and 960/1,680; the order-independence of volume across all 24 orderings); the essential-arc, budget-identity, and cap checks on the probed families; and the cascade depths at the committed cells (medians 2, maximum 4); and the character-system record of §9.5 (the rigidity census over 1,678 admissible keys with every key inside the committed census scope matching it exactly, 1,654/1,654, 0 mismatches; the per-assignment renormalization bijection, 801/801 images injective with per-key counts matching under the explicit σ; the falsifications of V' and V''; the character-system aggregate invariance on both full top orbits; the chain controls 8,880/4,680/29,760 reproduced by the same code path). The transfer-matrix route's record: the generic dichotomy and trace fingerprints (1,500 systems, 12,000 comparisons, zero mismatches); the 66 homogenized pair systems on the six committed families with the closed-form orbit inventories verified by full traversal and the finite-horizon survival counts verified by direct simulation (zero mismatches); the loop-composition identity (algebraic and numeric, 500 points per family); the per-step equality of the automaton's survivor counts with the committed μ-space cascade convention (zero mismatches, all steps, all families; the committed chain-family depth reproduced); the end-to-end fiber-covering ground truth (2,800/2,800); the reflection pairs (survival fraction exactly 1, all six families, all three primes); and the reproduction of the committed volume margins (876/108/186 against $95\times$ /1,207 $\times$ /3,803$\times$). The three-rung depth record of Table 11 is calibration from the earlier cascade runs and is labeled as such wherever it is used.

9.7 The epistemic status

The program is either one proof away—H1g and the sharp H2m, both reduced to concrete named objects, the strand census and the pair-line count—or one systematic obstruction away: a family class whose junction structure escapes the ov^{m-1} volume law. For the obstruction direction the $T = 8$ evidence points the other way: the measured margin *grows* with k, and no distinct family reaches even a tenth of the chain-class volume at equal ov beyond $k = 2$. The distinction will be settled by whether the strand census closes at $T = 8$ with uniform constants, not by further measurement.

One more disclosure belongs in the record. The sequence of named obligations has moved: the input structure, then the exhaustiveness of the census classification, then the scaling-closure lemma itself, then H1-general, now the strand census. Each closure has been real, and each new obligation has been at the same scale as the previous one. Whether this is a fractal obstruction—each gap revealing a new gap of the same size, indefinitely—or a sequence of genuinely decreasing obligations that will bottom out is not decidable from the present data; after five iterations of the pattern the defensible prior is that the reformulation has a stubborn residual, and Theorem 2.1 is stated so that the residual is exactly what the hypotheses name. What would change the verdict: a family at some $(T = 8, k)$ whose per-size volume exceeds ov^{m-1}—the measured pool collapse makes this increasingly unlikely, as admissibility itself dies at 13% by $p = 29$—or a strand-census proof at $T = 8$ with uniform constants, which would make H1g a theorem at the rung and supply the template for general T. No claim is made about the amount of work needed to close the open obligations; the failure modes are stated where the obligations are.

The record since that disclosure adds a sixth attack turn and a change in the pattern's shape. The strand census has now been attacked twice and has survived both attacks: the necessary-condition families of this section (the overlap budget, the spacing caps, the last-arc fit count—non-discriminating), and the marginal-rigidity route of §9.5 (the projection envelope measured full, the two naive coordinate readings false, the corrected character system closed but

silent on the count). Both closed as route-closures, and—for the first time in the sequence's six attack turns—the named obligation did not move: no successor lemma was minted; what moved was the obligation's coordinate system (Lemma 9.12) and its numerical form (Proposition 9.11). The reading has tightened accordingly: the residual is not a bookkeeping artifact awaiting the right census format; it is resistant to the toolkit that produced every proved result in this paper. A structural fact in the ledger explains why that may be: the proved results are closures, invariances, exhaustiveness statements, and machine verifications, while every remaining obligation—H1g's margin over the trivial bound, the volume law's unproved growth in k, the chain class's gap from measured to proved constants—is a growth-in-k margin over a trivial or measured baseline, and no turn of this program has proved a statement of that type. Both closed routes, moreover, were census-shaped attacks on what Proposition 9.11 shows is an inequality-shaped obstruction. Three candidate third routes are on record, none attempted here and none with a stated prior: pairwise-difference marginals in the corrected X coordinates; a combinatorial characterization of the joint support, given its measured sparsity (1.5%–7.4% median density); and a transfer-matrix or automaton model of the joint density under the arc-covering constraint.

The type-mismatch reading of the preceding paragraph has since been verified statement-by-statement against the paper's full proof inventory—every lemma, proposition, theorem, and machine record—and it holds, with the three statements that could be read as exceptions resolved explicitly. First, the spacing lemma (Lemma 9.3) is a packing bound whose slack over its own trivial baseline is the factor $b_{\min}$; at the cascade's operating point ($L = Rk$ against $p = Tk$) the bulk-fiber value $b_{\min} \gtrsim p/(2L+1) \approx T/(2R)$ is k-free, so the lemma is invariant under exactly the scaling in which the obligations' margins grow, and the "k-cancellation" (Proposition 9.4) is a k-invariance of the cut, conditional on the volume law—the cancellation of the dependence the obligations need to grow, not a proof that it grows. Second, part (d) of the comparison (Proposition 9.5) does state a margin growing linearly in k, but as a consequence of its hypothesis stack (H1g included), not as a proved statement about the measured world. Third, the chain-class volume law (Lemma 9.8) computes a count that grows in k, but its margin over the trivial box is the k-free threshold value of Proposition 9.11; the growth-in-k content of H1g is precisely the distinct class's extra thinness below the chain benchmark ($10\times/43\times/160\times$ at equal ov, measured), which the chain-class theorem does not touch, while the proved chain constants of Theorem 9.9 carry order factors ($4m\,s_{\max}$ with $s_{\max} \approx 2k$) whose slack grows in k—the measured-to-proved constant gap already listed among the obligations. With the reading verified, the three candidate third routes of the preceding paragraph are typed by output: pairwise-difference marginals in the corrected X coordinates and the joint-support characterization are census-shaped (classification outputs; the same type as the two routes already closed), while the transfer-matrix or automaton model is the one margin-shaped candidate—the natural output of a transfer-operator argument is a spectral quantity uniform in k, the missing statement type. The typing assigns output types, not success probabilities, and rests on the tool-reach half of the reading rather than a theorem; it is the reason the automaton route is the only one the record prioritizes, and the attempt itself is its test.

The attempt has since been run, under the discipline of a proof attempt: the two design specifications fixed beforehand, the success criterion pre-committed, and the verdict taken against it. The route closed by theorem *in its single-anchor scope* (Lemma 9.13 and the analysis following it): every spectral quantity the single-anchor automaton framework can produce is exactly 0 or 1, the cyclic product's trace is the cascade's own survivor count, and the survival-fraction deficit the criterion demanded translates exactly into the pair bound H2m and the volume law H1g—the instrument consumes the hypotheses it was commissioned to prove. The typing's prediction was therefore right about the output type and wrong about its shape: the route's natural output is not a margin in disguise but a closure about itself, and the expected spectral gap was an artifact of the classical analogy (sub-shifts of finite type earn their gaps through branching that a lossless, sufficient-statistic dynamics cannot have). Three routes into the strand census are

now closed—the necessary-condition families, the marginal-rigidity route, and the (single-anchor) automaton—the first two by measurement, the third by theorem *within its stated scope*. The empirical half of the reading (S1) is verified a second time by a second instrument: the one margin-shaped candidate, fully specified and executed, emitted exact census computations and a k-invariance, and no margin. The tool-reach half (S2) has its first proved instance—every automaton built on a bijective drift is census-shaped—which is a mechanism, not yet the general claim; instruments that branch (the general strand-census operator) or do not inherit the drift remain outside its scope, and nothing here is claimed about them. With this, the program's state is fixed, and it is the state the conditional theorem was stated to name: Theorem 2.1 with its ledger; the type-mismatch diagnosis, now with its mechanism *at the single-anchor level* (losslessness forbids generated decay for that operator); and a named residual—the growth-in-k margin, uniform over families, size tuples, and k—that the single-anchor toolkit is not the right shape to close, with the general strand-census route left open. The reformulation closes the covering obstruction exactly where classification suffices, and the conjecture-level consequence is the one stated in the introduction: within this framework the residual is a structural gap for the single-anchor toolkit, not a technical one for that toolkit; closing it in general is work for a different instrument, and the general strand-census route is left open.

The disposition of that verdict is now on record, and it fixes the program's final state. The natural external instrument for the missing margin—the Ehrhart theory of the Lonely-Runner zonotope, the third bridge of §11—was engaged as a full strand and executed to certificates. Its arithmetic is unimpeachable: the h^*-polynomials of both canonical families are real-rooted, log-concave, and Newton through $n = 10$, in exact rational arithmetic throughout. Its geometric reading is false: the deepest-hole form fails from $n = 5$ with exact witnesses (the harmonic instance closed *exactly* in the fourth revision, $\rho(1, 2, 3, 4, 5) = 16/47$; the two $n = 6$ boundary rows resolved as strict refutations, $\rho \geq 17/47$ at $m = 12$ and $\rho \geq 1793/5000$ at $m = 9$), and the modular law it suggested—deepest hole at the center exactly when $n \mid m$—holds exactly at $n = 3$ ($m \leq 24$), is certified at $n = 4$ (through $m = 16$, value $9/34$), breaks at $n = 5$ (the harmonic $m = 5$ and $m = 15$ refuted exactly, $\rho \geq 97/288$ and $6/19$; the surviving multiples $m = 10, 20, 25, 30$ certified at $7/22$, $13/42$, $4/13$, and $19/62$ respectively—$m = 20, 25, 30$ in the first certification and $m = 10$ closed by the companion paper's third attempt, which paired a heavily scaled random-local search with an exact Lipschitz box certificate and emptied the tree), and is extinct at $n = 6$ (every tested instance refuted exactly, through the $4n$ dilate, the narrowest margin $71/208$ against $17/50$). The instrument that most faithfully preserves the instance's arithmetic is again the one that cannot produce the margin: the type mismatch is a property of the lossless encodings themselves, not of one toolkit. The record is accordingly published in two parts—this monograph (the exact identities, the classifications, the conditional theorem with its ledger) and a companion paper [22] that works out the negative half: the four falsified strengthenings, the witness archive, the certification ladder above with its negative control, and the h^* addendum. The scope statement of the abstract applies to both parts unchanged.

10 Measured coverage and the decay curve

10.1 The battery and its status discipline

The measurement engine runs a fixed *battery* of certification flags on every pair of every speed set. The distinction that matters is between two kinds of flags:

Two batteries are reported. The *same* battery is the one committed at the $n = 4$ rung, ported verbatim (P4 at $\{7, 13, 17\}$). The *augmented* battery adds the classifications of §5.3 (P4 at $\{7, 13, 17, 19, 37\}$). Headline numbers are *ports* (P1+P2+P4) and *all proved* (adding P5); the verified tier (V1/V2) is reported separately and never silently merged.

Table 14: The battery. Proved lemmas are uniform in the modulus (N-free statements, §4–5); verified items are exact exhaustive checks for each fixed modulus or modulus-and-signature—rigorous for that modulus, but not uniform.

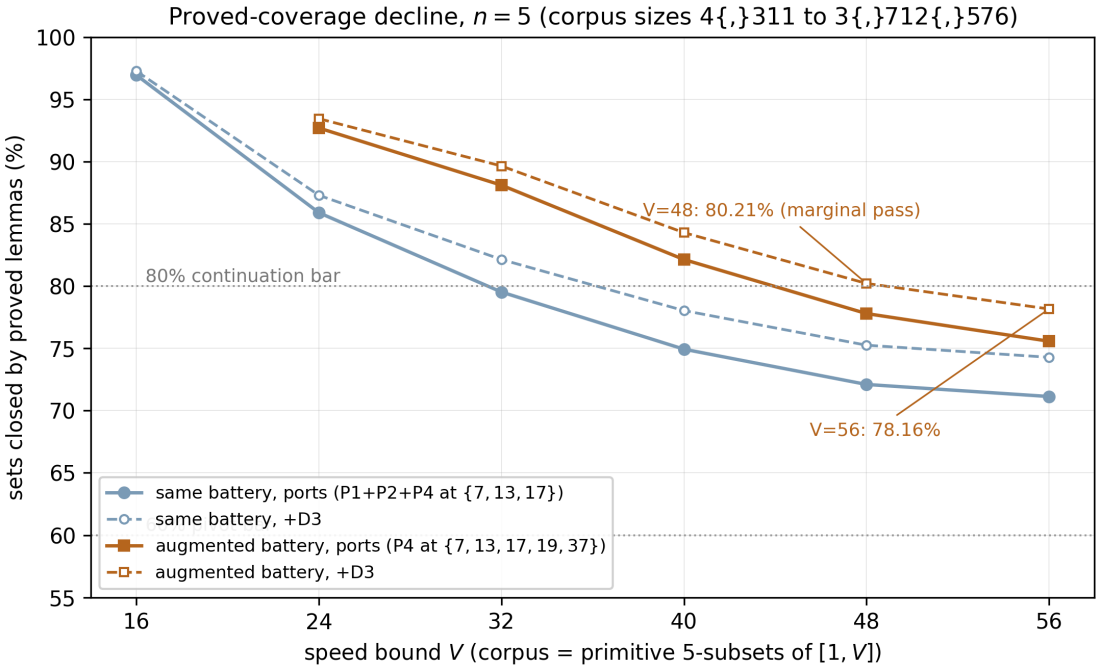
flag	statement	status
P1 polygon	$j \mid N$, $2 \le j \le T$, all residues $\not\equiv 0 \pmod{j} \Rightarrow k = N/j$ certifies	proved
P2 counting	$\sum_w \lvert B_w \rvert - (n-2) < N \Rightarrow$ certifies	proved
P4 classifications	at the rigid primes: covering iff the classified class condition holds	proved
P5 D_3	antipodal-coincidence counting (Lemma 4.4)	proved
V1 safe modulus	all residues invertible and N not unit-capable	per-modulus exhaustive
V2 safe signature	the pair's order signature never covers at N	per-(N, σ) exhaustive

10.2 The corpora and the coverage tables

The corpora are *all* primitive speed sets (gcd = 1) with $n = 5$ sspeeds bounded by V: 4,311 sets at $V = 16$, 41,656 at 24, 196,751 at 32, 641,166 at 40, 1,665,356 at 48, and 3,712,576 at 56—an 861-fold extension of the matched-range base. The $n = 4$ control corpus (all 1,745 primitive 4-sets at $V \le 16$) is run through the same engine with the $n = 4$ ports, reproducing the committed record exactly: 1,651 sets closed by proved ports (94.61%), 1,661 by all proved (95.19%), 1,744 with the verified tier, and exactly one open set, $V = (3, 5, 8, 13)$—identified, not merely counted. That set remains the single most instructive object of the base rung: its only live pairs are $(5, 13)$ and $(8, 13)$ with effective triples summing to $8 = 3 + 5$, a genuine kernel-arc covering at $N = 16$ that certifies at $N \in \{18, 21\}$.

Table 15: Proved coverage of the $n = 5$ corpora. The column ports means P1+P2+P4; the column $+D_3$ adds Lemma 4.4; the column $+V$ adds the verified tier. Open sets: zero at every range, for both batteries. The augmented battery is undefined at $V = 16$ (the committed base rung predates the K_{19}/K_{37} classifications).

		same battery (P4 at $\{7, 13, 17\}$)		augmented (P4 at $\{7, 13, 17, 19, 37\}$)		
V	sets	ports	$+D_3$	ports	$+D_3$	+V
16	4,311	96.94%	97.26%	n/a	n/a	100%
24	41,656	85.89%	87.29%	92.69%	93.44%	100%
32	196,751	79.52%	82.13%	88.10%	89.64%	100%
40	641,166	74.93%	78.03%	82.13%	84.29%	100%
48	1,665,356	72.10%	75.25%	77.79%	80.21%	100%
56	3,712,576	71.13%	74.28%	**75.57%**	**78.16%**	100%



Two framing statements, both true, must be carried together. At *matched range* ($V \leq 16$, the overlap region of 4,311 sets), the reformulation performs at least as well one rung up: 96.94% proved ports at $n = 5$ against 94.61% at $n = 4$—the strongest single datum against the worry that the base rung closed by accident of dimension. At *growing ranges* the proved rate drops, and the drop decomposes exactly: at $V = 56$ on the same battery, the 1,798,577 certifying pairs that carry no proved flag split into 593,148 all-invertible pairs at the then-unclassified primes $19, 37$, plus 1,205,429 kernel-world pairs at composite moduli (861,988 at $N \leq 47$, 267,331 at 48–79, 65,626 at 80–95, 10,484 at 96–111). After augmentation the first bucket vanishes identifiably—the self-check "unflagged all-invertible pair at 19 or 37" returns zero at every range—and the residual is *purely* the kernel world: mixed order signatures at composite moduli, one named class. The classifications close a shrinking share of the verified tier ($48.4\% \to 42.0\% \to 28.5\% \to 20.0\% \to 15.1\%$ of it over $V = 24 \to 56$) as the incidence of 19/37 pairs saturates; this is why the augmented curve decays even though the rigid zone itself never grows (§5).

10.3 The empirical pair-selection law

The verified tier closes everything, and it does so through a striking empirical law that deserves separate statement: *at every range measured, every single speed set has at least one certifying pair whose* $(N, \textit{order signature})$ *configuration is never-covering*—that is exactly why V2 closes 100%. On the full $V = 56$ corpus, all 3,712,576 sets have a certifying pair (the ground truth of (TAU-$_5$) itself, now verified on every primitive 5-set with speeds ≤ 56), the per-modulus capability and signature-capability tables leave no set unclosed, and the hard core—certifying pairs at *capable* configurations—exists (1,205,429 such pairs corpus-wide, all kernel-world) but is never load-bearing. Why a safe configuration is always available is not understood; it is the empirical face of the pair-selection law, the first open problem of §11.

10.4 Interpreting the decay curve

The reformulation's proved coverage declines as the corpus grows, and we report the decline in both directions—what it does not establish, and what it does. On the augmented battery, $+D_3$ coverage runs $93.44\% \to 89.64\% \to 84.29\% \to 80.21\% \to 78.16\%$ over $V = 24 \to 56$; on the same battery, $97.26\% \to 87.29\% \to 82.13\% \to 78.03\% \to 75.25\% \to 74.28\%$ from $V = 16$. In residual terms the augmented series reads $6.56\% \to 10.36\% \to 15.71\% \to 19.79\% \to 21.84\%$,

with increments +3.80, +5.36, +4.07, +2.06 points per 8 units of V: roughly linear growth with fluctuation. Five data points cannot distinguish asymptotic deceleration from a noisy linear trend—the last increment being the smallest is consistent with either reading—and we fit no trend and claim none.

What the series does establish is a level fact. The $V = 48$ point stood at 80.21%, a marginal pass of the program's 80% continuation threshold (by 0.21 points); the $V = 56$ point falls below it at 78.16% (1.84 points under, with augmented ports at 75.57%). The proved coverage of the present battery did not stabilize within the tested range, and the paper's framing follows that fact. What remains true at every range measured: open sets are zero, per-modulus verification closes everything, every set has a certifying pair at a never-covering configuration, and the residual is *purely* the kernel world—after the K_{19}/K_{37} classifications, not a single unflagged certifying pair lives at a classified prime (the self-check "unflagged all-invertible pair at 19 or 37" returns zero at every range, both batteries). Within the tested range the reformulation is productive; beyond it, the proved coverage of this particular battery decays. Closing the residual requires a kernel-world structure theorem—the analogue for composite moduli of what the class reduction and the K-classifications provide for the unit world—and that theorem is the single open mathematical problem the reformulation now has. Whether the decay is *fundamental* (an intrinsic scaling limit of the method) or an artifact of the missing lemma family is open problem 6 of §11; the present data does not decide it.

Remark 10.1 (the plain regime, measured correctly). Two notions of "plain" must not be conflated. A best-pair census (does the single highest-τ pair lack polygon structure?) put the plain share at 73% at $n = 5$; the battery-relevant any-pair census (does *no* pair of the set have polygon structure?) gives 34.3% at $V = 24$, 36.0% at $V = 48$, and 35.2% at $V = 56$, against 38.3% at $n = 4$. The plain share does not grow with the rung on the battery basis; and the plain regime is not monolithic—proved lemmas beyond the polygon one close a large part of it, and every set that needs the verified tier is plain. The difficulty at $n = 5$ lives in the plain regime, but "there" is a third of the corpus, not three-quarters of it.

11 Open problems

The reformulation sharpens several questions into finite, explicit form, listed in increasing distance from the present state of the framework.

1. **The three open hypotheses of Theorem 2.1.** The paper's own frontier, and now the program's closest obligation. H1g is reduced to the *strand census*: bound the number of arrangements and the phase freedom per arrangement, uniformly over the renormalization orbits (§9.5)—class-by-class work over resonance types, with the ratio-2 binary cascade the dominant first class and uniformity across classes itself a lemma. H2m's sharp form is the *pair-line count*: the Lemma 5.16 three-distance machinery applied to the junction draws, converting the cut law from hypothesis to theorem and pinning γ_0 (the chain-class bound of Theorem 9.9 has the right shape with constants loose by orders of magnitude; the route is to count on the $\sum o = \mathrm{ov}$ slice). H2r is the resonance-budget zone cutoff Λ_0 (data-chosen). If all three close, Theorem 2.1 becomes unconditional and yields a uniform-in-T closure of the p^2 frontier covering obstruction *as a covering statement*; the reading after the obligation sequence (§9.7) is that the reformulation has a stubborn residual, and the theorem is stated so that the residual is exactly what these hypotheses name.

2. **The pair-selection law.** (TAU-$_n$) asserts that some pair certifies; the empirical law of §10 says something stronger and unexplained: in every corpus measured, some pair certifies *at a configuration that never covers*—a never-capable $(N, \text{order signature})$. A structural reason for the ubiquity of safe configurations would be the first general theorem of the program

beyond the base lemmas, and the last ingredient needed to convert per-modulus verification into uniform proof on the unit side.

3. **The kernel world.** After the classifications, the entire residual of every corpus is one class: mixed order signatures at composite moduli (an order-d residue contributes a bad set of size N/d, so moat counting has slack exactly there). The kernel world already contains the base rung's single open set, $V = (3, 5, 8, 13)$, whose live pair obstruction is the covering of $\mathbb{Z}_{16}$ by the effective triple $(3, 5, 8)$. *Post-submission (Remark 5.10):* the structure theory now exists in proved form — the fibration theorem (kernel bad sets are pullbacks of unit bad sets at reduced moduli with the same threshold, so covering at composite N transfers to the divisor lattice of N), together with the no-unit-assistance theorems in their provable part. The p^2 side is closed *conditional on Input S* (the three-unit cell is empty at every prime, given the four-family classification input; see the next item and §6); the pq side is now carried by the tensor translation of §7 — verified empty to $N \leq 143$ and reduced to the same input structure, a single unbounded input after the discharge of Input E — with the rigid-zone lifts the remaining open piece.

4. **The composite micro-cases, restated.** The original form of this problem — "no covering exists at 49, 77, 91; the obstruction is unexplained" — is superseded (Remark 5.10): kernel-speed coverings exist there and are exactly the transferred rigid-zone configurations, and the unit safety of 77 is proved. The corrected open problem is the *no unit-assisted covering conjecture*: at $N = p^2$ and $N = pq$ with $p \geq 5$, no covering family of at most four distinct bad sets exists in which the kernel subfamily does not already cover. Proved: families with at most two unit members (fiber-cap counting, with the sharp closed-form cap $2\lfloor s/6 \rfloor + 1 + [s \equiv 5 \pmod 6]$ for fibers of prime size s); three unit members (an inclusion–exclusion identity forced to an exact partition, killed by the Dirichlet floor $|A \cap \lambda A| \geq 2$ on arc intersections); pure-unit families (reduced, via the p-multiples, to the rigid zone); and, post-submission, the entire p^2 side — the $(1K, 3U)$ cell is empty at every prime $p \geq 5$ via the trichotomy lemmas (proved for $p \geq 17$ resp. $p \geq 31$ *unconditionally*: the one-foot input is discharged by Lemma 5.16; finite-case-analyzed at the boundary primes; false and directly checked at $7, 13$) plus the shear-rigidity theorem for $\pm$-colliding units (linear per-fiber drift against pinned covering configurations; verified at every prime $11 \leq p \leq 113$ with exhaustive ground truth at $p \leq 43$) — and, the pq side of the two-unit and three-unit cells: the $(1, 1, 2)$ cell and both $(1K, 3U)$ orientations are verified empty at every $N = pq \leq 143$ (10 semiprimes, over a million family checks, margins 2–18), and the row translation (Theorem 7.1) reduces their closure to the same remaining input, with the row census and the pinning of §7.3–§7.4 as the measured base. Open: the rigid-zone lifts, and the all-p discharge of the single remaining named input. The boundary instance $N = 9$ — a genuine unit-assisted covering in exactly the closed p^2 cell shape — shows the hypothesis $p \geq 5$ is necessary. The named computational inputs are now precise objects (§6.4): the foot-count proposition *in its consumed form* (the one-foot trichotomy at the two size pairs $(2k{+}2, 2k{+}1)$ and $(2k{+}1, 2k{+}1)$, viable differences confined to $\{2, (p-1)/2\}$) is *proved* — Lemma 5.16, the three-distance foot count, clean at every prime $p \geq 31$ and cross-checked against the census to $p \leq 419$; the two-foot version is consumed by no current proof — leaving the pinning-size lemma (relative-offset sets of at most 6 elements with covering counts 28 and 68 independent of p, verified to $p \leq 113$, needed by the shear-rigidity closure, and now confirmed from the pq side by Proposition 7.3) as the programme's single unbounded input, dischargeable not by bounded search but by the four-family covering classification (Conjecture 6.7) — which is the programme's route to an unconditional-in-p statement.

5. **Rigid-zone termination.** Is the unit-capable set finite at each rung? Within the verified range (primes ≤ 150, composites ≤ 111) the $n = 5$ zone is exactly $\{7, 13, 17, 19, 37\}$, but no finiteness argument is known, and the $n = 4 \to n = 5$ transition (11 out, $\{17, 19, 37\}$

in) shows the zone is not monotone in the rung. Finiteness at one rung would make the classification program of §5 literally completable.

6. **The successor no-covering conjecture** (Conjecture 5.18): for prime $N \geq 41$, four translates of S_h never cover $\mathbb{Z}_{(N-1)/2}$. Counting never obstructs there ($4(2h+1)-3 \geq N$), so the statement is purely multiplicative, the exact analogue one rung up of the $n = 4$ no-covering statement. The classifications of Theorems 5.2–5.8 are the complete positive side of the ledger: every capable prime ≤ 37, classified.

7. **Is the decay fundamental?** The proved-coverage rate declines with range (§10): on the augmented battery the residual grows 6.56% → 21.84% over $V = 24 \to 56$, with increments +3.80, +5.36, +4.07, +2.06 points per 8 units of V—roughly linear growth with fluctuation, no established trend in either direction. The level fact is sharp: coverage fell below the program's 80% continuation threshold at $V = 56$ (78.16%). The alternatives remain: either the decay is intrinsic to the reformulation (proved coverage has a range-dependent ceiling), or it is an artifact of the battery—specifically of the absence of a kernel-world lemma, since the kernel world is 100% of the residual. Five data points do not decide the question; a kernel-world structure theorem would.

8. **Plain-regime growth.** On the any-pair basis the plain share is stable across rung and range (38.3% at $n = 4$; 34.3%–36.0% at $n = 5$), but the verified-only sets are all plain, and the polygon lemma's share shrinks structurally as T grows (a residue is polygon-certifiable only when some $j \mid N$ with $j \leq T$ avoids all residues). Does the plain share grow with n, and at what rate?

9. **Higher rungs.** At $n = 6$ ($T = 7$, five bad sets per pair) the rigid zone should be denser (more translates available) and the kernel world larger. The program's method—run the battery, map the new rigid zone, generate the classifications the new level demands, replace the ports that die—is exactly what was executed at $n = 5$; whether it survives a third rung is untested. *Post-submission* (§8): the census+shear method *does* survive the third and fourth rungs. At the critical cell of $T = 8$ the fiber census is exactly eleven families with placement counts constant in p (Proposition 8.2, seventeen primes); the $\pm$-colliding-only law holds at both rungs in the mechanism range; the drift law is T-free (Lemma 8.1); the shear step survives the growing pinned sets because the growth is dimensional (solution sets $\approx k^2$, projections saturating at 42 against $|U| = (T-2)k + \epsilon - 1$, exit time ≤ 2); and the frontier cells are empty — $(1K, 4U)$ at $T = 7$ for every prime $11 \leq p \leq 61$, $(2K, 4U)$ at $T = 8$ for every prime $17 \leq p \leq 43$ tested. The rigid-zone lift itself (pure-unit families) remains open at every rung, as does the $T = 8$ $(1K, 5U)$ cell, whose five-difference census is a zoo (840 $\pm$-distinct families at $k = 2$) — the boundary of the method, and now the object of the conditional scaling-closure attack of §9: Theorem 2.1 closes it under H1g and the H2 pair, with the strand census as the named remaining obligation.

10. **Bridges to neighboring methods.** Three external toolkits interface naturally with the reformulation's finite statements. (a) The covering lemmas now have a *proved* bridge: the group-ring translation (§6) turns them into sign conditions on a sparse identity in $\mathbb{Z}[\mathbb{Z}_p]$, proved as Theorem 6.1 and machine-verified on every covering in the $p \leq 61$ census; its census law (four difference families with p-independent placement counts, Conjecture 6.7) is the object the remaining two lemmas must prove, and it sits precisely in the Lam–Leung–de Bruijn–Schoenberg theory of sparse cyclotomic divisibility. (b) The no-covering statements (Conjecture 5.18 and its $n = 4$ ancestor) concern coverings of cyclic groups by *multiplicative arcs*—images of intervals under multiplication. Additive-combinatorial covering theory (Kneser-type theorems, Cauchy–Davenport and weighted variants) is the classical toolkit for covering questions in $\mathbb{Z}_N$; whether any of its forms applies to multiplicative arcs is open, and a positive answer at prime N would settle the successor conjecture. (c) Covering-radius

problems for lattice zonotopes are governed matroidally, and log-concave polynomial methods have turned matroidal counting statements into general theorems (Mason's ultra-log-concavity conjecture being the template); whether the pair-sum covering problems admit a zonotope–matroid encoding whose associated polynomial has a stability property is the corresponding question here—if they do, the kernel world would be its near-equality cases, and the sign-mixed sparse divisibility of §6.5 is the natural first candidate. The first bridge is exploited; the other two are recorded as directions, not claims. *Post-submission* (final revision): the third bridge was subsequently engaged as a full strand and is now closed. The zonotope–Ehrhart encoding exists—the Lonely-Runner zonotope of the instance at denominator N, its dilate family $mZ(V)$—and its arithmetic is entirely stable: the h^*-polynomials of both canonical families (the harmonic instances $v = (1, \ldots, n)$ and the rung-2 instances $v = (1, \ldots, n-1, 2n)$) are real-rooted, log-concave, and Newton at every $n \leq 10$, verified in exact rational arithmetic at 45 digits [22]. The geometric control the bridge asked for, however, is false in its natural form: the deepest-hole reading—the covering optimum attained at the zonotope's center, $\rho = \delta$—fails from $n = 5$ onward with exact rational witnesses (at the harmonic $n = 5$ instance $m = 5$, $\rho = 16/47 > \frac{1}{3}$ *exactly*, closed in the companion paper's fourth revision by a balance-family witness and two new box certificates—the search record $97/288 \to 607/1792 \to 4183/12288$ retained as provenance; the two $n = 6$ boundary rows resolved as strict refutations, $17/47$ and $1793/5000$), the modular law that would govern it (deepest hole at the center exactly when $n \mid m$) breaks at $n = 5$ (the odd multiples 5 and 15 fail exactly; the even and higher multiples $10, 20, 25, 30$ are certified, $m = 10$ closing with the companion paper's third revision) and is extinct at $n = 6$, and the strictness of h^* neither tracks nor predicts the covering gap (rank correlation at the noise level). The bridge's question is thus answered in the negative for the natural encoding, and with the same shape as the internal program's boundary: arithmetic fidelity, no margin. The witnesses, the exact certification ladder, and the h^* addendum are the subject of the companion paper [22].

12 Reproducibility and soundness record

Every empirical and classification claim in this paper rests on exact integer computation, and every computation is paired with a control or an independent reimplementation. This section records the checks; the committed scripts, logs, and experiment notes are bundled with the paper [17]. The record matters beyond convention here: a measurement program on 10^6-scale corpora is exactly the setting where silent bugs masquerade as findings, and the discipline below is what separates the two.

Table 16: The verification record. All counts are of exact checks with the stated outcome; "0 mismatches" means exact agreement of two independent computations.

check	scope	outcome
size law (Lem. 4.1)	all residues, $N \leq 111$ (T=6), $N \leq 31$ (T=5)	0 violations
classification predicate $\iff$ mask covering	exhaustive over unit multisets at $7, 11, 13, 17, 19, 37$ (126–82,251 per modulus)	0 mismatches
K_{19}/K_{37} class list $\iff$ covering	all unit 4-multisets, both directions (5,985 and 82,251)	0 mismatches
battery soundness (flag $\Rightarrow$ certify)	exhaustive $N \leq 31$ (46,371 at n=4; 324,626 at n=5); exhaustive $N \in \{37, 49, 81, 95\}$ (91,390; 270,725; 1,929,501; 3,612,280); exhaustive $N \in \{99, 111\}$ (4,249,575; 6,672,876); random 32–95 (256,000); random 96–111 (64,000)	0 violations
polygon constructive witness	sampled across all corpora	0 failures
reduced-optimum cross-check	direct grid maximization vs. bitmask, sampled	0 mismatches

Table 16 (continued)

check	scope	outcome
lift law (Lem. 4.5)	random lifts	0/2,727
equality theorem (Thm. 3.2)	fresh non-circular check (folklore breakpoints vs. pair lattices), 885/885; from-scratch independent reimplementation (time domain, differences included, $n = 2$–6, $v \le 200$), with exact anchors	2,870/2,870
ground truth of (TAU-$_5$)	every primitive 5-set, $v \le 56$	3,712,576/3,712,576
trichotomy proof, line by line	every intermediate claim of Lemma T (non-wrapping, span bounds, two-lattice descent, two-block identity) at all primes $p \le 250$, plus the run lemmas at $p \le 37$	0 violations
colliding-unit ground truth	all $\pm$-colliding $(1K, 3U)$ families, both collision topologies, $p \le 43$	3,3 M families, 0 coverings
shear linearity	per-fiber t-space offsets of colliding lifts, all v_2, m, j at $p \le 37$	0 violations of $\sigma_3 - \sigma_2 = -\bar{u}mj$
boundary finite case analyses	Lemma B at 11; Lemma A at 19: every viable configuration certified, independent brute force in agreement (and reproducing the known failures 50 at 7, 2 at 13 as witness-tuple counts; the $17, 31, 37$ analyses are redundant with the extended proof ranges)	exact
group-ring identity (Thm. 6.1)	36,308 checks: all 52 Lemma A exceptions (E-profiles exactly as classified), every covering of the census ($11 \le p \le 61$) plus boundary $p = 5, 7, 13$ (12,256), 24,000 random configurations incl. non-coverings (sharp signed form; sign $\Leftrightarrow$ covering; kernel $= \mathbb{Z}\overline{\Phi}_p$)	all pass
covering census (Prop. 6.4)	all critical coverings over all $(d_2, d_3) \in (\mathbb{Z}_p^*)^2$, normal form, $p \in \{5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61\}$	four families, counts constant in p; 0 $\pm$-distinct coverings at $p \ge 11$
foot census (Input E)	every prime $p \le 419$, both classes, every critical (s_1, s), $\pm$-reps $d \ge 2$: one/two-foot viable sets; the M4/M5 consumed pairs; Lemma B's contained census at $(6, 6)$	consumed form clean from $p \ge 31$; now *proved* (Lemma 5.16)
Lemma E machinery (Lemmas 5.11–5.16)	every (p, d, s) with $p \le 500$ (45 primes, 10,410 triples): column decomposition, band and block identities, the pair-sum identity as an *exact* equality, the auxiliary bounds; the theorem direct to $p \le 2000$ (136,806) and, via the identity, to $p \le 4000$ (367,986); endpoint/dirty-prime values exact; 76/76 viable-set agreement with the foot census	all hold
parity mechanism (Prop. 6.5)	every census $(2, 2)$ solution at $p \in \{23, 29, 41, 31, 37, 43\}$: integer-parity runs, non-wrapping, spills $\le$ ov in A_1; per-(multiset, family) counts	100%; counts = Table 8
pq translation (Thm. 7.1)	constants; footprint formula (4,394 set equalities, 8 moduli); row census (24 resp. 312, 14 primes); $(1, 1, 2)$ cells (144,138 checks, 11 moduli 35–143, including $N = 133$); $(1K, 3U)$ cells (989,000 checks); pinning (37,672 rows, ≤ 6 good rows); T-10 reproduction at $77, 91$	all pass; 0 coverings
higher-rung machinery (Lem. 8.1)	three-tier arc law vs the committed $T = 6$ tables ($p \le 37$, both classes) and at 19 primes of $T = 7/T = 8$ (all residue classes, every a_0); footprint formula vs direct Z_{p^2} enumeration; drift identity at $T \in \{7, 8\}$	957/957; 3,528/3,528
$T = 8$-critical census (Prop. 8.2)	17 primes, all four odd ϵ classes, $k \le 17$: exactly 11 families, counts 288 ($\epsilon \in \{3, 7\}$) resp. 2,784 ($\epsilon \in \{1, 5\}$) constant in p; 0 $\pm$-distinct in the mechanism range	exact
$T = 7$ census (Prop. 8.3)	18 primes, all six ϵ classes, $k \le 11$: core-eleven dominance ($\approx 90\%$ at $k \ge 6$), tail decay $183 \to 43$, 0 $\pm$-distinct at $k \ge 4/k \ge 2$	exact

Table 16 (continued)

check	scope	outcome
higher-rung cells (Prop. 8.4)	$(1K, 4U)$ at $T = 7$: census-restricted ground truth (distinctness-filtered) at every prime $11 \le p \le 61$ — 0 coverings, best exit 2; brute-force control at $p \in \{5, 7, 11\}$ agreeing exactly at the degenerate $p = 5$ (8 coverings both ways); $(2K, 4U)$ at $T = 8$ at $17 \le p \le 43$ — 0 coverings; sampled controls 20K families at four primes	exact
$n = 4$ control	full record incl. the open set $= (3, 5, 8, 13)$	exact
range controls, both batteries	$V = 16/24/32/40/48$, all tiers, unflagged pair totals and decompositions	exact vs. committed logs
capability controls	zones, counts 20/68/16/64/32, safety of 48–111	exact
independent capability recounts	set-based scanner at all danger moduli (27 moduli)	exact agreement

12.1 Controls

Every experiment run begins with hard assertions against previously committed numbers before any new number is produced: the $n = 4$ corpus reproduces the committed record (including the identity of the open set); the $n = 5$ corpora at $V = 16, 24, 32, 40, 48$ reproduce every committed tier count for both batteries, the corpus-wide unflagged-pair totals and decompositions included; the capability scan reproduces the committed zones, the per-modulus covering counts, and the safety of 48–111 (the composite range extended twice, both times with the new moduli re-derived by the independent set-based scanner); and the signature-capability maps for 48–79 and 80–95 match the committed dictionaries exactly. The classifications were additionally anchored by hand: the $\mathbb{Z}_{17}$ covering of Example 5.6 was expanded by hand, the D_3 falsifier $(1, 1, 2, 3)$ on $\mathbb{Z}_7$ and its exclusion $N \equiv 1 \pmod 6$ were checked by hand, and the K_{19}/K_{37} covering counts were reproduced by three independent implementations (bitmask, frozenset, exponent-space).

12.2 The cross-validation discipline

The program's standing rule is that every empirical claim must be checked against committed data or a reference implementation before being cited. The rule earned its keep: across the program's lifetime it caught, before any affected number was recorded, a divisibility test written as a bitwise OR; a sampling artifact in which a 720-point time grid silently missed denominators like 11 and briefly masqueraded as counterexamples to the equality theorem; an unsound counting bound for mixed-order residues (245 flagged-but-non-certifying pairs, all caught by the flag-implies-certify sweep); an off-by-one in the capability scanner that contradicted the committed $n = 4$ zone; a flag-parsing slip; a verified-tier flag leaking into a proved tier (caught by a too-good "100%"); a diagnostic accumulating corpus-wide instead of per-tier (caught by an impossible count); a mislabeled canonicalization of cyclic translate classes (contains-0 representatives are not orbit representatives on $\mathbb{Z}_m$); and an unsound attempt to quotient the classification by exponent inversion (class reflection $e \mapsto -e$, which fixes the bad set only through $B_{-w} = B_w$ and is not a covering symmetry); an off-by-one in the minimal-arc length of the census fit test (caught by the assertion that every enumerated covering has $E \ge 0$, fixed, and then confirmed against a no-prune brute force); and a count-semantics reconciliation forced by the census run (the earlier 50/2 at $p = 7/13$ are witness-tuple counts, one per (d_2, d_3, a_2) with a completing placement, while the true solution counts are 272 normalized configurations $= 68$ set-triples and $8 = 2$ — the brute force confirms both readings, and the identity check ran on the full solution set). Each fix was verified by the control that caught it. We report these not as penance but as evidence that the numbers above survived an adversarial process.

12.3 Artifacts

The bundle [17] contains the equality-theorem verifications (`V1_equality_fresh_check.py` and the from-scratch `EQ_independent_verify.py`); the independent capability cross-checks (`N5_prime_scan_crosscheck.py` and the set-based scanners); the literature-search scripts and raw results; the experiment notes summarizing each stage; and the corpus and battery engines with their controls, one per stability extension, each with its full run log:

- `N5_leverage_experiment.py` (the committed $n = 5$ battery and rigid-zone scan),
- `N6_stability_experiment.py` (the $V = 32/40$ extension and the composite capability completion to $N \leq 79$),
- `N7_classify_19_37.py` (the K_{19}/K_{37} classifications and the augmented re-measurement),
- `N8_v48_extension.py` (the $V = 48$ data point, both batteries, capability to $N \leq 95$),
- `N9_v56_extension.py` (the $V = 56$ data point, both batteries, capability to $N \leq 111$; staged execution with per-stage state persistence and hard controls re-asserted at every stage),
- `ap_lemmas_verify.py` (the trichotomy/Lemma A/B verification of the companion note: exact lemma search, censuses, capacity structure, cell anchors; $p \leq 250$),
- `trichotomy_linecheck.py` (the line-by-line machine check of the trichotomy proof, including the general-arc variant),
- `pm_colliding_close.py` (the $\pm$-colliding closure: ground truth, fiber structure, shear linearity, pinning tables, pigeonhole),
- `boundary_finite_cases.py` (the boundary-prime finite case analyses with margin certificates and the independent brute-force cross-check),
- `groupring_bridge.py` (the group-ring translation verification: identity checks on all exceptions and census coverings, sharp-form sampling, kernel checks, the four-family census with per-multiset counts, defect statistics, and the witness/solution count reconciliation; output `out_groupring_bridge.json`, log `groupring_run.log`),
- `foot_census.py` (the Input E boundary census: one/two-foot viable sets at every prime $p \leq 419$, the consumed-pair cleanliness, and the Lemma B $p = 17$ form check; output `out_foot_census.json`),
- `lemmaE_master.py` (the three-distance foot-count verification, discharging Input E: column decomposition, band and block identities, the pair-sum identity as an exact equality, the auxiliary bounds of Lemma 5.15, the theorem of Lemma 5.16 direct to $p \leq 2000$ and via the identity to $p \leq 4000$, the endpoint and dirty-prime values of Remark 5.17, and the viable-set cross-check against the foot census; output `out_lemmaE.json`),
- `pq_side.py` / `pq_side2.py` / `parity_check.py` (the pq translation battery: constants and footprint formula, row censuses, $(1, 1, 2)$ and $(1K, 3U)$ cell enumeration with margins, pinning and drift analysis, and the parity-mechanism structural check; outputs `out_pq_side.json`, `out_pq_side2.json`),
- `higher_rung_step1.py` with the resumable `drive_step1.py` (the general-T machinery controls and the higher-rung censuses: the $T = 7$ four-difference census at 18 primes with per-family projections, the $T = 8$-critical census at 17 primes, the $T = 8$ five-difference probe; output `out_higher_rung_step1.json`, log `higher_rung_run1.log`),
- `higher_rung_step2.py` (the shear side at the higher rungs: the drift-law verification, the census-restricted and brute-force ground truths for the $(1K, 4U)$ cell at $T = 7$ and the $(2K, 4U)$ cell at $T = 8$, the exit-time measurements; output `out_higher_rung_step2.json`),

- `higher_rung_step3.py` (the sampled family controls for the two frontier cells; output `out_higher_rung_step3.json`),
- `pool_sat.c` with `gt_pool_brute.py` (the pool saturation censuses of §9.2: the m-general admissibility checker in two cross-validated implementations plus dump/resume stride sampling, and the independent pure-enumeration brute force agreeing on 21/21 sampled tuples across $p = 17/19/23/31$; dumps `dump_p{17,19,23}.tsv`, `pool_T8_p31.tsv`, `dump_T9_p{29,31}_sample.tsv`),
- `h2_pair_check.py` (the zone-resolved cut-quality measurement of Table 12: per-pair γ on the committed cells, 764 pairs; output `out_h2_pair_check.json`),
- `chain_structure_check.py` (the chain-class machine checks of Lemmas 9.7–9.8 and Theorem 9.9: 123,240 solutions, the sum identity, the reach bound, and the count reconciliations),
- `scaling_closure_extract.py` and `scaling_closure_verify.py` (the read-only calibration from the committed JSONs and the targeted lemma verification of §9.3: solution-volume probes, two-fiber dilate law, resonance zones, cascade depths at the committed cells, sign robustness; outputs `out_scaling_closure_extract.json`, `out_scaling_closure_verify.json`),
- `h1_t8_census.py`, `h1_t8_structure.py`, `h1_t8_prime.py`, `h1_t8_path.py` (the $T = 8$ volume census, mechanism probes, and prime extensions of §9.5, with the independent brute-force ground truth (9/9 keys) and the renormalization-orbit verification; states `h1_t8_state.json` / `h1_t8_prime_state.json`, outputs `out_h1_t8_census.json`, `out_h1_t8_structure.json`, `out_h1_t8_prime.json`).

All runs are deterministic; every number in §10–12 appears verbatim in a committed log.

12.4 Data and code availability

The complete artifact bundle (scripts, committed logs, JSON records, and the figures) accompanies this paper and is deposited for public access together with the PDF; the sources build with `tectonic` and a single command. The verification scripts have no dependencies beyond CPython 3.12 with `numpy`; each re-derives its controls from first principles and asserts them before producing any new number, so any single experiment can be re-verified in isolation. This document and the bundle are released under the Creative Commons Attribution 4.0 license (CC BY 4.0), the better to allow the community to re-run, refute, or extend the record.

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